

Lectures on Planetary Atmospheres

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Disclaimer: These lecture notes are a work in progress. Arguments might be unclear, mistakes might be present, and references and credit to other work might not be given.

LECTURE 1

Terrestrial Atmospheres

Home is where the wind blows.
Fred Hoyle (autobiography), 1994.

1.1 PLANETS MODOKI

Planets come in many sizes, but generally only one shape, and a short taxonomy follows.

Planets. A planet is defined to be a body that directly orbits its host star and is massive enough to be in hydrostatic equilibrium (effectively meaning it has formed under its own gravitational force and has a spheroidal shape) and to dominate its own orbit, clearing it of other bodies. In so far as this definition is official (i.e., as defined by the International Astronomical Union, or IAU) it does not apply to bodies in other solar systems.

Dwarf planets are bodies that are large enough to form under their own gravity but are not able to clear their orbit of other bodies. Pluto is the most famous example, and Eris (in an orbit beyond Neptune) and Ceres (in the asteroid belt) are two others. More generally, *minor planets* are objects that orbit around a star that need not have formed under their own gravity, including dwarf planets and asteroids but excluding true planets and comets.

Exoplanets are planets in other solar systems. A *rogue planet* is a planetary body that orbits the galactic center and not a particular star.

Terrestrial or telluric planets. These are planetary bodies that, like Earth, have an atmosphere with a distinct lower boundary, often a rocky surface but also possibly an ocean or other distinct change of character. They are usually composed of silicate rocks and metals, and are sometimes called rocky planets. Telluric is from the Latin 'tellus', meaning Earth, and terrestrial and telluric are synonyms in the planetary context.

Giant planets. A giant planet may be defined as any planet at least ten times more massive than Earth (although other definitions may differ slightly), including ice giants, gas giants (both defined below) and massive terrestrial planets.

Modoki is a Japanese word sometimes translated to 'the same but different'. It became a popular English adjective when describing different types of El Niño event. (Modoki is also the name of a comic character who mocks the main character in Japanese performing arts.)

When discussing other planets a 'day', without a qualifying adjective, still refers to 86,400 seconds, and a 'year' refers to 365 days. One may of course refer to such things as 'Venusian day' or a 'Jovian year', in which case the terms apply to the planet in question.

Gas giants. These are giant planets, like Jupiter and Saturn, that are composed mainly of hydrogen and helium and that do not have a sharp interface between atmosphere and solid planet. Gas giants are usually more than 90% hydrogen and helium, although not all of it is gaseous: much of the hydrogen may be in liquid form.

Ice giants. Giant planets that are composed of elements heavier than hydrogen and helium are called ice giants. (Here, ice has a more general definition than usual, and does not only refer to water-ice.) Uranus and Neptune (about 15 times more massive than Earth) are ice giants and have less than 20% hydrogen and helium, the rest being such elements as oxygen, nitrogen and carbonic compounds such as ammonia and methane.

Super-Earths. A super-Earth is a planet with a mass between that of the Earth and that of a giant planet. A super-Earth might in principle be a terrestrial or a gaseous planet, and sometimes they are called ‘mini-Neptunes’.

Hot Jupiters. These are gas-giant exoplanets that are in close proximity to their host stars and that as a consequence may be tidally locked (with one side permanently facing their sun), and so can be expected to have very high surface temperatures on one side, low on the other. Their orbital period may be of order tens of Earth days, and if they are tidally-locked this is also their rotation period around their own axis.

Brown dwarfs. These are hybrids between small stars and gas-giant planets, and may at some stage have undergone nuclear fusion and have elements much heavier than hydrogen and helium.

1.2 EQUATIONS OF MOTION

Equations of motion are:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + \omega \frac{\partial \mathbf{u}}{\partial z} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla p - r\mathbf{u}, \quad (1.1a)$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (1.1b)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) + \frac{\partial \rho w}{\partial z} = 0, \quad (1.1c)$$

$$\frac{\partial \theta}{\partial t} + \mathbf{u} \cdot \nabla \theta + \omega \frac{\partial \theta}{\partial z} = Q_{[\theta]}, \quad (1.1d)$$

where $\mathbf{u}$ is the horizontal velocity and the ∇ operator is taken to be horizontal, meaning at constant pressure and we omit viscous and diffusion terms. The term $-r\mathbf{u}$ parameterizes surface drag and r is large only very close to the surface, and is negligible in the free atmosphere. The term $Q_{[\theta]}$ represents heating due to radiative forcing, and θ is the potential temperature, $\theta = T(p_R/p)^\kappa$ where p_R is the reference pressure, which we take to be the mean surface pressure, p_s , and $\kappa = R/c_p$.

In the hydrostatic approximation the vertical acceleration terms (Dw/Dt) are assumed small compared to the pressure gradient force and we have

$$\frac{\partial p}{\partial z} = -\rho g \quad (1.2)$$

It turns out that this is generally a good approximation when horizontal scales are much larger than vertical scales. It is an *aspect ratio* approximation.

1.2.1 Boussinesq Approximation

In the Boussinesq approximation we ignore the variations of density except when associated with gravity. Thus, we let $\rho = \rho_0 + \delta\rho$ and $p = p_0 + \delta p$ where $\partial p_0 / \partial z = -\rho_0 g$ so that $p_0 = -\rho_0 g z + p_s$, and we obtain:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + \omega \frac{\partial \mathbf{u}}{\partial z} + \mathbf{f} \times \mathbf{u} = -\nabla \phi - r\mathbf{u}, \quad (1.3a)$$

$$\frac{Dw}{Dt} = -\frac{\partial \phi}{\partial z} + b, \quad (1.3b)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \quad (1.3c)$$

$$\frac{\partial b}{\partial t} + \mathbf{u} \cdot \nabla b + w \frac{\partial b}{\partial z} = Q_b, \quad (1.3d)$$

where $\phi = \delta p / \rho_0$ and $b = -g \delta \rho / \rho_0$ is the buoyancy, sometime loosely called the ‘temperature’ in the AFD community!

1.2.2 Pressure Co-ordinates

Using pressure as a vertical co-ordinate is useful when the flow is hydrostatic. Rather loosely, variables are replaced as follows: $z \rightarrow p, w \rightarrow \omega, p \rightarrow \phi$ where $\omega = Dp/Dt$ is the equivalent of vertical velocity and $\phi = gz$ is the geopotential. The hydrostatic equation and the mass continuity equation become, respectively,

$$\frac{\partial \phi}{\partial p} = -\frac{RT}{p} = -\frac{R\theta}{p_R} \left(\frac{p_R}{p} \right)^{c_v/c_p}, \quad (1.4)$$

$$\nabla_p \cdot \mathbf{u} + \frac{\partial \omega}{\partial z} = 0. \quad (1.5)$$

where ∇_p is the gradient operator at constant pressure. The thermodynamic equation can also be written as (cf the first law is $c_p dT - V dp = dQ$ and V is like $1/\rho$ and here $Q = 0$).

$$c_p \frac{DT}{Dt} - \frac{1}{\rho} \frac{Dp}{Dt} = 0 \quad \text{or} \quad c_p \frac{DT}{Dt} - \frac{RT}{p} \omega = 0. \quad (1.6)$$

From this

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T - \omega S_p = 0 \quad \text{where} \quad S_p = \frac{\kappa T}{p} - \frac{\partial T}{\partial p} = -\frac{T}{\theta} \frac{\partial \theta}{\partial p}. \quad (1.7)$$

where $\kappa = R/c_p$. This is useful because the vertical temperature gradient is determined by different processes than the horizontal one.

1.2.3 Hydrostatic Balance

Summarizing, we ignore vertical acceleration.

$$\text{Full equations:} \quad \frac{\partial p}{\partial z} = \rho g, \quad (1.8a)$$

$$\text{Boussinesq :} \quad \frac{\partial \phi}{\partial z} = b, \quad (1.8b)$$

$$\text{Pressure co-ordinates:} \quad \frac{\partial \phi}{\partial p} = -\frac{RT}{p}, \quad \text{or} \quad \frac{\partial \phi}{\partial Z} = \frac{RT}{H} \quad (1.8c)$$

$$(1.8d)$$

1.2.4 Geostrophic and thermal wind balance

Geostrophic balance occurs when the Coriolis term balances the pressure gradient. In component form, in Cartesian co-ordinates:

$$fv = \frac{1}{\rho} \frac{\partial p}{\partial x}, \quad fu = -\frac{1}{\rho} \frac{\partial p}{\partial y}. \quad (1.9)$$

In the Boussinesq approximation we ignore the variation of ρ except where associated with g , so this becomes

$$fv = \frac{\partial \phi}{\partial x}, \quad fu = -\frac{\partial \phi}{\partial y}. \quad (1.10)$$

where $\phi = p/\rho_0$. This is almost the same as the pressure co-ordinate version,

$$fv = \left(\frac{\partial \phi}{\partial x} \right)_p, \quad fu = -\left(\frac{\partial \phi}{\partial y} \right)_p. \quad (1.11)$$

with a slightly different meaning for ϕ (i.e., gz and not p/ρ_0).

Thermal wind: Thermal wind balance combines hydrostasy with geostrophy. It is easiest to do this in pressure co-ordinates or the Boussinesq approximation. In the former we get

$$f \frac{\partial u}{\partial p} = \frac{R}{p} \frac{\partial T}{\partial y} \quad (1.12)$$

or

$$f \frac{\partial u}{\partial Z} = -\frac{R}{H} \frac{\partial T}{\partial y}, \quad (1.13)$$

where $Z = -H \log p$ and H is some chosen height scale (arbitrary in this equation). A similar equation arises for $f \partial v / \partial p$.

In the Boussinesq approximation we get:

$$f \frac{\partial u}{\partial z} = - \frac{\partial b}{\partial y}. \quad (1.14)$$

Without making the Boussinesq approximation, in height coordinates we get the rather complicated expression,

$$f \frac{\partial u}{\partial z} = \frac{g}{\rho} \frac{\partial \rho}{\partial y} - \frac{\partial p}{\partial y} \frac{\partial}{\partial z} \left(\frac{1}{\rho} \right). \quad (1.15)$$

If the horizontal variations are primarily due to temperature and not pressure changes, and neglecting the second term on the right-hand side, we obtain,

$$f \frac{\partial u}{\partial z} = - \frac{g}{T} \frac{\partial T}{\partial y}. \quad (1.16)$$

This expression is useful for ball-park estimates. Note that if we use the hydrostatic relation in (1.12) we obtain

$$\frac{f}{\rho g} \frac{\partial u}{\partial z} = - \frac{1}{\rho T} \left(\frac{\partial T}{\partial y} \right)_p, \quad (1.17)$$

or

$$f \frac{\partial u}{\partial z} = - \frac{g}{T} \left(\frac{\partial T}{\partial y} \right)_p, \quad (1.18)$$

(now making explicit the constancy of p in the horizontal derivative), which is very similar. (In many instances you can go from height to pressure coordinates by replacing gH with RT .)

The main point is that the vertical shear is proportional to the horizontal temperature gradient.

1.3 DIMENSIONAL AND NONDIMENSIONAL PARAMETERS

Point of view of someone who can take measurements, mainly of thermal properties, from afar, without knowing the flow in advance.

Rossby number:

The conventional Rossby number is the ration of the inertial force (acceleration) to the Coriolis force (acceleration). Thus

$$Ro = \frac{|\mathbf{u} \cdot \nabla \mathbf{u}|}{|\mathbf{f} \times \mathbf{u}|} \sim \frac{U}{fL} \quad (1.19)$$

But what is U and what is L ? We take U to be the wind that is in thermal wind balance with radiative equilibrium temperature between the equator and pole, and L to be the radius of the planet. Using $f \partial u / \partial p = (R/p) \frac{\partial T}{\partial y}$ gives

$$U = \frac{R \Delta T}{\Omega a} \quad (1.20)$$

so that

$$Ro = \frac{R\Delta T}{\Omega^2 a^2}. \quad (1.21)$$

This is an *external Rossby number* and in particular it is a *thermal Rossby number*. (In many instances you can go from height to pressure coordinates by replacing gH with RT .)

Height co-ordinates We can get the same result in a very ad hoc way by starting with geostrophic balance, namely $\rho f u = -\partial p / \partial y$. Then if we suppose the variation of pressure in y is associated mainly with temperature we get $\rho f U \sim \rho R \Delta T / L$ or, for planetary scales, $U \sim R \Delta T / (a \Omega)$, as before. The use of temperature in lieu of pressure is an implicit use of thermal wind.

1.3.1 Deformation radius

The deformation radius is the distance that gravity waves can travel before feeling the effect of rotation. In shallow water, gravity waves travel at a speed $\sqrt{gH}$ where H is a depth of the fluid; the deformation scale is then:

$$L_d = \frac{\sqrt{gH}}{f} \quad (1.22)$$

To obtain the external deformation radius for a planetary atmosphere we replace f by Ω and must then choose a value for H .

Simplest way: Let H be the density scale height, so $H = RT/g$. Then

$$L_d = \sqrt{RT}/\Omega. \quad (1.23)$$

Stratified fluid: In a stratified fluid then there are internal gravity waves that travel at a speed $\sqrt{g'H}$, where, in a liquid like water, $g' = g\Delta\rho/\rho_0$ is the 'reduced gravity'. In a gas like air we use potential temperature and the usual measure of deformation radius is NH/f where N is the buoyancy frequency. Then

$$N^2 = \frac{g}{\theta_0} \frac{\partial \theta}{\partial z}, \quad \text{so that} \quad N^2 H^2 \sim H \frac{\theta}{\theta_0} \sim R \theta_0 \quad (1.24)$$

and we obtain $L_d = \sqrt{R\theta}/\Omega$ much as before. (Note that, for a stratified fluid at large horizontal scale, $\omega^2 = k^2 N^2 / m^2$ so that $c = N/m \sim NH$. Hence $L_d \sim NH/f \sim \sqrt{R\theta}/\Omega$.)

To do: Check vertical scales and difference between T and ΔT .

Burger number

The Burger number is defined as the square of the ratio of the deformation radius to the scale of motion of the fluid. For the external Burger number

we may define

$$Bu = \left(\frac{L_d}{a} \right)^2 = \frac{RT/\Omega^2}{a^2} = \frac{RT}{\Omega^2 a^2}. \quad (1.25)$$

This is similar to the external Rossby number, except for a difference in the temperature to be used: the Rossby number uses a horizontal temperature difference and the Burger number a temperature associated with the vertical profiles. (xxx Check with Read et al).

1.3.2 Notes on the Thermal Rossby Number

The thermal Rossby number is an external Rossby number, since it depends on the forcing and composition and not the solution. It differs from planet to planet for three main reasons:

- (i) The distance from and the strength of the host star — the further a planet is from its host sun, the smaller will be θ_H . This is because $\theta_H = \bar{\theta} \Delta_H$, and Δ_H is essentially a geometric factor associated with the sphericity of a planet. Thus, as the mean temperature $\bar{\theta}$ falls, so do θ_H and Ro_E . A planet like Earth with a mean temperature of 250 K may have a 50 K difference between equator and pole, but a planet with a mean temperature of only 100 K may have a proportionally smaller equator to pole temperature difference.
- (ii) The planet's rotation rate and...
- (iii) The planet's size. The size and the rotation rate enter only in the combination Ωa , so that a smaller planet may behave similarly to a larger planet with a slower rotation rate. To see this, imagine a soccer ball that is rotating once every 24 hours, like Earth. We can then certainly imagine that the air flow over the ball's surface hardly knows that the ball is rotating. As with the usual Rossby number, U/fL , larger scales feel the rotation more than smaller scales.

With $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$, $a = 6,371 \text{ km}$, $\theta_H = 50 \text{ K}$ and $\Omega = 7.27 \times 10^{-5} \text{ rad/s}$, the external Rossby number of Earth is about 0.07. Venus has a similar radius and similar net incoming solar radiation but a rotation rate that is over 200 times less. Its atmosphere is CO_2 , with $R \approx 189 \text{ J kg}^{-1} \text{ K}^{-1}$, and, taking $\theta_H = 50 \text{ K}$, the external Rossby number for Venus is about 2000! Mars is an intermediate case: its rotation rate is almost the same as Earth but it is much smaller, with a radius of about 3,400 km. Taking $\theta_H = 40 \text{ K}$ gives $Ro_E(\text{Mars}) = 0.18$. Jupiter has a much bigger radius, about 70,000 km, and a rotation period of only 10 hours ($\Omega = 1.76 \times 10^{-4} \text{ rad/s}$). It is hard to estimate a proper value for θ_H , since much of the heating for Jupiter's atmosphere is internal, but let us suppose that $\theta_H = 10 \text{ K}$ (most likely a significant overestimate). We then find $Ro_E(\text{Jupiter}) \approx 2 \times 10^{-5}$, far smaller than for Earth and a factor of 10^8 smaller than for Venus.

What might the effects of changing the external Rossby number be? We may hypothesize the following effects:

- A Hadley Cell width that increases as the as rotation rate decreases. In the axi-symmetric theory of Section ?? the width of the Hadley

Table 1.1: Planetary and orbital parameters for terrestrial-style solar system bodies with substantial atmospheres, ordered by rotation rate. The values given for the obliquities of Titan and Triton are those of Saturn and Neptune, respectively, with the obliquity of Titan and Triton with respect to their orbits around those planets given in brackets.

Body	Mean radius (km)	Rotation period (hr)	Solar day (hr)	Obliquity (°)
Earth	6371	23.9	24.0	23.45
Mars	3396	24.6	24.7	25.2
Pluto	1151	154	154	119.6
Triton	1353	141	141	29.56 (< 1)
Titan	2575	383	383	26.73 (< 1)
Venus	6051	5833	2802	177.4

Table 1.2: Estimates of the main dimensionless parameters for Earth, Mars, Pluto, Triton, Venus and Titan, including the dimensionless super-rotation parameter s_{max} .

Body	Bu	Ro_T	N_J	s_{max}
Earth	0.02	0.06	1.26	≤ 0.04
Mars	0.04	0.20	0.92	0.16
Pluto	163	4.8	0.16	0.75 - 1.1
Triton	22.4	1.05	0.35	0.3 - 0.6
Titan	11.8	18	0.06	8.5 - 15
Venus	140	370	0.009	55 - 65

Cell scales with the square root of the external Rossby number — if H in (1.39) is taken as a scale height the expression is essentially the same as (??). If the Hadley Cell terminates because of baroclinic instability, then (??) predicts a Hadley Cell width of $(NH/2\Omega a)^{1/2}$. Using (??) one may easily show this scales as the fourth root of the external Rossby number. Thus, in any event, the Hadley Cell should expand as Ω falls.

- The number of midlatitude jets will increase as the external Rossby number decreases.

1.4 EARTH

- The temperature falls monotonically with height to an altitude of about 16 km (in the tropics) or 8 km at high latitudes, before increasing with height. The lower region is called the *troposphere*, above which lies the *stratosphere*, and the boundary between the two is the *tropopause*.
- The temperature also falls monotonically from equator to pole, at the surface falling from about 300 K to 240 K at the pole. The tropopause temperature is more uniform, varying from about 230 K

to 210 K.

- (iii) The surface winds are easterly (i.e., westward) in low latitudes (from about 30°S to 30°N), westerly in mid- and high latitudes, with weak polar easterlies in some seasons.
- (iv) The winds increase in height, especially in midlatitudes, with pronounced westerly jets in both hemispheres centred at about 40° latitude in both hemispheres.
- (v) In the meridional (y - z) plane the circulation in each hemisphere (Fig. ??) is characterized by:
 - (i) A 'direct' *Hadley Cell*, with warm air rising near the equator and sinking in the tropics. The winter Hadley Cell is much stronger and has greater latitudinal extent than the summer one, with the warm air rising at low latitudes in the summer hemisphere, sinking in the tropics in the winter hemisphere.
 - (ii) An 'indirect' *Ferrel Cell*, with cool air apparently rising around 60° and sinking in the tropics.

A direct cell is one that is thermally driven, with warm, buoyant fluid rising and cold fluid descending. An indirect cell may be mechanically driven, as we discuss later.

1.5 HADLEY CELL

1.5.1 Axi-symmetric theory, simplified

Consider a rotating planet with zero obliquity, or consider the annual average situation on a planet with moderate obliquity. The planet is heated at the equator causing air to rise and then move poleward just below the tropopause (which on Earth is a height of about 15 km in the tropics, falling to about 8 km at high latitudes).

Equations of motion

$$\frac{\partial u}{\partial t} - (f + \zeta)v + w \frac{\partial u}{\partial z} = 0. \quad (1.26)$$

The variables in this equation are functions of latitude and height (ϑ and z) only, and not longitude, λ , and $\zeta = -(a \cos \vartheta)^{-1} \partial_{\vartheta}(u \cos \vartheta)$. If the vertical advection is small then a steady solution obeys

$$(f + \zeta)v = 0. \quad (1.27)$$

Presuming that the meridional flow v is non-zero then $f + \zeta = 0$, or equivalently, on the sphere,

$$2\Omega \sin \vartheta = \frac{1}{a} \frac{\partial u}{\partial \vartheta} - \frac{u \tan \vartheta}{a}. \quad (1.28)$$

At the equator we may assume that $u = 0$, because here parcels have risen from the surface where the flow is weak (or mirror symmetry across the

The Hadley Cell is named for George Hadley (1685–1768), a British meteorologist who put forward perhaps the first scientific model of Earth's overturning circulation, in which the air rose near the equator and sank near the pole.

equator). Equation (1.28) then has a solution of

$$u = \Omega a \frac{\sin^2 \vartheta}{\cos \vartheta}. \quad (1.29)$$

This gives the zonal velocity of the poleward moving air in the upper branch of the (model) Hadley Cell, above the frictional boundary layer. Evidently the zonal velocity increases rapidly with latitude: at 20° and 40° the values of u are about 59 and 256 m s^{-1} respectively (look ahead to Fig. 1.1 or Fig. ??), becoming far larger than the observed values at midlatitudes.

Alternate: The angular momentum per unit mass of a parcel of air with zonal velocity u is

$$m = (u + \Omega a \cos \vartheta) a \cos \vartheta, \quad (1.30)$$

and if $u = 0$ at $\vartheta = 0$ then $m = \Omega a^2$ at the equator. If m is conserved on a poleward moving parcel, then (1.30) leads directly to (1.29).

Thermal wind, pressure and Boussinesq systems

Boussinesq:

$$f u = -\frac{\partial \phi}{\partial y}, \quad \frac{\partial \phi}{\partial z} = b \quad (1.31)$$

where $\phi = p/\rho_0$ and $b = g\delta\rho/\rho_0$ is buoyancy. Thus,

$$f \frac{\partial u}{\partial z} = -\frac{\partial b}{\partial y}. \quad (1.32)$$

Pressure:

$$f u = -\frac{\partial \Phi}{\partial y}, \quad \frac{\partial \Phi}{\partial p} = -\frac{RT}{p} \quad \text{or} \quad \frac{\partial \Phi}{\partial Z} = \frac{RT}{H} \quad (1.33)$$

where $\Phi = gz$ and $Z = -H \ln(p/p_R)$. Thus

$$f \frac{\partial u}{\partial Z} = -\frac{R}{H} \frac{\partial T}{\partial y}. \quad (1.34)$$

If $H = RT_0/g$ then this becomes

$$f \frac{\partial u}{\partial Z} = -\frac{g}{T_0} \frac{\partial T}{\partial y}. \quad (1.35)$$

Generally we will use Boussinesq, but we can think of $b \rightarrow gT/T_0$ or $b \rightarrow g\theta/\theta_0$

1.5.2 Temperature distribution in Hadley Cell

$$2\Omega \sin \vartheta \frac{\partial u}{\partial z} = -\frac{1}{a} \frac{\partial b}{\partial \vartheta}, \quad (1.36)$$

This becomes

$$T = T(0) - \frac{T_0 \Omega^2 y^4}{2gHa^2}, \quad (1.37)$$

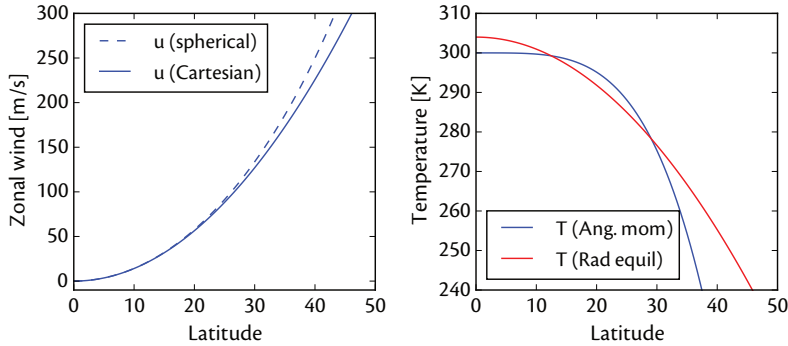


Fig. 1.1: Left: Upper tropospheric zonal wind computed for an angular momentum conserving Hadley Cell. The solid line makes the small-angle approximation. Right: The blue line shows temperature of the angular momentum conserving wind, computed using (1.37). The red curve shows the radiative equilibrium temperature, using (1.38). The Hadley Cell extends to the latitude of the second crossing, at about 30°.

Thermodynamic constraint: Radiative equilibrium temperature given by

$$T_E = T_{E0} - \Delta T \left(\frac{y}{a} \right)^2, \quad (1.38)$$

Equate

Equating (1.37) with (1.38) and solving for y gives the latitudinal extent of the Hadley Cell, y_M , or its equivalent latitude ϑ_M , namely

$$\vartheta_M = \frac{y_M}{a} = \left(\frac{2\Delta T g H}{\Omega^2 a^2 T_0} \right)^{1/2}, \quad (1.39)$$

Main point: Hadley Cell extends further in a low rotation planets – Mars, Venus, Titan.

1.6 NUMERICAL SIMULATIONS

Some results of a numerical integration of (1.1), using a thermal forcing similar to that given by (??), are shown in Fig. ???. The following features are apparent, all consistent with the earlier theoretical explanations.

- The Hadley cells extend further at higher values of the external Rossby number. This property is consistent with arguments in Sections ?? and ??. Concomitantly, the eastward jets at the Hadley Cell edge move to higher latitude as rotation rate decreases. At the lowest rotation rates the Hadley Cell extends almost to the pole, as do the Hadley Cells on Venus and Titan, both of which are very slowly rotating. (In a sense, the atmospheric circulation of these planets is wholly tropical.) The Hadley Cell is very efficient at transporting heat (because its mixing length is so large) and at low rotation rates the equator to pole temperature gradient is small and no baroclinic instability develops.

The more accurate calculation shown in the right-hand panel of Fig. 1.1 computes the equatorial temperature using the requirement that the circulation in the Hadley Cell does not create energy, so that the areas between the red and blue lines within the Hadley Cell sum to zero. This consideration does not change the scaling of (1.39).

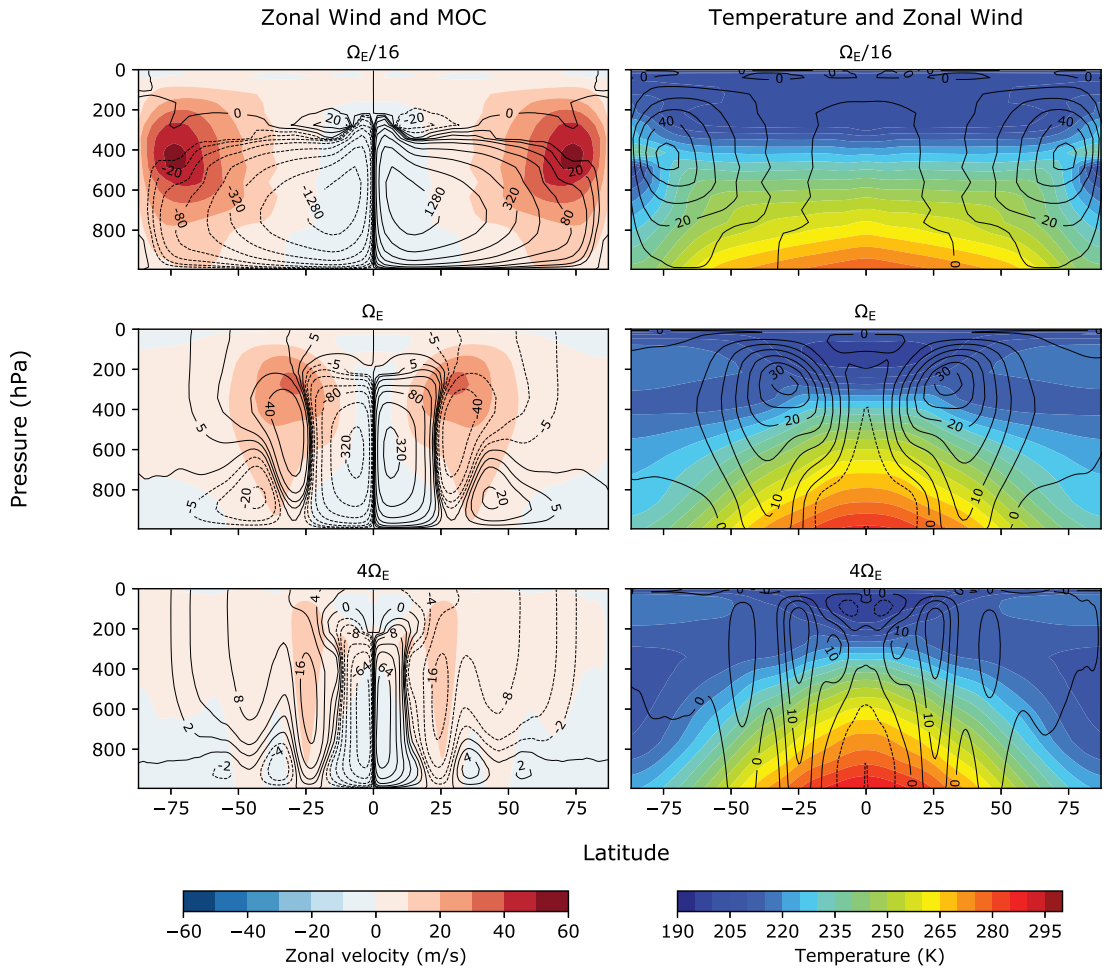


Fig. 1.2: Numerical simulations of a moist, Earth-like planet at varying rotation rates, where Ω_e is equal to the rotation rate of Earth. The thermal Rossby number ranges from about 20 (top) to 0.004 (bottom). The left column shows zonal-mean zonal wind (colours) and mean meridional mass streamfunction (contours). The right column shows temperature (colours) and zonal mean zonal wind (contours).

- At higher rotation rates a distinct extratropical zone develops, with baroclinic eddies and jets that are distinct from the subtropical jets. The eddies in zones transport heat poleward less efficiently than does the Hadley Cell, and the equator to pole temperature gradient increases with rotation. These jets are driven by an eddy momentum convergence (Section ??), which in turn comes from the propagation of Rossby waves away from the baroclinic zones.
- The scale of the eddy motion in midlatitudes decreases as the rotation rate increases, as is apparent in Fig. 1.3. The underlying reason is that, in the absence of changes in stratification, we would expect the deformation radius, NH/f , which is a measure of the linear instability scale, to decrease with increasing rotation.
- At the highest rotation rates, multiple eddy-driven jets can form, rather like the multiple jets discussed in Section ?? and, perhaps, similar to the jets seen on Jupiter and Saturn discussed later. Multiple jets arise because the external Rhines scale decreases as the external Rossby number decreases, and so the jet width decreases.

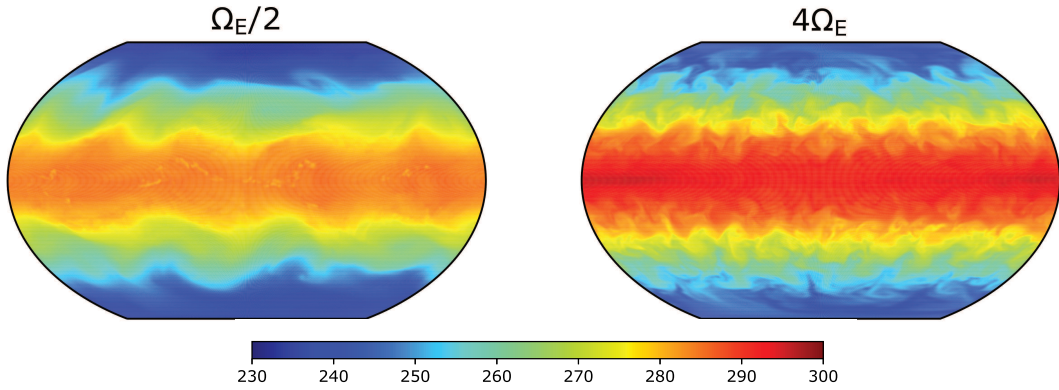


Fig. 1.3: Snapshots of surface temperature (K) in two of the simulations of Fig. ??, with rotation rates half that of earth and four times that of Earth. Two effects can be seen. First, in the simulation at lower rotation, the equator to pole temperature gradient is smaller. Second, the eddy scale is much smaller in the simulation at high rotation. (Simulation and figure by Jake Eager, and similar to a figure in Kaspi and Showman 2015.)

1.7 PARAMETER DEPENDENCIES

1.7.1 Gravity and Atmospheric Mass

Suppose we increase the gravity on a planet like Earth. (For example we make the core out of a denser material.) What happens to the atmosphere, supposing that we keep its mass the same? And what happens if we change the mass?

- (i) If the planet's atmosphere obeys the dry, ideal gas, primitive equations, with simple 'physics', then nothing!
- (ii) If the atmosphere is non-hydrostatic, or deep, invariance is lost (but effects may be small or moderate).
- (iii) If the planet's atmosphere is moist and/or has radiative effects, a lot!

Scaling the dry primitive equations

The momentum equations are:

$$\frac{Du}{Dt} - 2\Omega v \sin \vartheta + \frac{uv \tan \vartheta}{a} = -\frac{1}{\rho a \cos \vartheta} \frac{\partial p}{\partial \lambda}, \quad (1.40)$$

$$\frac{Dv}{Dt} + 2\Omega u \sin \vartheta + \frac{u^2 \tan \vartheta}{a} = -\frac{1}{\rho a} \frac{\partial p}{\partial \vartheta}, \quad (1.41)$$

$$\frac{\partial p}{\partial z} = -\rho g. \quad (1.42)$$

The mass continuity and adiabatic thermodynamic equations are, respectively,

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0, \quad (1.43)$$

and

$$c_v \frac{DT}{Dt} + \frac{p}{\rho} \nabla \cdot \mathbf{v} = 0, \quad \text{or} \quad \frac{D\theta}{Dt} = 0. \quad (1.44)$$

Let us change gravity by the factor α and the surface pressure of the atmosphere by the factor γ . These equations remain invariant under the

following transformation:

$$\begin{aligned} g &\rightarrow \alpha g, & p &\rightarrow \gamma p, & \rho &\rightarrow \gamma \rho, & (T, \theta) &\rightarrow (T, \theta), \\ t &\rightarrow t, & (x, y) &\rightarrow (x, y), & z &\rightarrow z/\alpha, \\ (u, v) &\rightarrow (u, v), & w &\rightarrow w/\alpha. \end{aligned} \quad (1.45)$$

Here α is the change in gravity and γ is the change in surface pressure. If $\gamma = \alpha$ then the mass of the atmosphere has not changed.

Example: the only place where g appears explicitly is in the hydrostatic equation:

$$\frac{\partial p}{\partial z} = -\rho g \rightarrow \frac{\partial(\gamma p)}{\partial(\alpha^{-1}z)} = -(\gamma \rho)(\alpha g) \rightarrow \frac{\partial p}{\partial z} = -\rho g. \quad (1.46)$$

Also:

$$w \frac{\partial u}{\partial z} \rightarrow \left(\frac{w}{\alpha}\right) \frac{\partial u}{\partial(\alpha^{-1}z)} = w \frac{\partial u}{\partial z}. \quad (1.47)$$

The case $\alpha = \gamma$ is equivalent to changing gravity but keeping the mass of the atmosphere fixed. In general, the mass of the atmosphere changes by the amount proportional to $\gamma - \alpha$. [Check this xxx.]

Main point: any changes that might arise in a planet with higher gravity or different mass arise because of changes in the ‘physics’ — i.e. radiation or moist processes — not the fluid dynamics.

Note: It is sometime said that Mars has much higher topography than Earth, so the atmosphere is like a collection of lakes. But because gravity is much less than Earth (1/3rd), vertical distances must be scaled accordingly so that really isn’t so.

Non-hydrostatic equations

In the nonhydrostatic equations the invariance of the gravity scaling fails. Suppose the vertical momentum equation is, for example

$$\frac{\partial w}{\partial t} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g \quad (1.48)$$

Then

$$\frac{\partial w}{\partial t} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g \rightarrow \left(\frac{1}{\alpha}\right) \frac{\partial w}{\partial t} = -\frac{\alpha}{\rho} \frac{\partial p}{\partial z} - \alpha g \quad (1.49)$$

so evidently the non-hydrostatic term scales differently by a factor α^2 . The additional metric terms in the full equations, such as uw/r and the other terms in Dw/Dt , are also not invariant with respect to changes in α . However, the importance of these metric terms (aside perhaps from Titan) is relatively small on the terrestrial planets in the Solar System, since they depend on the ratio of the thickness of the atmosphere to the radius of the planet.

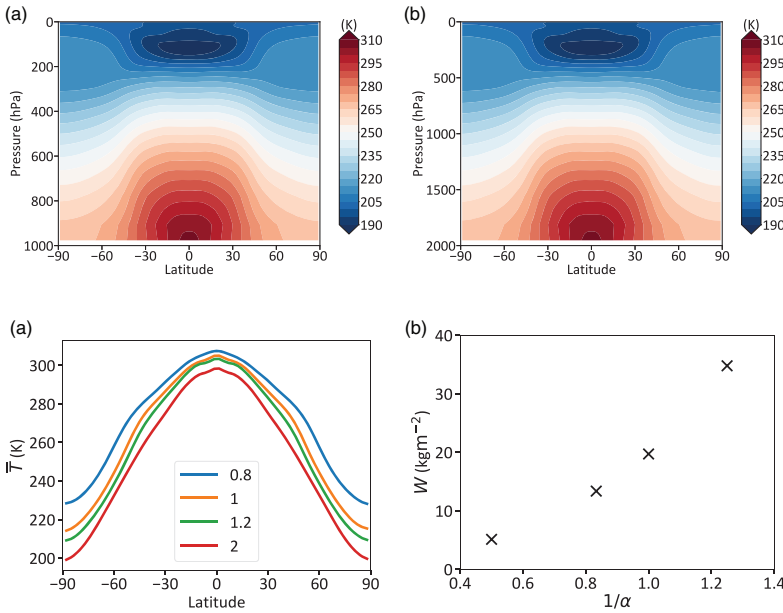


Fig. 1.4: (a) The zonal-mean temperature as a function of pressure and latitude in numerical integration of a dry primitive equation model run with normal Earth gravity. (b) The same but with twice Earth gravity.

Fig. 1.5: ((a) Temperature changes in a moist model as gravity is changed (the inset shows the values of α). (b) Total water content in the same runs.

Moisture and radiative effects

The moisture (or other condensable, such as methane in some atmospheres) will change if gravity changes, and this can have considerable effects on the circulation and temperature, as we now discuss.

If gravity increases then the rate of change of both pressure and temperature increase with height; that is to say, temperature falls off more rapidly with height than before. Specifically, if $g \rightarrow \alpha g$ then

$$\frac{\partial T}{\partial z} \rightarrow \alpha \frac{\partial T}{\partial z} \quad (1.50)$$

where z is the original measure of vertical distance, and we have not scaled this for the new system. For $\alpha > 1$, $\partial T / \partial z$ increase. At a fixed relative humidity the vapor pressure is a function *only* of temperature so that vapor pressure falls *more* rapidly with height than before, and so the total water vapor content of the increased-gravity atmosphere is less than before.

The total water vapour content of a column of atmosphere, W , is given by, using the ideal gas relation for water vapor $e = \rho_v R_v T$,

$$W = \int_0^\infty \frac{e}{R_v T} dz. \quad (1.51)$$

where e is the vapour pressure of the water vapour, R_v is its specific gas constant and T is the temperature. Suppose that we increase the gravity of a planet by a factor α , without initially changing the temperature. The vapour pressure is a strong function of temperature, and at fixed relative humidity is only a function of temperature (since saturation vapour pressure is a function only of temperature in an ideal gas). If gravity increases

These effects were discussed and explored in Thomson and Vallis (2019) and some of the references therein.

by a factor α then the lowest order effect is for temperature to fall with height by a factor α more rapidly than before. That is, *at any given height z the value of e will be lower than before*, and the total water content of the atmosphere will fall roughly by a factor α . Changes in relative humidity can quantitatively alter this conclusion but unless relative humidity also changes by a factor α , which is in most circumstances very unlikely, that change will be small and total water content will fall by the factor α . To see the above argument another way, we transform (1.51) into an integral over pressure and, using the hydrostatic ($dp = -\rho g dz$ and ideal gas ($p = \rho R_d T$) relations for a dilute ideal gas, obtain

$$W = -\frac{1}{g} \int_0^\infty e \frac{R_d}{R_v} d \log(p/p_s). \quad (1.52)$$

where R_d is the specific gas constant for air, p_s is the surface pressure, and the limits of the integral are the same even as g and p_s change. Now, as noted above, e is determined largely by temperature, and the value of temperature at any given value of $\log(p/p_s)$ is, to lowest order, unaltered (as in Fig. 1.4). Thus, the integrand is unaltered by the transformation, but the factor of $1/g$ outside the integral indicates that the total water content will scale by a factor of $1/\alpha$. Of course, once the water content changes the temperature changes because of radiative effects, which causes the water content to change again, so the effect is not a simple one. Nonetheless, the most basic effect that can be expected is that if gravity increases water vapour content will fall. This can have two effects: (i) Since water vapour is a potent greenhouse gas, temperature will then further fall. (ii) the dynamical effects caused by a release of latent heat of condensation will be smaller.

There are additional radiative effects due to pressure broadening. We don't discuss those here.

1.7.2 Moist effects when increasing atmospheric mass

Suppose we increase atmospheric mass, keeping gravity and radiative effects the same but in a moist atmosphere. What happens? In fact, nothing! Note that from (1.52) the total mass of water vapor does not depend on the mass of the atmosphere (assuming that the atmosphere is dilute and that the water vapor pressure is much smaller than the ambient pressure.)

1.7.3 Effects of Changes in Humidity

There are two main effects that a reduction in water vapor content can have:

- (i) A reduction in the greenhouse effect, and a consequent cooling of the atmosphere.
- (ii) A reduction in the dynamical effects arising from the condensation of water vapor.

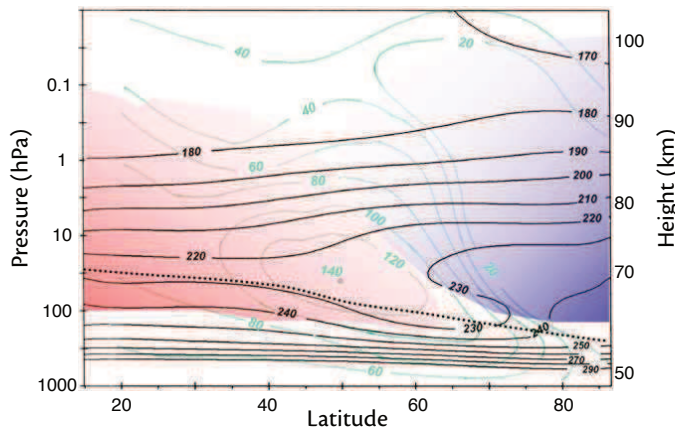


Fig. 1.6: Temperature (black contours, in K) and thermal wind field (green, m/s) as a function of latitude and height for the middle atmosphere of Venus, obtained from an orbital infrared radiometer during the Pioneer mission. The dotted line shows the approximate cloud-top height, and pink and blue colours indicated radiative heating and cooling, respectively. (Adapted from Titov et al. 2013.)

An increase in water vapor content will have the opposite effect. There may also be changes in cloud amount, but we won't deal with those here. The first effect is fairly plain whereas the second is a little subtle.

Changes in Radiative Effects

Specific Humidity

1.8 EXTRATROPICAL STRUCTURE AND ZONAL JETS

We leave this until our discussion of giant planets.

1.9 PLANETS IN OUR SOLAR SYSTEM

We now look at other planets to see if and how the radiative and dynamical discussions of the previous sections and chapters are reflected in their atmospheric structure. We focus on Venus and Jupiter, since these two planets are both quite different, in different ways, from Earth and many other planets in the universe are likely to lie parametrically between them.

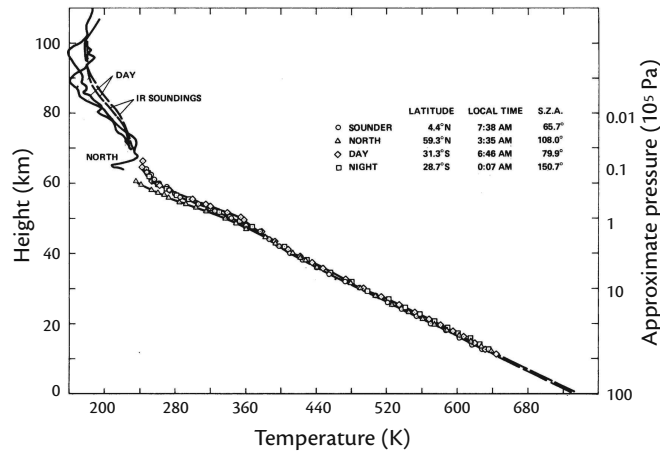
1.9.1 Venus, the Slowly Rotating Hothouse

Venus is a planet that is very similar to Earth in some respects, very dissimilar in others. It is the second planet from the Sun, orbiting it every 225 days. It is thus closer to the Sun (a factor of 0.72) but it is covered by a thick layer of cloud and has a high albedo (about 0.7 vs 0.3 for Earth), with the consequence that its net incoming solar radiation, and hence its emitting temperature, are not very different from those of Earth. Venus is a similar size to Earth, with a radius of about 6,000 km, or 0.95 that of Earth, and a mass of 82% of Earth, and it also has a rocky surface and a gaseous atmosphere. So far, so similar. However, Venus differs in the following striking ways:

- (i) Venus has a much slower rotation rate, with a sidereal rotation period (i.e., the time it takes to rotate about its own axis, relative to the

A sidereal day is the time a planet takes to spin round its own axis, whereas a solar day is the time taken for its sun to appear at the same place in the sky. On Earth the two are almost the same (approximately 23 hours 56 minutes and 24 hours, respectively) but on other planets they may be quite different. The solar day of a particular planet is sometimes called a 'sol'.

Fig. 1.7: Four temperature profiles on Venus obtained by probes as part of the Pioneer mission. The probes were dropped at various latitudes and times but give remarkably similar results. The lapse rate is close to that of the dry adiabat up to tropopause, at about 60 km. (Adapted from Seiff et al. 1979.)



distant stars) of about 243 Earth days. (A Venusian sol, namely the period taken for the Sun to return to the same point in the sky, is a combination of the sidereal day and the rotation around the sun, and is about 117 Earth days.)

- (ii) Venus has a very massive atmosphere, with a surface pressure of about 92 bars (92×10^5 Pa) compared to the 1 bar of Earth.
- (iii) Venus's atmosphere is 95% CO_2 , which is a potent greenhouse gas. This, combined with the large extent of the atmosphere, means that its optical depth is over 100 times higher than that of Earth.

Venus also has a small obliquity (about 3°) so that seasonal effects are small. Given the above, what can we posit about Venus and how does that compare to observation?

Temperature

A soothing exercise in geometry will reveal that the length of a solar day, Sol , and the length of a sidereal day, Sid , are related by

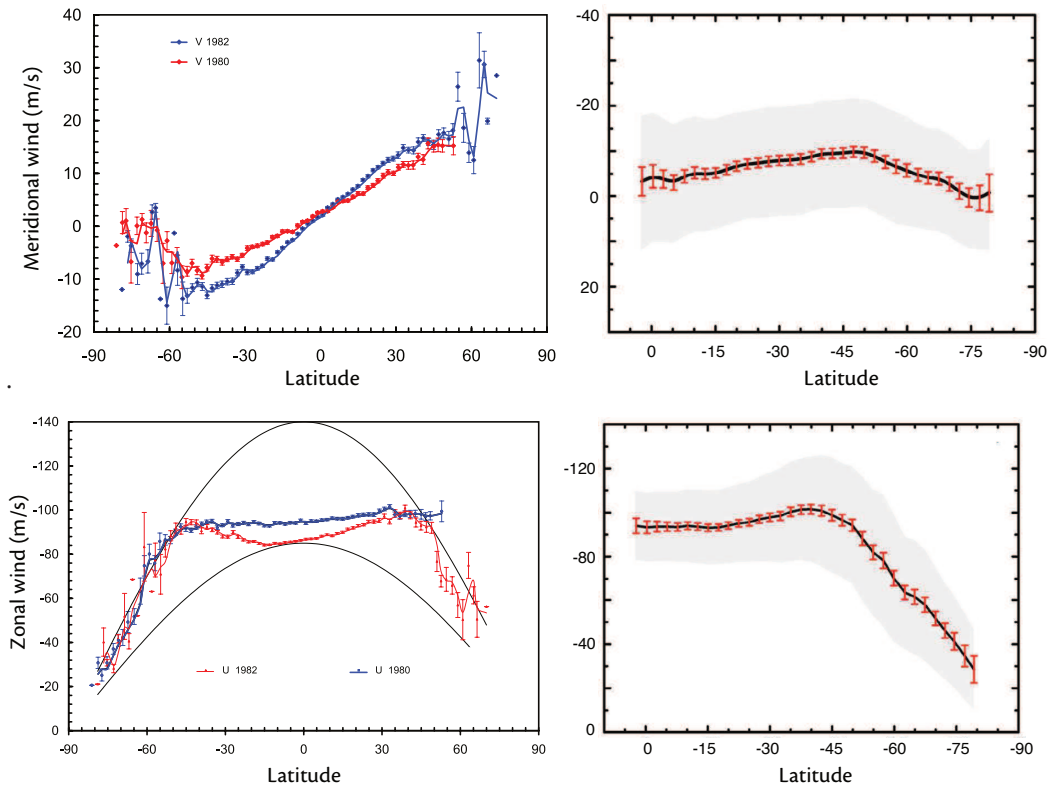
$$Sol = \frac{Sid}{1 \pm Sid/OP}$$

where OP is the orbital period and the minus or plus sign is taken depending on whether the planet rotates in a prograde or retrograde fashion, respectively. Venus has a sidereal day of 243 Earth days, a sol of 117 Earth days and a year of 225 Earth days.

The temperature structure of Venus's atmosphere is illustrated in Fig. 1.9 and Fig. 1.7. Three obvious features are:

- (i) The very small variation of temperature in the horizontal, both in the upper troposphere and the surface.
- (ii) The almost uniform lapse rate, with a tropopause at about 60 km, some six times higher than that of Earth.
- (iii) The very high surface temperature compared to Earth.

The first feature is a consequence of the slow rotation and Hadley Cell extending very far poleward (as discussed below), for the Hadley Cell is a region of almost uniform temperature, as found in Chapter ?? . The second and third features are consequences of the high optical depth of the Venusian atmosphere: the tropopause temperature, which is similar to the emitting temperature, is about 240 K, but the surface temperature is much higher, about 700 K, because of the strong greenhouse effect. The dry adiabatic lapse rate, g/c_p varies from about 7.8 K km^{-1} to about 11 K km^{-1} at



50 km ($g = 8.86 \text{ m s}^{-2}$ and c_p varies from about $1130 \text{ J kg}^{-1} \text{ K}^{-1}$ at the surface to about $800 \text{ J kg}^{-1} \text{ K}^{-1}$ at the tropopause, because the temperature varies), and to connect the surface and tropopause with such a lapse rate requires 50 km or more (and in fact the lapse rate is slightly less than, or more stable than, the dry adiabat in the upper troposphere).

Winds

As noted in Section 1.3.2, the slow rotation period of Venus leads to an external Rossby number of about 2000 and, given this, we expect that the Hadley Cell will extend much further poleward than on Earth and that its meridional temperature gradient will be much less. These expectations are largely borne out by the observations, as shown in Fig. 1.8, Fig. 1.9 and Fig. 1.7. Observations of the wind are most easily obtained at the cloud layer and these show motion away from the equator, extending to about 50° on either side, which we can interpret as being the edge of the Hadley Cell. We note that the axisymmetric theory of the Hadley Cell extent given in Section ?? does not apply without modification to Venus, because the planetary rotation rate of Venus is so slow that geostrophic balance in its usual form is not applicable. Nevertheless, the same general principles apply, with a slower rotation rate leading to a greater poleward extent, and the extended Hadley Cell keeps temperature gradients small.

Fig. 1.8: Meridional winds (top) and zonal winds (bottom) on Venus obtained by cloud tracking. Left panels: the wind on the day-side only for 1980 and 1982. Right panels: zonally-averaged winds from the Venus Express orbiter over about 10 Venusian years. The two solid lines in the bottom left panel show constant angular velocity profiles (i.e., $u / \cos \vartheta = \text{constant}$), with two different values at the equator. (Adapted from Limaye (2007) and Khatuntsev et al. (2013).)

The zonal winds are notable in two ways:

- (i) They show a distinct fall off at latitudes beyond about 50° , that is beyond the inferred Hadley Cell.
- (ii) They are large (and negative) at low- and midlatitudes. The negative values just mean the winds are in the same direction as the solid surface, since Venus's rotation is retrograde. The fact that the zonal winds are large at the equator means that the atmosphere is *super-rotating*, with the atmosphere spinning faster than the planet itself.

The fall-off in winds beyond the edge of the Hadley Cell is to be expected, given the Hadley Cell theory of Sections ?? and ?. The high values of the zonal wind in the Venusian midlatitudes are analogues of the subtropical jets on Earth, rather than the eddy-driven midlatitude jets. Baroclinic instability is very weak on Venus because of the small pole-to-equator temperature gradient and because the radius of deformation, NH/f , is very large, and by and large the zonal mean flow is much stronger than the eddy flow.

The equatorial super-rotation (item (ii) above) on Venus is a striking feature with no real analogue on Earth. The velocity of the ground at the equator is about 1.8 m s^{-1} , so that a zonal wind of 90 m s^{-1} means the atmosphere is rotating about *50 times faster than the ground*, and in the same direction. Super-rotation is also apparent in the top two left-hand panels of Fig. ??, and is in fact a common feature of numerical simulations at low rotation rates.

The remarkable aspect of this phenomenon is that, even though the eddies on Venus are weak, they *must* be involved in the production of super-rotation, for a rather simple reason. If the atmosphere is zonally symmetric then, except for viscous effects, the angular momentum of a ring of air around the axis of rotation, $\bar{m}$, is conserved, as discussed in Section ??, and no internal maximum (such as super-rotation) can be produced. To see this explicitly, after a little manipulation we find that, if the flow is zonally-symmetric, we can write the zonal momentum equation as

$$\frac{Dm}{Dt} = \nu \frac{\partial^2 m}{\partial z^2}, \quad (1.53)$$

where $m = (u + \Omega a \cos \vartheta)a \cos \vartheta$ as in (1.30) and the right-hand side represents viscous effects, keeping only the vertical derivative as it is much larger than the horizontal one. Now, the advective terms certainly cannot create angular momentum, and the viscous terms can only smooth it out, destroying any pre-existing maximum, as discussed in Section ?. Thus, no maximum of angular momentum, and therefore no super-rotation, can exist in a steady zonally-symmetric flow. If we allow for the possibility of zonal asymmetries then additional terms in the equation arise (such as $\partial_y(\overline{u'v'})$) and an eddy flux of angular momentum can maintain the super-rotation.

The fact that super-rotation must involve eddy motion is known as Hide's theorem, after Raymond Hide.

1.10 RADIATIVE TRANSFER AND VERTICAL STRUCTURE

We now look at the *vertical* structure of planetary atmospheres, and this involves a significant diversion into radiation and its interaction with convection. We begin with an elementary introduction to radiative transfer.

1.10.1 Elements of Radiative Transfer

Radiative transfer is a complex subject and in order to make progress we will simplify it in two main ways:

- (i) Rather than treating an entire spectrum of wavelengths, we suppose that the radiation exists only in two well-separated bands, namely solar (or shortwave) radiation and infra-red (or longwave) radiation. In each band the absorption is not a function of wavelength, and this is called the grey approximation (or the semi-grey approximation, considering that there are two bands of radiation). We will suppose that the planet only emits longwave radiation, and its host star only emits shortwave radiation.
- (ii) We suppose that the radiation travels only vertically, up or down, through a planetary atmosphere. This is the *two-stream* approximation.

Consider a beam of radiation propagating through a thin slab of gas, as in Fig. 1.9. Some of the incoming radiation may be absorbed, some reflected, and the slab may emit radiation of its own. In the three-dimensional problem some radiation may also be scattered into the beam from radiation travelling in other directions. Let us neglect scattering and reflection, a good approximation for longwave radiation. The difference between the outgoing and incoming radiation is then

$$dI = I^{\text{out}} - I^{\text{in}} = -d\tau I + dE, \quad (1.54)$$

where I is the irradiance or radiant intensity (with units of power per unit area, or W m^{-2}), the term $-d\tau I$ is the absorption and dE is the thermal emission. The quantity $d\tau$ is the nondimensional *optical depth*; it may be written as $d\tau = e_L ds$ where ds is the slab thickness, and e_L is the *emissivity* of the medium. The minus sign on $d\tau$ in (1.54) is appropriate when τ increases in the direction of the beam. The emission of radiation is, in thermal equilibrium, the Planck function, B , such that $dE = B d\tau$. If the atmosphere is grey (as in assumption (i) above) then $B = \sigma T^4$ where $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ is Stefan's constant. With all this, (1.54) becomes

$$dI = -d\tau(I - B) \quad \text{or} \quad \frac{dI}{d\tau} = -(I - B). \quad (1.55)$$

In planetary atmospheres it is common to choose τ increasing downwards, from 0 at the top of the atmosphere. The downwards (D) and upwards (U) irradiances are then

$$\frac{dD}{d\tau} = B - D, \quad \frac{dU}{d\tau} = U - B. \quad (1.56a,b)$$

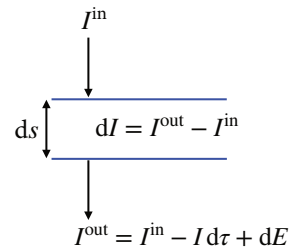


Fig. 1.9: Radiative transfer across a thin slab, with no scattering or reflection and the radiation travelling in the vertical direction only.

These equations are sometimes known as the *Schwarzschild equations*. The net flux of radiation is $N = U - D$ and the radiative heating is proportional to the net flux divergence, $-\partial N/\partial z$.

The grey assumption is not particularly accurate for many planetary atmospheres but for conceptual or approximate calculations it is often useful to suppose that the atmosphere is grey in the infra-red, in which case (1.56) applies to infra-red radiation. Similar but separate equations (that would normally include reflection, but not thermal emission) are then used for solar radiation. In Earth's atmosphere the absorption of solar radiation in the atmosphere is quite small, and although there is considerable reflection due to clouds most of the net incoming solar radiation is absorbed at the surface.

1.10.2 Radiative Equilibrium

A *radiative equilibrium* state has, by definition, no radiative heating. If the atmosphere is transparent to solar radiation then the condition implies that the vertical divergence of the longwave radiation (denoted with a subscript L) is zero:

$$\frac{\partial(U_L - D_L)}{\partial z} = 0 \quad \text{implying} \quad \frac{\partial(U_L - D_L)}{\partial \tau} = 0. \quad (1.57a,b)$$

This condition is normally *not* satisfied in the atmosphere because the air moves. But if it were satisfied then (1.56) and (1.57b) form three equations in three unknowns, U_L , D_L and B , and a solution can be found as follows.

Consider an atmosphere with net incoming solar radiation S_{net} and suppose the planet is in radiative equilibrium with the incoming solar radiation balanced by outgoing infra-red radiation. That is, $U_{Lt} \equiv U_L(\tau = 0) = S_{\text{net}}$ where U_{Lt} is the net outgoing longwave radiation (OLR) at the top of the atmosphere. The downward infra-red radiation at the top of the atmosphere is zero, so that the boundary conditions on the radiative transfer equations at the top of the atmosphere are

$$D_L = 0, \quad U_L = U_{Lt} \quad \text{at} \quad \tau = 0. \quad (1.58)$$

A little algebra reveals that a solution of (1.56) and (1.57b) that satisfies (1.58) is

$$D_L = \frac{\tau}{2} U_{Lt}, \quad U_L = \left(1 + \frac{\tau}{2}\right) U_{Lt}, \quad B = \left(\frac{1 + \tau}{2}\right) U_{Lt}, \quad (1.59a,b,c)$$

as can be easily verified by substitution back into the equations.

It remains to explicitly relate τ to z , and one approximate recipe is to suppose that τ has an exponential profile,

$$\tau(z) = \tau_0 \exp(-z/H_a), \quad (1.60)$$

where τ_0 is the optical depth at $z = 0$ and H_a is the scale height of the absorber. In the Earth's atmosphere the optical depth is determined by the

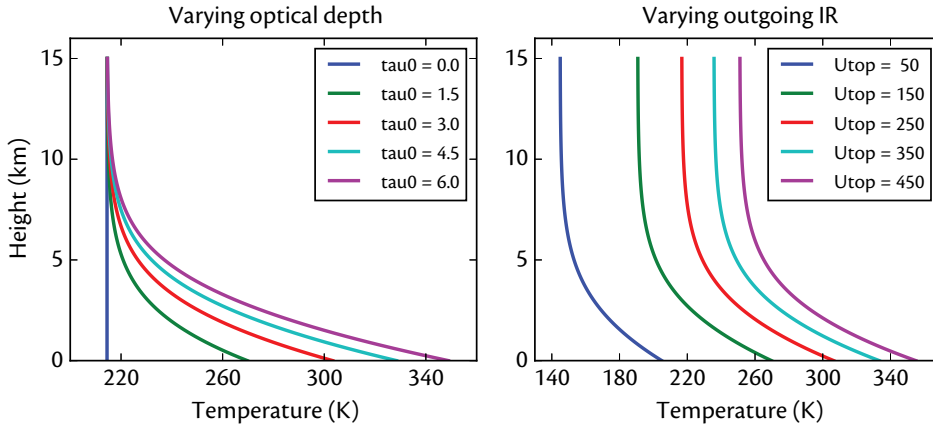


Fig. 1.10: Radiative equilibrium temperature calculated using (1.61), with $H_a = 2$ km. Left: outgoing IR (and incoming solar) radiation is 240 W m^{-2} and surface optical depth varies from 0 to 6. Right: the optical depth is 3.0 and outgoing IR radiation varies from 50 to 450 W m^{-2}

concentrations of water vapour (primarily) and carbon dioxide (secondarily), and τ_0 (the scaled optical depth) typically varies between 2 and 4, depending on the water vapour content of the atmosphere, and $H_a \approx 2$ km, this being a typical scale height for water vapour. Using (1.59c) with $B = \sigma T^4$ the temperature then varies as

$$T^4 = U_{Lt} \left(\frac{1 + \tau_0 e^{-z/H_a}}{2\sigma} \right), \quad (1.61)$$

as illustrated in Fig. 1.10. We note the following aspects of the solution:

- (i) Temperature increases rapidly with height near the ground.
- (ii) The upper atmosphere, where τ is small, is nearly isothermal.
- (iii) The temperature at the top of the atmosphere, T_t is given by

$$\sigma T_t^4 = \frac{U_{Lt}}{2}. \quad (1.62)$$

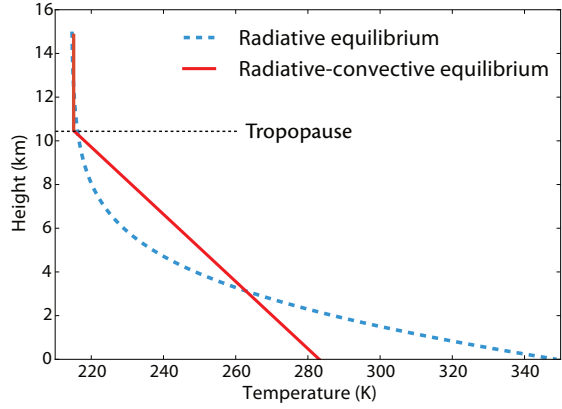
Thus, if we define the emitting temperature, T_e , to be such that $\sigma T_e^4 = U_{Lt}$, then $T_t = T_e/2^{1/4} < T_e$; that is, the temperature at the top of the atmosphere is *lower* than the emitting temperature.

- (iv) Related to the previous point, $B_t/U_{Lt} = 1/2$. That is, the upwards longwave flux at the top of the atmosphere is twice that which would arise if there were a black surface at a temperature T_t . The reason is that there is radiation coming from lower down in the atmosphere where the temperature is higher.

1.10.3 Radiative–Convective Equilibrium

In the radiative equilibrium solution the temperature gradient near the ground varies so rapidly that $-\partial T/\partial z$ may well exceed even the dry adiabatic lapse rate. If so, the radiative equilibrium solution is convectively unstable and convection will be triggered. The convection will then establish a convectively neutral lapse rate, Γ typically given by the dry adiabatic or,

Fig. 1.11: Radiative and radiative–convective equilibrium profiles. The initial radiative equilibrium temperature, dashed blue line, adjusts to a specified profile (red line, in this instance a constant lapse rate) that extends to a finite height, beyond which radiative equilibrium holds. The break in the slope of the radiative–convective temperature profile is the tropopause.



if moisture or another condensate is present, the saturated adiabatic lapse rate. The convection cannot proceed to arbitrary height of course — it will cease when the lapse rate is no longer unstable, and at higher altitudes the temperature profile will more-or-less be that of the radiative equilibrium profile, as illustrated in Fig. 1.11. The break in the temperature profile is nothing other than the tropopause, above which lies the stratosphere. The temperature in the real stratosphere is not exactly constant with height, because there may be absorption of solar radiation within it (as by ozone in Earth’s stratosphere) and because of the presence of fluid motion. However, it is much more constant than in the troposphere, and a temperature structure similar to that of the red line in Fig. 1.11 is a common feature of many other planets too.

We can obtain a very rough estimate of the average height of the tropopause of a terrestrial planet if we know the optical depth of the atmosphere as well as the lapse rate, Γ . The tropopause temperature, T_T is given by the requirement that the outgoing longwave radiation, U_{Lr} equal the net incoming solar radiation, S_N . For a stratosphere in radiative equilibrium the tropopause temperature, T_T is equal to that at the top of the atmosphere and using (1.62), it must then satisfy

$$\sigma T_T^4 = \frac{S_N}{2} = \frac{\sigma T_e^4}{2} \quad \text{or} \quad T_T = \frac{T_e}{2^{1/4}}. \quad (1.63)$$

The surface temperature, T_S , in radiative equilibrium is given by (1.61) evaluated at $z = 0$, giving

$$\sigma T_S^4 = S_N \left(\frac{1 + \tau_0}{2} \right) \quad \text{or} \quad T_S = T_T (1 + \tau_0)^{1/4}. \quad (1.64)$$

The surface temperature in radiative–convective equilibrium is usually considerably lower than this, because convection tends to cool the lower atmosphere and heat the upper atmosphere, but nevertheless (1.64) gives a first estimate. The height of the tropopause, H_T , is then such that $(T_S - T_T)/H_T = \Gamma$ giving

$$H_T = \frac{T_S - T_T}{\Gamma} = \frac{T_T}{\Gamma} \left((1 + \tau_0)^{1/4} - 1 \right). \quad (1.65)$$

This expression is an overestimate because we have used the radiative equilibrium surface temperature for T_S and to properly account for that requires a complicated calculation (the result of which is the red line in Fig. 1.11). However, (1.65) qualitatively captures three of the main factors determining tropopause height on a terrestrial planet, namely:

- (i) The increase of the tropopause height with decreasing lapse rate. With a smaller lapse rate, the tropopause must extend higher in order that its temperature falls sufficiently so that the outgoing infra-red radiation equals the incoming solar radiation.
- (ii) The increase of tropopause height with optical depth. Thus, a planet like Venus with a thick, CO_2 rich atmosphere can be expected to have a higher tropopause than Earth. This effect arises because an increased optical depth raises the surface temperature, but the tropopause temperature, being proportional to the emissions temperature, remains the same. Given a larger value of $T_S - T_T$ the tropopause height must be larger.
- (iii) The increase in tropopause height with increased incoming solar radiation. This effect arises because the difference between the surface and tropopause temperatures in radiative equilibrium is proportional to the tropopause temperature itself, as we can see from (1.61) and in the right-hand panel of Fig. 1.11. If we were to move the Earth further from the Sun, or increase its albedo, the tropopause height would fall.

Item (ii) has a robust consequence for Earth. As we add greenhouse gases to the atmosphere causing global warming, the optical depth increases, and consequently the tropopause height must increase. Detailed calculations show that the increase will occur at the rate of about 300 m for each degree of warming, and this is one of the most robust predictions about how the atmosphere will change in the future as the planet warms.

We should add some caveats to the above discussion. First, we have treated the atmosphere as if it were a single column. The horizontal transfer of heat may make a significant difference, and on Earth the tropical tropopause is about twice as high as the polar one. Relatedly, it is not always convection that maintains the lapse rate, and in baroclinic zones the stratification may be maintained by baroclinic waves and not convection. However, if we know the lapse rate that the baroclinic eddies produce the above arguments may still be applied. Finally, most planetary atmospheres are not grey, even in the infra-red, and may have clouds, but the complications that these entail is a story for another day.

LECTURE 2

Tidally-Locked Planets

Time and tide wait for no man.
Proverb, uncertain origin, c. 13th century.

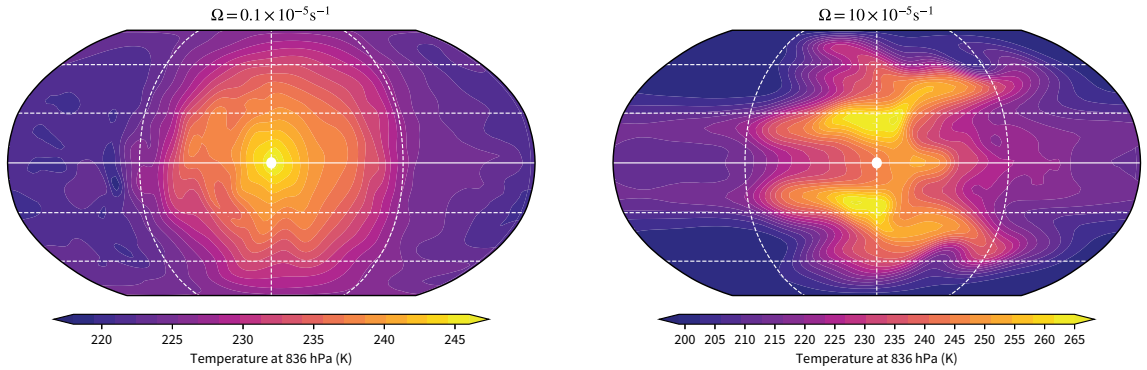
2.1 EXOPLANETS AND TIDAL LOCKING

When a planet is close to its host star it feels a strong tidal force that drags the planet into a spin-orbit alignment. In this configuration the orbital period and rotation period are equal so that the same face of the planet always points to the star (just as the same face of the Moon always points toward Earth) creating permanent day and night hemispheres. The time it takes for a planet to become tidally locked varies inversely as the distance from the star. For Earth it will be billions of years but for a close-in planet it may only be millions, implying that many of those that have been detected are tidally locked, with a rotation period of a few days.

Figure 2.1 shows the temperature in the lower troposphere in two numerical simulations of tidally-locked Earth-like planets. In both cases the heating is stationary with respect to the planet's surface, centered on the equator and diminishing with both latitude and longitude away from the substellar point (marked with a small white dot). The planet on the left rotates slowly, in about 73 (Earth) days, whereas the planet on the right completes an orbit in 18 hours. For the slowly rotating planet there is efficient advective heat redistribution from day side (the two central quadrants) to night accomplished by a large-scale overturning circulation spanning the entire planet, and the difference in temperature between the day and night side is about 30 K. In the high rotation case the day-night temperature contrast is double that, about 60 K. In this case the circulation is much more localized to the tropics and to the day side of the planet. The heating generates both Kelvin waves and Rossby waves that propagate east and west from the disturbance, respectively, and which are damped by friction forming trapped, quasi-stationary patterns.

The atmosphere becomes hottest in the 'Rossby gyres', which are the cyclonic lobes in both hemispheres west and slightly poleward of the substellar point and which inhibit meridional heat transport. In the more slowly rotating case the deformation radius, NH/f , is larger than the planet itself and there is no such trapping, and although the hotspot is more-or-less coincident with the substellar point the temperature is much

Many exoplanets thus far detected are giant planets close to their host star, and are likely to be tidally locked. Such hot Jupiters are relatively easy to detect because when they pass between their host star and us (or behind their host star) there is a detectable change in the solar radiation that we receive. However, most planets in the Universe are almost certainly not tidally-locked hot Jupiters and solar systems with a variety of different planets within them have been discovered. For example, at least five planets of different types are in orbit in the Cancri system and seven terrestrial planets are in orbit in the TRAPPIST-1 system.



more uniform over the entire planetary surface. Because of the weak Coriolis effect the entire planet is in some ways a tropical one obeying the weak temperature gradient approximation discussed in Section 2.6.

2.2 RESPONSE TO A STATIONARY HEATING AT THE EQUATOR

What is the response of an exoplanet to a stationary thermal forcing, centered at the equator? This problem has a lot of similarities to one considered by Adrian Gill (Gill, 1980). His concern was related to how the atmosphere responded to a localized sea-surface temperature anomaly, but many aspects of the problem are the same. The full solution to what has become known as the Gill Problem is quite algebraically complicated, so we'll take a simpler approach that still captures the essence.

2.2.1 The Gill Problem, Simplified

Consider the steady response of a linearized shallow-water system on the equatorial beta-plane, thermally forced at the equator. In our standard notation the equations of motion are

$$-fv + \frac{\partial \phi}{\partial x} = -\alpha u, \quad fu + \frac{\partial \phi}{\partial y} = -\alpha v, \quad c^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = -\alpha \phi - Q. \quad (2.1a,b,c)$$

These are, respectively, the steady zonal-momentum equations in the x - and y -directions and the height equation, which is the thermodynamic equation for this system. The Coriolis parameter varies as $f = \beta y$, and $c^2 = gH$ is the gravity wave speed. The parameter α is a damping rate, assumed for algebraic convenience to take the same value in the momentum equation (where it represents a linear drag) and the thermodynamic equation (where it represents radiative damping). Thermal forcing is represented by Q , which we assume to be localized in both x and y and centered at the equator and at, say, $x = 0$, but we won't be concerned about its exact form, just the steady response away from the forcing.

Fig. 2.1: Snapshots of temperature field in the lower troposphere in numerical simulations of two tidally-locked planets, each with the same radius, gravity and atmospheric mass as Earth. The planet on the left has a rotation period of 73 days, and that on the right a period of 18 hours. The substellar point (the position on the planet where the host star is overhead) is stationary, denoted by the small white dot in the centre of each plot. Note the different scales on the colour maps.

If the friction is relatively weak, such that $\alpha v \ll fu$, then the zonal flow will be in geostrophic balance with the meridional pressure gradient and (2.1b) is simply

$$fu = -\partial\phi/\partial y \quad (2.2)$$

However, we keep the friction term in the x -momentum equation. From (2.1a,c) and (2.2) we can construct a single equation for v , namely

$$\frac{f^2}{c^2}\alpha v - \alpha \frac{\partial^2 v}{\partial y^2} - \beta \frac{\partial v}{\partial x} = \frac{\alpha}{c^2} \frac{\partial Q}{\partial y} - \frac{f}{c^2} \frac{\partial Q}{\partial x}. \quad (2.3)$$

For particular forms of Q it is possible to solve this equation exactly (and this is what Gill did) although the algebra is a little complicated. Since, and certainly for an exoplanet, we cannot expect to know the exact form of the forcing we will seek only the solution away from that forcing. That is, we will assume only that the forcing exists and is fairly localized, but not much else. In the absence of forcing (2.3) becomes

$$\frac{f^2}{c^2}\alpha v - \alpha \frac{\partial^2 v}{\partial y^2} - \beta \frac{\partial v}{\partial x} = 0 \quad (2.4)$$

This equation may be compared with the one that results if time derivatives are kept in the x -momentum and height equations, but dissipative terms are eliminated, and that equation is

$$\frac{f^2}{c^2} \frac{\partial v}{\partial t} - \frac{\partial}{\partial t} \left(\frac{\partial^2 v}{\partial y^2} \right) - \beta \frac{\partial v}{\partial x} = 0. \quad (2.5)$$

We now construct the solution by considering the Kelvin wave and Rossby wave responses separately.

The Kelvin wave solution

We know from our previous analysis that there exist wave solutions with $v = 0$ everywhere, and these are Kelvin waves. In the corresponding stationary case the equations of motion are

$$\alpha u + \frac{\partial \phi}{\partial x} = 0, \quad fu + \frac{\partial \phi}{\partial y} = 0, \quad \alpha \phi + c^2 \frac{\partial u}{\partial x} = 0. \quad (2.6a,b,c)$$

Combining (2.6a) and (2.6c) gives

$$\alpha^2 \phi = c^2 \frac{\partial^2 \phi}{\partial x^2} \quad \text{and} \quad \alpha^2 u = c^2 \frac{\partial^2 u}{\partial x^2}, \quad (2.7)$$

which may be compared to the usual wave equation $\partial^2 \phi / \partial t^2 = c^2 \partial^2 \phi / \partial x^2$, and similarly for u .

The solution to (2.7) is

$$\phi = A(y)e^{kx} + B(y)e^{-kx} \quad \text{and} \quad u = -\frac{A(y)}{c}e^{kx} + \frac{B(y)}{c}e^{-kx}. \quad (2.8)$$

where $k = \alpha/c$. To determine the unknown functions A and B we use (2.6b) and, with $f = \beta y$, we have

$$\beta y \left[-\frac{A(y)}{c} e^{kx} + \frac{B(y)}{c} e^{-kx} \right] = -\frac{\partial}{\partial y} [A(y) e^{kx} + B(y) e^{-kx}]. \quad (2.9)$$

and so

$$-\frac{A(y)\beta y}{c} = \frac{\partial}{\partial y} A(y) \quad \text{and} \quad \frac{B(y)\beta y}{c} = -\frac{\partial}{\partial y} B(y). \quad (2.10)$$

The solutions are

$$A(y) = A_0 \exp(\beta y^2/2c), \quad B(y) = B_0 \exp(-\beta y^2/2c), \quad (2.11)$$

where A_0 and B_0 are constants. In order to satisfy the condition of boundedness at $y = \pm\infty$ we must have $A_0 = 0$ and the solution is therefore

$$\phi = B_0 \exp(-\beta y^2/2c) \exp(-\alpha x/c). \quad (2.12)$$

Furthermore, the solution must decay away from the wave source and if this is at $x = 0$ the solution is non-zero only *east* of the source, and decays over a distance c/α . This solution is plotted in Fig. 2.2. Note that we have not considered how boundary conditions are satisfied in this derivation, nor can arbitrary boundary conditions be satisfied because the Kelvin wave is not the complete solution to the equations.

The Rossby wave solution

We are going to look at Rossby waves from the perspective of potential vorticity. Keeping (for now) both time dependence and dissipation, the linear shallow water equations may be combined to give

$$\left(\frac{\partial}{\partial t} + \alpha \right) \left(\zeta - \frac{f\phi}{c^2} \right) + \beta v = 0, \quad (2.13)$$

where, as before, α is a linear dissipation coefficient and $f = \beta y$.

Let us now suppose that the divergence is small and that the flow is close to geostrophic balance so that the velocity, vorticity and height fields can all be written in terms of a streamfunction,

$$u = -\frac{\partial \psi}{\partial y}, \quad v = \frac{\partial \psi}{\partial x}, \quad \zeta = \nabla^2 \psi, \quad \phi = f\psi. \quad (2.14)$$

This is similar to what is done in the quasi-geostrophic approximation, except that here the Coriolis parameter is allowed to vary (i.e., $f = \beta y$).

Using (2.14) in (2.13), and dropping the time derivative. gives

$$\alpha \left(\nabla^2 \psi - \frac{(\beta y)^2 \psi}{c^2} \right) + \beta \frac{\partial \psi}{\partial x} = 0. \quad (2.15)$$

We seek solutions to this equation in the form

$$\psi = \tilde{\psi}(y) e^{kx}, \quad (2.16)$$

and (2.15) becomes

$$\frac{d^2\tilde{\psi}}{dy^2} + \left(k^2 + \frac{\beta k}{\alpha} - \frac{\beta^2 y^2}{c^2} \right) \tilde{\psi} = 0. \quad (2.17)$$

At his point it is useful to nondimensionalize the variables and we do so by choosing the length and time scales

$$T_{eq} = (c\beta)^{-1/2}, \quad L_{eq} = \left(\frac{c}{\beta} \right)^{1/2}, \quad (2.18)$$

and defining

$$\hat{y} = \frac{y}{L_{eq}}, \quad \hat{k} = kL_{eq}, \quad \hat{\alpha} = \alpha T_{eq}. \quad (2.19)$$

Equation (2.17) then becomes

$$\frac{d^2\tilde{\psi}}{d\hat{y}^2} + \left(\hat{k}^2 + \frac{\hat{k}}{\hat{\alpha}} - \hat{y}^2 \right) \tilde{\psi} = 0. \quad (2.20)$$

This equation may be put into a standard form by using the substitution $\tilde{\psi} = \Psi \exp(-\hat{y}^2/2)$ (and we may note the connection to the solution for the stationary Kelvin wave). With this substitution, (2.20) becomes

$$\frac{d^2\Psi}{d\hat{y}^2} - 2\hat{y}\Psi + \left(\hat{k}^2 + \frac{\hat{k}}{\hat{\alpha}} - 1 \right) \Psi = 0. \quad (2.21)$$

Equation (2.21) is known as Hermite's equation. If we let $\lambda = (\hat{k}^2 + \frac{\hat{k}}{\hat{\alpha}} - 1)$ then the solution are Hermite polynomials $\Psi(\hat{y}) = H_m(\hat{y})$ with $\lambda = 2m$ and

$$(H_0, H_1, H_2, H_3, H_4) = (1, 2\hat{y}, 4\hat{y}^2 - 2, 8\hat{y}^3 - 12\hat{y}, 16\hat{y}^4 - 48\hat{y}^2 + 12). \quad (2.22)$$

The values of k are quantized by this condition; that is to say

$$\left(\hat{k}^2 + \frac{\hat{k}}{\hat{\alpha}} \right) = 2m + 1, \quad (2.23)$$

or, in dimensional form,

$$\left(c^2 k^2 + \beta c^2 \frac{k}{\alpha} \right) = (2m + 1)\beta c, \quad (2.24)$$

This is analogous to the dispersion relation for low frequency waves when time-dependence but not dissipation is included, namely

$$-c^2 k^2 - \beta \frac{kc^2}{\omega} = (2m + 1)\beta c. \quad (2.25)$$

The difference is that in (2.25) is generally regarded as an equation for the frequency, given the spatial dependence. In (2.24) the decay rate, α is a

given physical parameter and the equation then yields the spatial rate of decay.

The solutions are symmetric or antisymmetric around $y = 0$ according as to whether m is even or odd, respectively. If we recall that $\psi = \phi/f$ then mirror symmetry (i.e., $\phi(y) = \phi(-y)$) obtains for odd values of m . The solution for the gravest mode, $m = 1$, then has the form,

$$\phi = \phi_0 \exp(-\beta y^2/2c) y^2 \exp(kx), \quad (2.26)$$

where k is given by the solution of (2.25), and $\phi = 0$ for $x > 0$ to satisfy boundedness at infinity. If we further simplify to the small-wavenumber case, and neglect the term in k^2 (2.25), then, for any m have $k = (2m + 1)\alpha/c$ and for the gravest mode $k = 3\alpha/c$. The solution is then

$$\phi = \begin{cases} \phi_0 y^2 \exp(-\beta y^2/2c) \exp(3\alpha x/c), & x < 0 \\ 0, & x > 0. \end{cases} \quad (2.27)$$

In this case the decay rate is precisely three times as rapid as it is for the decaying Kelvin mode given by (2.12). Also, and importantly, the solution decays in the *westward* direction, reflecting the westward propagation of Rossby waves. East of the wave source there is no disturbance.

Taken together, (2.27) and (2.12) comprise the Gill solution away from the source region; that is, they are the solution to the homogeneous problem. These, and as the full solution obtained by adding the particular solution with a particular source confined about the origin, are illustrated in Fig. 2.2.

2.3 EXO-PLANET CONSIDERATIONS

The situation on tidally-locked exoplanets differs in a number of ways from that considered above (and from the original problems considered by Matsuno and Gill). The main differences are likely to be:

- The thermal forcing is quite large scale. Heating can be expected to be at a maximum at the sub-stellar point, but extending considerably in both the zonal and meridional directions from there, as illustrated in Fig. ??.
- Most tidally-locked planets, unless they are in an extremely close orbit to their host star, will have a slower rotation rate than Earth, with a correspondingly large radius of deformation (variously NH/f , c/f or $\sqrt{c/\beta}$, depending on context and as discussed previously).
- The momentum dissipation may be noticeably weaker than the thermal dissipation. On a gas giant the fluid in the gaseous weather layer overlays another fluid (or simply gets more dense with depth) rather than overlaying a solid or liquid with a rough surface. The frictional dissipation may then be relatively small whereas the thermal dissipation may remain large. This difference can lead to noticeable differences in the phase of the Rossby wave response, as

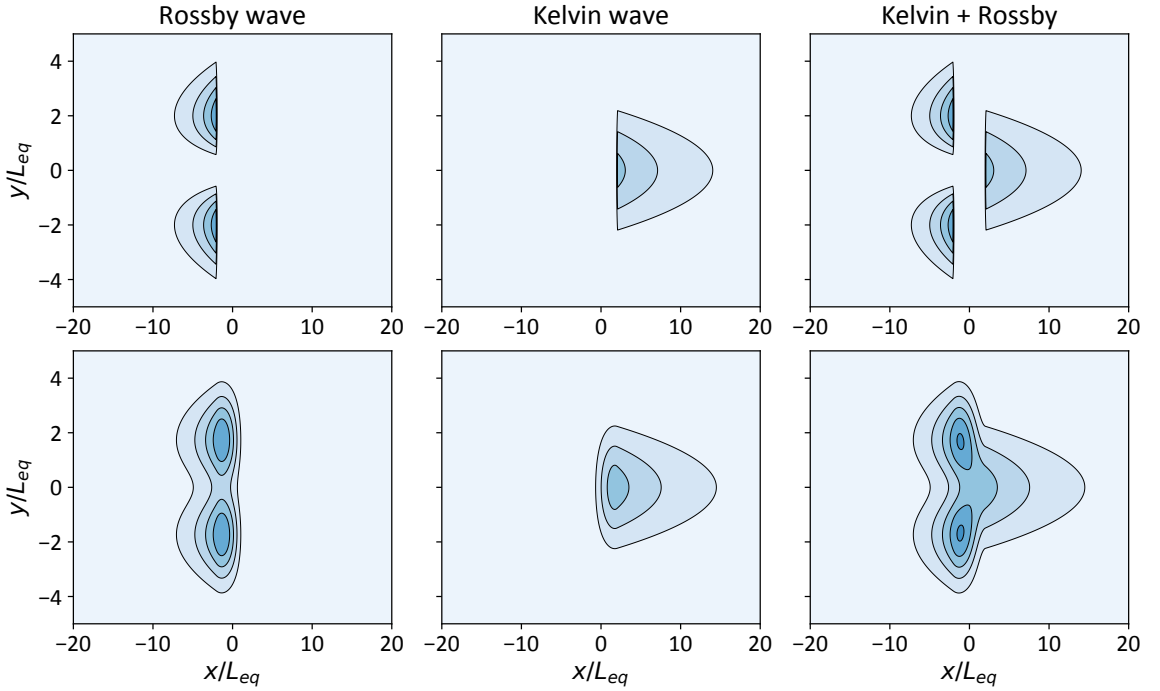


Fig. 2.2: Stationary Kelvin and Rossby wave solutions to a localized heating at the equator ($y = 0$) that is nonzero only for $|x| < 2$ and that varies in y as $\exp(-y^2/2)$. The top row shows the solutions away from the heating as given by (2.12) and (2.27). The bottom row shows the full solution, including the particular solution with the heating term included. The solutions are nondimensional and the contour interval is 0.3.

discussed below. In some contrast the Kelvin wave response may only quantitatively altered, as can be seen from inspection of (2.6): if α is replaced by α_m (a momentum damping coefficient) in (2.6a) and by α_ϕ in (2.6c) then the only change in the Kelvin-wave solution is that α^2 is replaced by $\alpha_\phi \alpha_m$.

- The stratification of exo-planetary atmospheres is not known and will depend on both the level at which solar heating occurs and the composition of the atmosphere. This then affects the deformation radius.

We will now discuss some of these effects.

Approximate Analytic Solution

To explore the above effects generally requires a numerical approach but we can make some headway by obtaining an approximate analytic solution. We will assume that the zonal wind is in geostrophic balance with the meridional pressure gradient (this is called the longwave approximation, for reasons we do not delve into), but we allow different thermal and momentum damping rates. The governing equations are then, for a steady state and in the linear approximation.

$$\alpha_m u - f v + \frac{\partial \phi}{\partial x}, \quad f u + \frac{\partial \phi}{\partial y} = 0, \quad \alpha_\phi \phi + c^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = Q, \quad (2.28a,b,c)$$

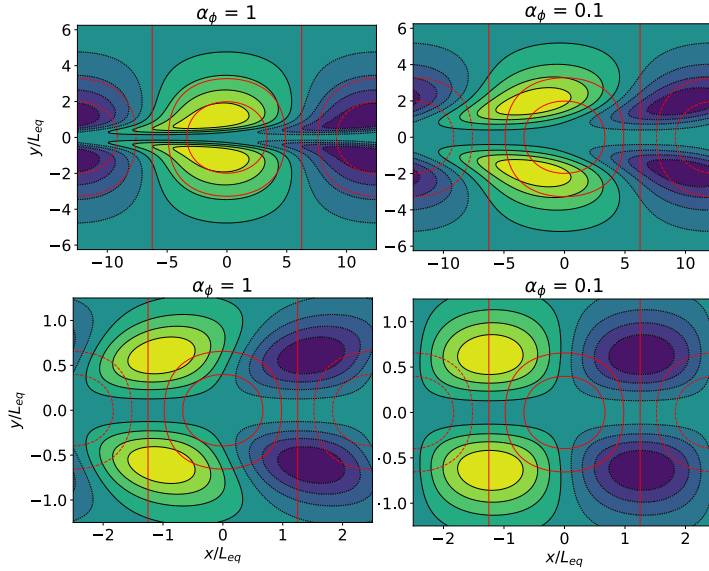


Fig. 2.3: Analytic solutions of the linear shallow water equations showing the steady response of the height field to a stationary, tidally-locked, thermal forcing with zero momentum damping and various values of thermal damping coefficient, α_ϕ and deformation radius L_{eq} , relative to the size of the planet. Length scales are nondimensionalized by L_{eq} , so that a larger values in the x and y axes indicate a larger planet or smaller deformation radius. The red contours show the location of the heating, with solid contours positive, and the lighter shading (the yellows) correspond to higher temperatures

where α_m and α_ϕ are the momentum and thermal damping rates respectively and we imagine $Q(x, y)$ to be a heating term provided by the planets star, and thus centered at the substellar point which we take to be at $x = y = 0$. With a little bit of algebra we obtain a single equation for ϕ , namely

$$\frac{f^2}{c^2} \alpha_\phi \phi - \beta \frac{\partial \phi}{\partial x} - \alpha_m \frac{\partial^2 \phi}{\partial y^2} + \alpha_m \frac{2\beta}{f} \frac{\partial \phi}{\partial y} = \frac{f^2}{c^2} Q. \quad (2.29)$$

On the equatorial β -plane $f = \beta y$ and $k_{eq} = (\beta/c)^{1/2} = L_{eq}^{-1}$ so that the above equation may be written

$$k_{eq}^4 y^2 \alpha_\phi \phi - \beta \frac{\partial \phi}{\partial x} - \alpha_m \frac{\partial^2 \phi}{\partial y^2} + \alpha_m \frac{2\beta}{f} \frac{\partial \phi}{\partial y} = k_{eq}^4 y^2 Q. \quad (2.30)$$

This equation is hard to solve in full generality, but we can at least eliminate the explicit x dependence by expanding ϕ and Q as Fourier modes; that is, we let

$$\phi(x, y) = \text{Re } \tilde{\phi}(y) \exp(ikx), \quad Q(x, y) = \text{Re } \tilde{Q}(y) \exp(ikx), \quad (2.31)$$

or a sum of such Fourier modes, whence (and henceforth omitting the Re decorator)

$$(\alpha_\phi k_{eq}^4 y^2 - ik\beta) \tilde{\phi} + \alpha_m \frac{\partial^2 \tilde{\phi}}{\partial y^2} + \alpha_m \frac{2\beta}{f} \frac{\partial \tilde{\phi}}{\partial y} = k_{eq}^4 y^2 \tilde{Q}(y). \quad (2.32)$$

At this stage we make a big simplification by neglecting the friction terms involving α_m , giving

$$(\alpha_\phi k_{eq}^4 y^2 - ik\beta) \tilde{\phi} = k_{eq}^4 y^2 \tilde{Q}(y). \quad (2.33)$$

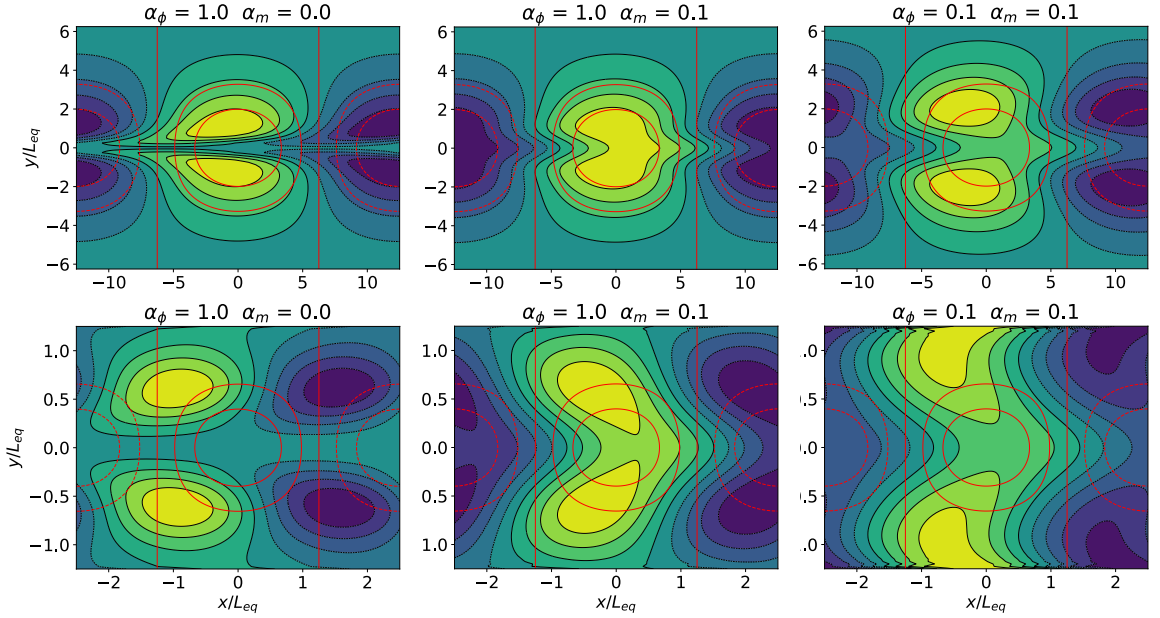


Fig. 2.4: As for Fig. 2.3, but now showing full numerical solutions to the linear shallow water equations for various values of thermal and momentum damping coefficients α_ϕ and α_m . The panels with $\alpha_m = 0$ have the same parameters as the equivalent panels in Fig. 2.3.

with solution

$$\tilde{\phi} = \frac{\tilde{Q}_\phi(y) k_{eq}^4 y^2}{\alpha_\phi k_{eq}^4 y^2 - i k \beta}, \quad (2.34)$$

or with nondimensional variables

$$\hat{\phi} = \frac{\tilde{Q}_\phi(y) \hat{y}^2}{\hat{\alpha}_\phi \hat{y}^2 - i \hat{k}}, \quad (2.35)$$

$$(2.36)$$

Approaching the equator the variations thus become completely out of phase with the forcing, although at the same time the response weakens. For small enough thermal damping the height field will be in quadrature with the thermal forcing over most of the planet. Various solutions with different damping can be seen in Fig. 2.3 and Fig. 2.4. The length-scales are nondimensionalized by the equatorial deformation radius $\sqrt{c/\beta}$, so solutions with an apparently smaller scales can be considered to be either more slowly rotating, or to be smaller, or to be more strongly stratified. For comparison, Earth with a rotation period of about 24 hours and a stratification corresponding to a Brunt–Väisälä frequency of about 10 minutes has a gravity wave speed of about 20 m s^{-1} and an equatorial deformation radius of about 1000 km, giving an equatorial circumference of about 40 such radii.

Numerical solutions

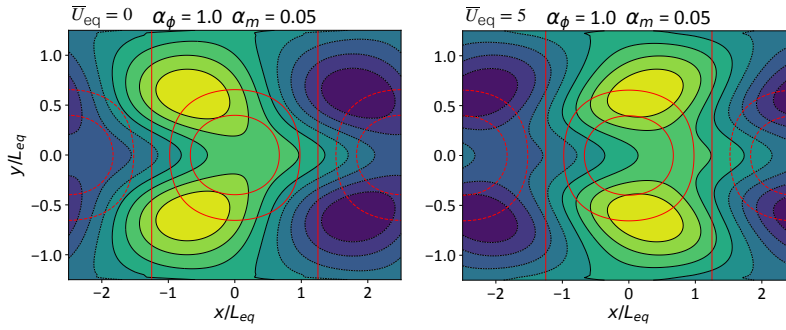


Fig. 2.5: Numerical solutions to the linear shallow water equation with and without an added superrotating wind.

2.4 SUPERROTATION

The other effect that can give rise to an offset between the substellar point and the location of maximum temperature is superrotation. Let us first discuss what it is, then the necessary conditions under which it may occur, and then the dynamics that gives rise to it.

Figure Fig. 2.5

2.4.1 What is it?

Superrotation in planetary atmospheres occurs when the zonal flow of the atmosphere in the direction of planetary rotation has an angular momentum that exceeds that which the planet itself has at the equator. To be specific, in the shallow-atmosphere approximation the axial component of the specific absolute angular momentum, m , of a parcel is given by

$$m = a \cos \vartheta (\Omega a \cos \vartheta + u), \quad (2.37)$$

where a is the radius of the planet. In most circumstances we would be concerned with the zonal average flow and replace u by $\bar{u}$ in the above. The axial angular momentum of the planet itself at the equator is given by

$$m_p = \Omega a^2. \quad (2.38)$$

If $u > 0$ at the equator then the atmosphere is superrotating. Using (2.37) and (2.38) we can see that at some latitude ϑ the atmosphere is superrotating if

$$u > \Omega a^2 \sin^2 \vartheta. \quad (2.39)$$

It would be unusual for this criterion to be satisfied at some off-equator latitude and not at the equator, for that would imply the atmospheric angular momentum is increasing away from the equator, a situation that is likely to be unstable.

2.4.2 Necessary conditions for superrotation

Consider first the evolution of a scalar, φ that is advected and diffused and obeying the equation

$$\frac{\partial \varphi}{\partial t} + \mathbf{v} \cdot \nabla \varphi = \nabla \cdot (\kappa \nabla \varphi). \quad (2.40)$$

Discussions of superrotation can be found in Showman and Polvani (2011), Read and Lebonnois (2018), Imamura et al. (2020) and various other articles.

We suppose the flow is steady and incompressible, thus obeying

$$\nabla \cdot (\mathbf{v}\varphi) = \nabla \cdot (\kappa \nabla \phi). \quad (2.41)$$

Let us suppose there is some maximum in the flow interior and that this is surrounded by a surface of constant φ . If we integrate the left-hand side of (2.41) over the volume enclosed by that isosurface we obtain

$$\int_V \nabla \cdot (\mathbf{v}\varphi) dV = \int_S \mathbf{v}\varphi dS = \varphi \int_S \mathbf{v} \cdot d\mathbf{S} = \varphi \int_V \nabla \cdot \mathbf{v} dV = 0, \quad (2.42)$$

having used the divergence theorem twice, the fact that φ is a constant on the bounding surface and that $\nabla \cdot \mathbf{v} = 0$. But if we integrate the right-hand side of (2.41) over that same volume we obtain

$$\int_V \nabla \cdot (\kappa \nabla \varphi) dV = \int_S \kappa \nabla \varphi \cdot d\mathbf{S} \neq 0, \quad (2.43)$$

since the gradient of φ will in general be nonzero over a surface surrounding an extremum. Thus we have a contradiction, meaning that there can be no isolated interior extremum of a quantity that is advected and conserved, if the flow is steady. Essentially, the diffusion is trying to dissipate away the extremum and the advection cannot compensate for this. The result applies whether or not the tracer, φ , is passive or active, and a very similar result can be found for a compressible flow.

Application to angular momentum

The axial angular momentum of a fluid obeys an equation that is similar but not identical to (2.40). We begin with the zonal momentum equation which after a little manipulation may be written as

$$\frac{Dm}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial \lambda} + \mathcal{D}(u), \quad (2.44)$$

where $m = (u + \Omega r \cos \vartheta)r \cos \vartheta$ is the axial angular momentum (per unit mass) and $\mathcal{D}(u)$ represents viscous terms and these act on velocity, not angular momentum. With the help of the mass continuity equation, (2.44) becomes

$$\frac{\partial \rho m}{\partial t} + \frac{1}{r \cos \vartheta} \frac{\partial(\rho u m)}{\partial \lambda} + \frac{1}{r \cos \vartheta} \frac{\partial}{\partial \vartheta}(\rho v m \cos \vartheta) + \frac{1}{r^2} \frac{\partial}{\partial r}(\rho m w r^2) = -\frac{\partial p}{\partial \lambda} + \mathcal{D}, \quad (2.45)$$

or, in the shallow-atmosphere approximation,

$$\frac{\partial \rho m}{\partial t} + \frac{1}{a \cos \vartheta} \frac{\partial(\rho u m)}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta}(\rho v m \cos \vartheta) + \frac{\partial}{\partial z}(\rho m w) = -\frac{\partial p}{\partial \lambda} + \mathcal{D}. \quad (2.46)$$

If we take the zonal average, consider only the steady state and explicitly write out the viscous terms, we obtain

$$\begin{aligned} \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta}(\rho v m \cos \vartheta) + \frac{\partial}{\partial z}(\rho m w) &= \frac{v}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} \left(\cos^2 \vartheta \frac{\partial}{\partial \vartheta} \frac{u}{\cos \vartheta} \right) \\ &\quad + v a \cos \vartheta \frac{\partial^2 u}{\partial z^2}. \end{aligned} \quad (2.47)$$

This equation evidently is *not* of the same form as (2.41). However, in shallow planetary atmospheres the vertical derivative on the right-hand side will dominate the horizontal derivative, and since $\partial^2 u / \partial z^2 = r \cos \vartheta \partial^2 m / \partial z^2$ we have

$$\nabla \cdot (\rho m \mathbf{v}) = \nu a \cos \vartheta \frac{\partial^2 m}{\partial z^2}, \quad (2.48)$$

where $\nabla_\lambda \cdot$ is the divergence in the meridional plane (i.e., at constant λ). This equation has the form of a divergence balanced by a diffusion and thus by the same reasoning as we employed in Section 2.4.2 we conclude that, under the same conditions, there can be no maximum of the angular momentum — and so no superrotation — in the fluid, a result known as Hide's theorem.

If the viscosity were so large that the viscous term were dominant in (2.47), and so with the horizontal term now important, then the fluid would evolve toward a state of solid body rotation, this being the fluid state with no internal stresses. In that case, there would be a maximum of angular momentum at the equator — a state of superrotation. (Related mechanisms have been proposed for the maintenance of super-rotation on Venus (Gierasch, 1975).)

Some atmospheres *do* superrotate. They can do so if the flow is not axisymmetric — essentially if there are eddies in the flow.

2.5 MECHANISMS OF SUPERROTATION

Prograde jets can be generated by Rossby waves propagating away from the region of their generation. We will look more into this mechanism later on, but we may note that we need to generate Rossby waves for this to be a viable mechanism. In midlatitudes on Earth-like atmospheres Rossby waves arise as a product of baroclinic instability. In slowly-rotating planets it seems that superrotation should because of an interaction between Rossby waves and Kelvin waves that generates an instability that converges eastward momentum toward the equator. The stationary patterns illustrated in Fig. 2.4 suggest that off equatorial eddies may have a tilted structure and thus might converge momentum toward the equator. However, the solutions as they stand do not produce superrotation as a simple argument shows.

The shallow water momentum equation is, neglecting friction terms

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \frac{\partial \mathbf{u}}{\partial x} + \mathbf{f} \times \mathbf{u} = -g \nabla h, \quad (2.49)$$

or, with approximation,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{f} + \boldsymbol{\omega}) \times \mathbf{u} = -\nabla (gh + \mathbf{u}^2/2) \quad (2.50)$$

where $\boldsymbol{\omega} = (\partial v / \partial x - \partial u / \partial y) \mathbf{k}$. The zonal component of this is

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} - f v = -g \frac{\partial h}{\partial x}, \quad (2.51)$$

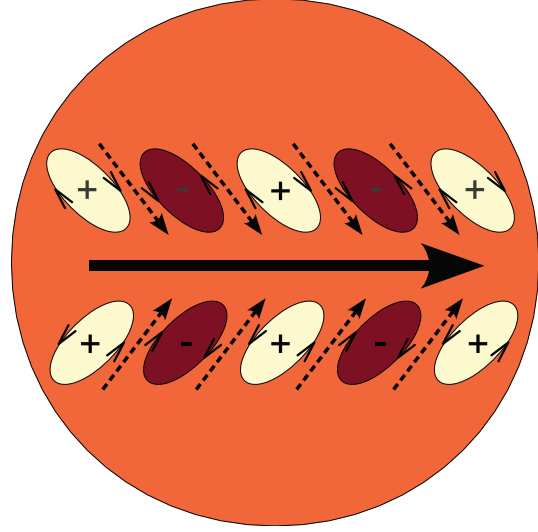


Fig. 2.6: A putative mechanism for the generation of superrotation in a tidally-locked planet. The tilted off-equatorial eddies, perhaps similar to those in Fig. 2.4, converge eastward momentum toward the equator, so producing superrotation, provided that nonlinear effects are included. However, in a single-layer shallow-water model this mechanism is insufficient. (From Showman and Polvani (2011).)

or, without approximation,

$$\frac{\partial u}{\partial t} - v(\zeta + f) = -g \frac{\partial h}{\partial x} - \frac{1}{2} \frac{\partial}{\partial x} (u^2 + v^2). \quad (2.52)$$

Zonally averaging gives

$$\frac{\partial \bar{u}}{\partial t} - v(\zeta + f) = \overline{v' \zeta'}. \quad (2.53)$$

Note that

$$\overline{v' \zeta'} \neq -\frac{\partial \overline{u' v'}}{\partial y} \quad (2.54)$$

except in the divergence-free limit. Rather,

$$\overline{v' \zeta'} = -\frac{\partial \overline{u' v'}}{\partial y} + \overline{u \frac{\partial v}{\partial y}}. \quad (2.55)$$

If the flow is divergence-free in two dimensions then $\overline{u \frac{\partial v}{\partial y}} = -\overline{v \frac{\partial u}{\partial x}} = 0$. In a Gill-like solution, if it is steady and symmetric across the equator, $v = 0$. Thus, $\overline{v' \zeta'} = 0$ and there is no eddy-induced acceleration at the equator.

2.5.1 Vertical Transfer of Mass and Momentum

Showman and Polvani (2011), following Shell and Held (2004) argue that if (2.50) and (2.51) are meant to represent the upper layer in a two-layer atmosphere then they are incomplete and should be written as:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{f} + \boldsymbol{\omega}) \times \mathbf{u} = -\nabla(g h + \mathbf{u}^2/2) - \mathbf{R} \quad (2.56)$$

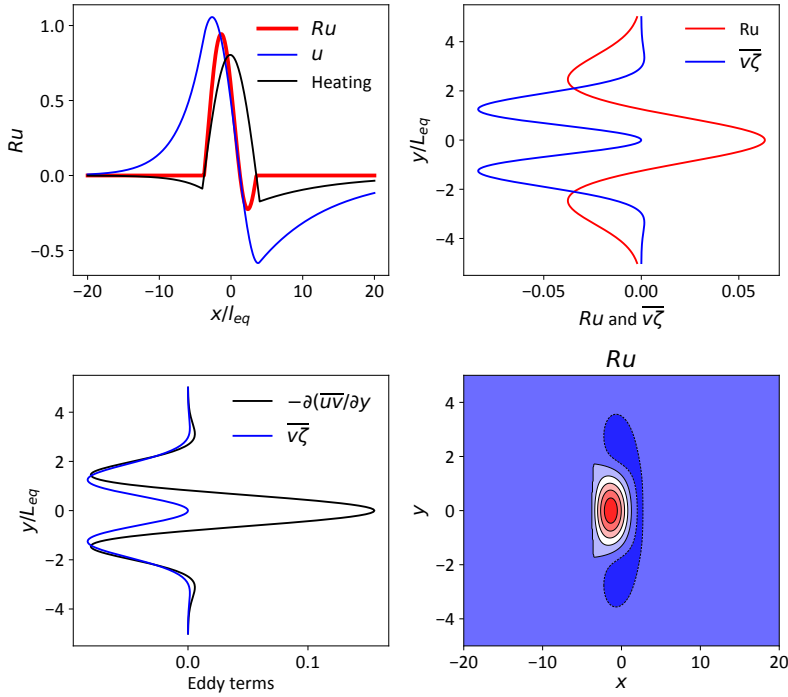


Fig. 2.7: Various acceleration terms in a shallow-water model of a tidally-locked planet.

where

$$\mathbf{R} = \begin{cases} -\mathbf{Q}\mathbf{u}/h, & Q > 0, \\ 0, & Q < 0. \end{cases} \quad (2.57)$$

The term $\mathbf{R}$ represents mass exchange between the layers. The idea is that if $Q > 0$ then air transported from the lower layer into the upper layer, carrying its momentum with it. If $Q < 0$ then fluid is transported from the upper layer to the lower layer, which does not affect the upper layer specific momentum budget so that $\mathbf{R} = 0$.

Derivation: if u_l and h_l represent lower layer values, the change in upper layer momentum is given by $h\partial u/\partial t = \dot{h}_l u_l = Qu_l$, so that the extra term is Qu_l/h . If $u = -u_l$ (the flow is baroclinic) then the extra term is $-Qu/h$.

In the Gill model the lower-layer flow is mostly eastward in the region where the source is positive. This brings eastward momentum up from the lower layer to the upper layer, accelerating the flow eastwards and generating superrotation.

2.6 TEMPERATURE VARIABILITY ON PLANETS WITH LOCALIZED HEATING

Tidally-locked planets have a heating that is concentrated in one area and cooling in another, and so might be expected to be extremely hot on one side and very cold on another. However, atmospheric circulation will attempt to even out the temperature and how well it can do that will depend

on things like the rotation rate and the stratification as well as the intensity of the forcing, and whether the ‘weak temperature gradient’ (WTG) approximation is satisfied. This approximation arises when rotation is weak and/or in the tropical circulation of a more rapidly rotating planet like Earth. The scaling goes back to Charney (1963, 1975) and the idea was used by various authors [Browning, Neelin, Pierrehumbert], and extended by Sobel et al (2001). When WTG is satisfied there are small temperature gradients in the free troposphere, meaning weak compared the temperature gradients in mid-latitudes (on an Earth-like planet), or compared to a similar more rapidly rotating planet, or compared to the surface temperature or radiative-equilibrium temperature.

2.6.1 Scaling

The dynamics relies on the weakness of the Coriolis force and the absence of a strong geostrophic or thermal wind balance. We will use the familiar shallow water equations in conventional form and the momentum equation is

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + \mathbf{f} \times \mathbf{u} = -g \nabla h, \quad \frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}h) = Q, \quad (2.58a,b)$$

where h is the surface height which is a proxy for temperature. (This statement can be formalized by considering the shallow water equations as the first baroclinic mode of the primitive equations.) If we let $h = H + \eta$ then (2.58) becomes, without approximation,

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + \mathbf{f} \times \mathbf{u} = -g \nabla \eta, \quad \frac{\partial \eta}{\partial t} + \nabla \cdot (\mathbf{u}\eta) + H \nabla \cdot \mathbf{u} = Q, \quad (2.59)$$

We can regard H as being maintained by some overall, non-spatially dependent forcing and we will ultimately be looking for the conditions under which $\eta \ll H$.

If geostrophic balance holds then the scaling relating velocity and pressure is the familiar one,

$$fU \sim \frac{g\mathcal{H}}{L} \quad (2.60)$$

where $\mathcal{H}$ is the scaling for η . If Rossby number is large we have

$$U^2 \sim g\mathcal{H} \quad (2.61)$$

If U is similar in the two cases then $\mathcal{H}$ is much smaller in the second case, by a factor $U/(fL)$ - i.e., by a Rossby number factor, and hence temperature variations are much smaller. This is the essence of WTG. However, we need to know how big the velocity is in order that the height variations are small.

In the nonrotating regime we evidently have

$$\frac{\mathcal{H}}{H} = \frac{U^2}{gH} \equiv Fr \quad (2.62)$$

where $Fr \equiv U^2/(gH)$ is the Froude number. So for small Froude number height variations are small. For Earth, $H \sim 7$ km and $g \sim 10 \text{ m s}^{-2}$ so that $Fr < 0.1$ holds for velocities up to about 80 m s^{-1} . In the rotating case we have instead

$$\frac{\mathcal{H}}{H} = \frac{fUL}{gH} \quad (2.63)$$

and $\mathcal{H}/H < 1$ only holds for velocities up to about 10 m s^{-1} if $L = 1000$ km.

To translate back to an ideal gas atmosphere, make the substitution:

$$\frac{\mathcal{H}}{H} \rightarrow \frac{\delta\theta}{\theta_0} \quad \text{and} \quad gH \rightarrow RT \quad (2.64)$$

so that

$$\text{Nonrotating:} \quad \frac{\delta\theta}{\theta_0} \sim \frac{U^2}{RT} \quad (2.65)$$

$$\text{Rotating:} \quad \frac{\delta\theta}{\theta_0} \sim \frac{fUL}{RT} \quad (2.66)$$

Since $R = 286 \text{ J kg}^{-1}$ then $RT = gH$ for $H = 7$ km and $T = 245$ K, and for $H = 8$ km and $T = 280$ K.

2.6.2 Scaling with a Diabatic Source

If the height variations are small then (2.59b) suggests

$$U = \frac{QL}{H} \quad (2.67)$$

and is the same in the rapidly-rotating and non-rotating cases. The associated Rossby number is then just

$$Ro = \frac{U}{fL} = \frac{Q}{fH}. \quad (2.68)$$

If the velocity is determined by the source term, and so about the same in the rapidly-rotating and non-rotating cases, then using (2.60) and (2.61) the height variations will be smaller in the non-rotating case by a factor of Rossby number. That is, if we let

$$\mathcal{H}_r = \frac{fUL}{g} \quad \text{and} \quad \mathcal{H}_{nr} = \frac{U^2}{g} = \frac{U}{fL} = \frac{Q}{fH} \quad (2.69a,b)$$

then

$$\frac{\mathcal{H}_{nr}}{\mathcal{H}_r} = \frac{U}{fL} = \frac{Q}{fH} < 1. \quad (2.70)$$

(Note that the rapidly-rotating scaling assumes that $fL > U$.)

Conditions for diabatic scaling

In the rotating case we use (2.60) and (2.67) to give

$$\frac{\mathcal{H}_r}{H} = \frac{QfL^2}{gH^2}, \quad (2.71)$$

and this is the condition under which weak temperature variations occur (assuming a steady state or slow time variations). We can put this into a perhaps more familiar form using (2.68)

$$\frac{\mathcal{H}_r}{H} = Ro \frac{f^2 L^2}{gH} = Ro \frac{L^2}{L_d^2} \quad (2.72)$$

where $L_d = \sqrt{gH}/f$ is the deformation radius (and $(L_d/L)^2$ is the Burger number). In the rotating case the Rossby number is small (by definition) so that the main condition for weak temperature variations (i.e. $\mathcal{H}/H \ll 1$) is that the scale of the motion be not significantly larger than the deformation radius. (The deformation radius may be considered to be the distance a gravity wave propagates before feeling the effects of rotation, and the velocity falling into geostrophic balance.)

If the planet is slowly rotating, or if the heating source is particularly strong, the Rossby number will not be small and we must use (2.61) and (2.67), giving

$$\frac{\mathcal{H}_{nr}}{H} = \frac{Q^2 L^2}{gH^3}. \quad (2.73)$$

Since $\mathcal{H}_{nr}/\mathcal{H}_r = Q/fH < 1$, it temperature gradients are likely to be smaller on slowly rotating planets than on rapid-rotators, in so far as this is determined by a divergent wind that responds to a diabatic source. In fact, the wind on a slowly rotating planet contains a superrotating component that is not divergent, and this tends to equalize the temperatures on the day side and night side more efficiently than the divergent wind.

2.7 SURFACE TEMPERATURE DISTRIBUTION

The surface temperature can have a stronger temperature variation than the atmosphere above it, because heating and cooling affect it more directly. Thus, even though the atmospheric circulation equalizes the temperature in the atmosphere, the surface temperature can have a significant contrast, as illustrated in Fig. 2.8.

Notes and References

Showman and Polvani (2011) and Pierrehumbert and Hammond (2019) provide discussions of some of the topics in this chapter.

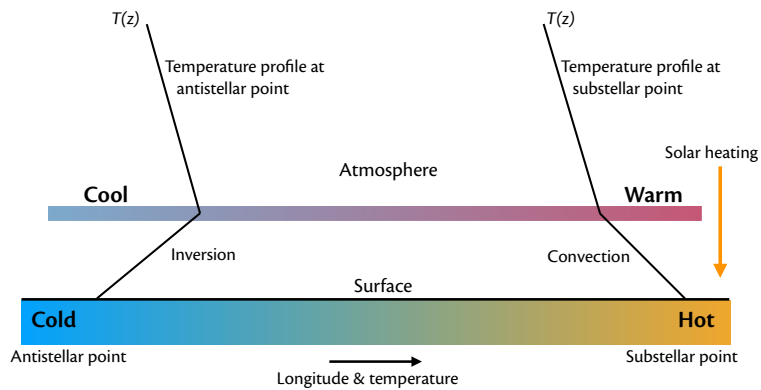


Fig. 2.8: Sketch of the possible temperature distribution on a tidally-locked planet. The temperature of the atmosphere is much more uniform than that of the surface, leading to an inversion at the anti-stellar point.

LECTURE 3

Giant Planets

In this chapter we will discuss the dynamics of 'cold' giant planets, such as Jupiter and Saturn in our own Solar System. (We won't discuss so-called ice giants in these lectures.) We'll focus quite a bit on the jets, as these are very visible and a nice GFD problem, although still quite controversial and not well understood. But first an informal overview of the entire structure of giant planets, using Jupiter as our main example, although the general principles apply also to Saturn and, most likely, to cold giants in other solar systems.

3.1 INTERIOR STRUCTURE

A posited interior structure of Jupiter is shown in Fig. 3.1.

3.2 JETS IN THE WEATHER LAYER

Jets in the weather layer can form through the action of differential rotation, or the β -plane. The simplest case is to take the limit of stratified quasi-geostrophic flow when it becomes two-dimensional flow, obeying the potential vorticity equation

$$\frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q = F - D, \quad (3.1)$$

where $q = \zeta + f$ and $f = \beta y$ and F is some forcing and D is the dissipation which acts only at small scales. We may write (3.1) as

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = F - D. \quad (3.2)$$

The very simplest argument suggestive of the production of zonal flow is a simple scaling one. Suppose we nondimensionalize everything in (3.1) so that the forcing is $\mathcal{O}(1)$ and is balanced by the terms on the left-hand side, which are all $\mathcal{O}(1)$, and the flow is in a statistically steady state. Then we increase the value of β ; the value of v must diminish, or else there is nothing to balance the βv term, unless there were cancellation between the advective term and the β term, but that would be non-generic.

This type of structure goes back to Stevenson, Salpeter and colleagues, c. 1980, with recent updates by Debras & Chabrier and colleagues.

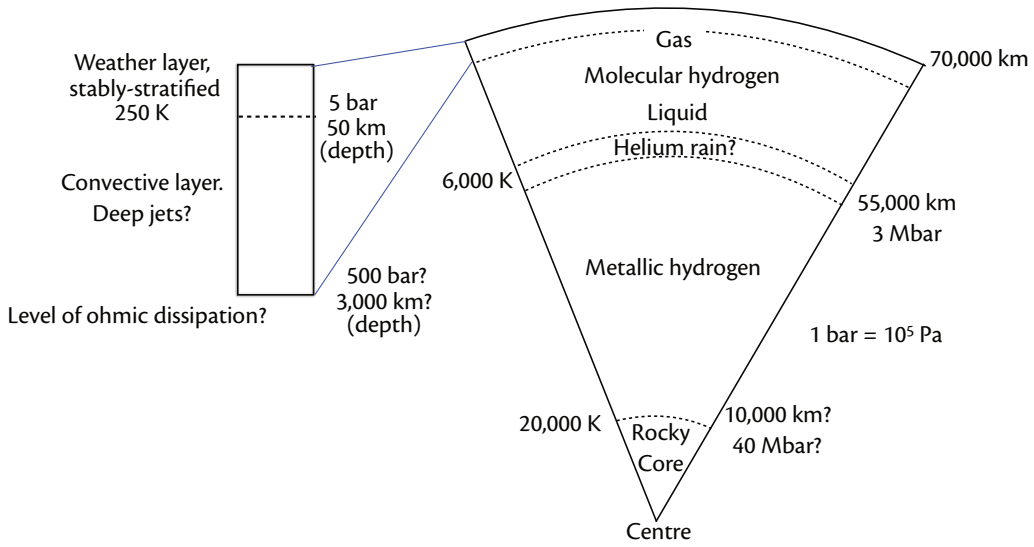


Fig. 3.1: A possible radial structure of Jupiter. Nearly all the aspects illustrated are estimates based on a combination of limited observations and uncertain theoretical models. All the numbers have significant error bars and, for example, the level of significant ohmic dissipation might be much deeper, the presence of helium rain is speculative and the rocky core could be more extensive.

3.2.1 Rhines Scale

Omitting the forcing term we write the equation of motion as

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = 0. \quad (3.3)$$

If $\zeta \sim U/L$ and if $t \sim T$ then the respective terms in this equation scale as

$$\frac{U}{LT} \quad \frac{U^2}{L^2} \quad \beta U, \quad (3.4)$$

and for now we do not differentiate between U and V . If the nonlinear term dominates then the flow will be turbulent and display all the characteristics of two-dimensional turbulence. If the beta term dominates then Rossby waves will dominate and the flow will be more linear. The cross-over scale, denoted L_R , is called the *Rhines scale* and is given by

$$L_R \sim \left(\frac{U}{\beta} \right)^{1/2}. \quad (3.5)$$

At scales larger than L_R Rossby waves are likely to dominate whereas at smaller scales advection and turbulence dominate and the flow is more isotropic.

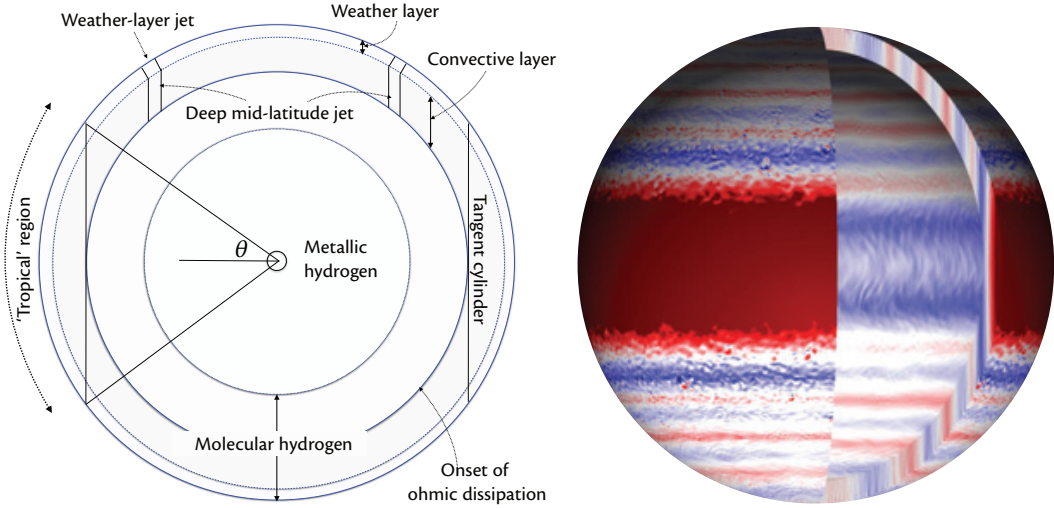


Fig. 3.2: Left: Sketch of the potential structure of Jupiter's atmosphere (not to scale) with the jets in a convective layer between a layer of ohmic dissipation and the very thin weather layer. Right: The zonal velocity on Jupiter obtained from a numerical simulation (from Heimpel et al. 2016) of a spherical shell with inner radius equal to 0.9 of the planetary radius, giving $\theta = 23^\circ$. Red colors denote eastward flow and blue colors westward.

3.2.2 Rossby waves and Jets

If a stirring is present in midlatitudes then Rossby waves will be generated there before propagating away where they dissipate. To the extent that the waves are quasi-linear, then just away from the source region each wave has the form

$$\psi = \text{Re } C e^{i(kx + ly - \omega t)} = \text{Re } C e^{i(k(x-ct) + ly)}, \quad (3.6)$$

where C is a constant, with dispersion relation (now back to the β -plane)

$$\omega = ck = \bar{u}k - \frac{\beta k}{k^2 + l^2}, \quad (3.7)$$

provided that there is no meridional shear in the zonal flow. The meridional component of the group velocity is given by

$$c_g^y = \frac{\partial \omega}{\partial l} = \frac{2\beta kl}{(k^2 + l^2)^2}. \quad (3.8)$$

Now, the direction of the group velocity must be away from the source region, because Rossby waves transport energy away from the disturbance. Thus, northward of the source kl is positive and southward of the source kl is negative. That the product kl can be positive or negative arises because for each k there are two possible values of l that satisfy the dispersion relation (3.7), namely

$$l = \pm \left(\frac{\beta}{\bar{u} - c} - k^2 \right)^{1/2}, \quad (3.9)$$

assuming that the quantity in parentheses is positive.

The velocity variations associated with the Rossby waves are

$$u' = -\text{Re } C i l e^{i(kx + ly - \omega t)}, \quad v' = \text{Re } C i k e^{i(kx + ly - \omega t)}, \quad (3.10a,b)$$

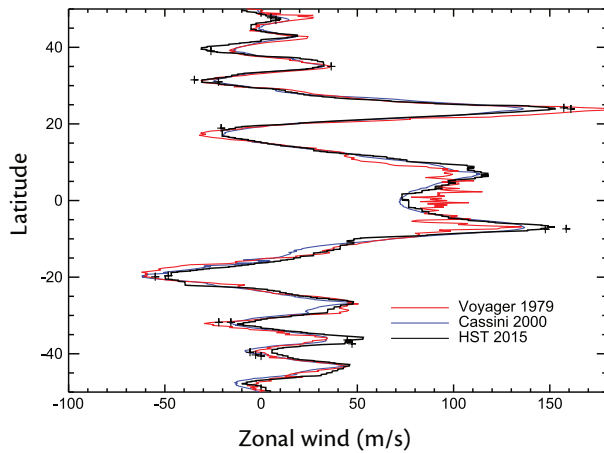


Fig. 3.3: Left: Zonal-average zonal winds on Jupiter at cloud top. The blue line is the Cassini profile in late 2000 (Porco et al., 2003), the red line is from Voyager in 1979 (Simon-Miller and Gierasch, 2010). The black line is from the Hubble Space Telescope in 2015 (Simon et al. 2015, from which the figure is adapted). Right: An image of Jupiter from the Cassini mission in visible light showing ammonia clouds. The belts (dark bands) are relatively warm and are areas of downwelling while the 'zones' (light bands) are cooler and areas of upwelling.

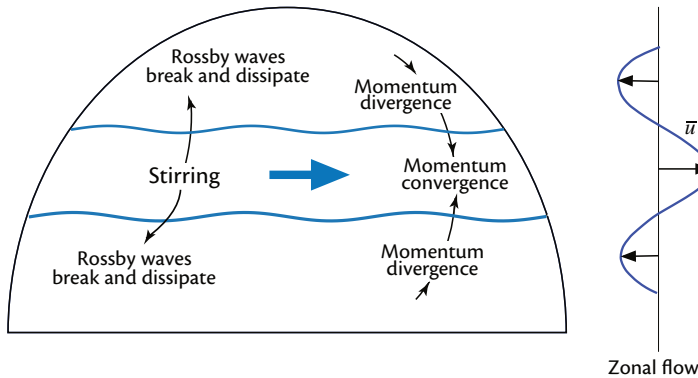


Fig. 3.4: Generation of zonal flow on a rotating sphere. Stirring in mid-latitudes (by baroclinic eddies) generates Rossby waves that propagate away. Momentum converges in the region of stirring, producing eastward flow there and weaker westward flow on its flanks.

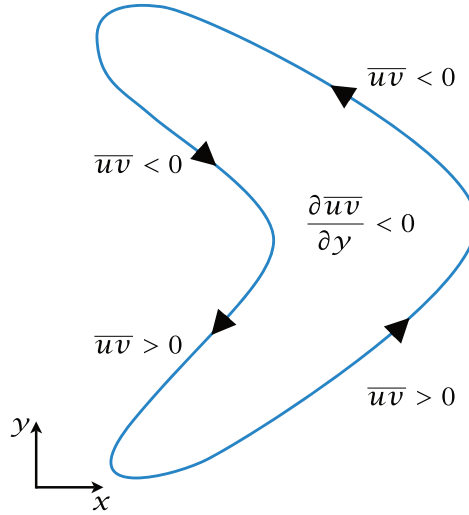
and the associated momentum flux is

$$\overline{u'v'} = -\frac{1}{2}C^2kl. \quad (3.11)$$

Thus, given that the sign of kl is determined by the group velocity, northward of the source the momentum flux associated with the Rossby waves is southward (i.e., $\overline{u'v'}$ is negative), and southward of the source the momentum flux is northward (i.e., $\overline{u'v'}$ is positive). That is, the momentum flux associated with the Rossby waves is *toward* the source region. Momentum thus converges in the region of the stirring, producing net eastward flow there and westward flow to either side (see Fig. 3.4).

If we think of this effect in physical space, then if kl is positive lines of constant phase ($kx + ly = \text{constant}$) are tilted north-west/south-east, and the momentum flux associated with such a disturbance is negative

Fig. 3.5: The momentum transport in physical space, caused by the propagation of Rossby waves away from a source in midlatitudes. The ensuing bow-shaped eddies are responsible for a convergence of momentum in the centre of the bow, which in turn accelerates the mean flow eastward. If the arrows were reversed the momentum transport would still have the same sign.



(that is, $\overline{u'v'} < 0$). Similarly, if kl is negative then the constant-phase lines are tilted north-east/south-west and the associated momentum flux is positive ($\overline{u'v'} > 0$). The net result is a convergence of momentum flux into the source region; that is, $\partial(u'v')/\partial y < 0$ and from (??) the flow accelerates eastward. In physical space the momentum flux is reflected by having eddies that are shaped like a boomerang, or a bow, as in Fig. 3.5.

Although our argument is barotropic, it captures the pattern of the eddy momentum fluxes in Earth's atmosphere, as illustrated in the bottom panel of Fig. ???. In the Northern Hemisphere the eddy momentum flux is in fact negative ($\overline{u'v'} < 0$) poleward of about 55°N and positive equatorward of that, thus converging momentum into midlatitudes and producing eastward surface winds. The same pattern (accounting for a change in sign of v) occurs in the Southern Hemisphere.

3.2.3 Potential Vorticity Staircase

Basic idea is that, because PV is an adiabatic invariant (i.e., conserved except for the actions of forcing and dissipation) it is mixed; the stirring properties of advection combined with viscosity acting at small scales give rise to homogenized regions. However, homogenizing requires energy and so the entire domain cannot necessarily be homogenized, leading to regions of homogenization separated by sharp jumps — a staircase. (This phenomena is related to the Prandtl–Batchelor effect, and to potential vorticity homogenization in ocean gyres.)

Suppose we start out with some random field of vorticity, ζ , existing in a uniform background gradient of PV, βy , so that $q = \zeta + \beta y$. We suppose that PV is to homogenized over some meridional scale L_h , so that in the homogenized region $\zeta = -\beta y + C$, or $u = \beta y^2 - Cy$ where C is a constant. If we take $C = 0$ then the energy in the homogenized region is

$$\frac{1}{2} \int u^2 dx dy \approx \int (\beta y^2)^2 dx dy, \quad (3.12)$$

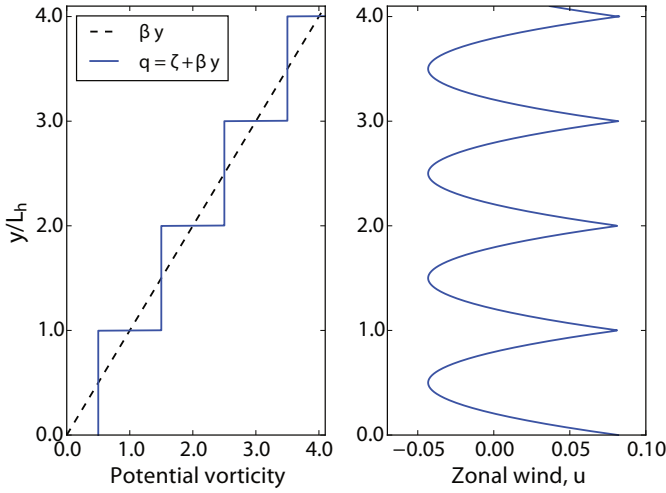


Fig. 3.6: An idealized potential vorticity staircase on a β -plane with $\beta = 1$. The left panel shows homogeneous regions of potential vorticity of meridional extent L_h separated by jumps, as calculated using (3.14). The right-hand panel shows the corresponding zonal flow, calculated using (3.17) with u_0 chosen to be such that the average zonal flow is zero.

giving $U^2 L_h \approx \beta^2 L_h^5$ where U is some average value. The value of U and the value of L_h will be determined by the amount of energy initially in the flow, but if we take its value as given, then the potential vorticity will become homogenized over the scale

$$L_h \sim \left(\frac{U}{\beta} \right)^{1/2}. \quad (3.13)$$

The value of U will jump at the edge of the homogenized region and form another homogenized region; the value of u will remain continuous, with the potential vorticity going through a series of jumps, as in Fig. 3.6.

The upshot is that potential vorticity is piecewise continuous and given by

$$q = \zeta + \beta y = q_0, \quad 0 < y < L_h, \quad (3.14a)$$

$$q = \zeta + \beta y = q_0 + \beta L_h \equiv q_1, \quad L_h < y < 2L_h, \quad \text{and so on.} \quad (3.14b)$$

The value of u is then given by, in each region,

$$u = \frac{1}{2} \beta (y - L_h/2)^2 + u_0, \quad 0 < y < L_h, \quad (3.15)$$

$$u = \frac{1}{2} \beta (y - 3L_h/2)^2 + u_0, \quad L_h < y < 2L_h, \quad (3.16)$$

where u_0 is a constant. In general we have

$$u = \frac{1}{2} \beta (y - (n - 1/2)L_h)^2 + u_0, \quad (n - 1)L_h < y < nL_h, \quad n = 1, 2, 3 \dots \quad (3.17)$$

3.3 CONVECTION AND JETS IN PLANETARY INTERIORS

Three groups of investigators: fluid dynamicists/applied mathematicians, astrophysicists, meteorologists.

We will come back to convection in the

Applied Mathematicians – ‘Pure’ RB convection: Convection between two plates each held at a fixed temperature. Heat can only enter or leave via conduction (i.e. diffusion) at the boundaries. These are a bottleneck. What happens to the flux as diffusion $\rightarrow 0$? Does the heat flux go to a limit independent of diffusivity (the ‘ultimate regime’)? Or go to zero? Or keep on increasing? This is an unanswered question (Charlie Doering knew more about this than anyone.)

Astrophysicists: Tend to argue that diffusion and viscosity are unimportant for their problems because the boundary layers are not the those of a rigid surface. Always gaseous, and radiation is important providing a heat source too.

Meteorologists Also tend to ignore viscous and diffusive effects because, ‘obviously’, viscosity and diffusivity are negligible. Strong focus on moisture and radiative-convective equilibria. They don’t care about fluxes, just the end-state of convection.

3.3.1 Taylor–Proudman in Spheres

Taylor–Proudman Effect: If the flow is in hydrostatic-geostrophic balance, then the flow is independent of ‘height’, meaning the distance in the direction of the rotation axis, even if gravity is in a different direction. Begin with the three-dimensional momentum equation

$$\mathbf{f}_0 \times \mathbf{v} = -\nabla\phi - \nabla\chi, \quad (3.18)$$

where $\mathbf{f}_0 = 2\mathbf{\Omega}$, $\phi = p/\rho_0$, and $\nabla\chi$ represents other potential forces, for example gravity. Gravity need not be aligned with the rotation axis.

On taking the curl of the above equation, the terms on the right-hand side vanish and the left-hand side becomes

$$(\mathbf{f}_0 \cdot \nabla)\mathbf{v} - \mathbf{f}_0 \nabla \cdot \mathbf{v} - (\mathbf{v} \cdot \nabla)\mathbf{f}_0 + \mathbf{v} \nabla \cdot \mathbf{f}_0 = 0. \quad (3.19)$$

But $\nabla \cdot \mathbf{v} = 0$ by mass conservation, and because $\mathbf{f}_0$ is constant both $\nabla \cdot \mathbf{f}_0$ and $(\mathbf{v} \cdot \nabla)\mathbf{f}_0$ vanish. Equation (3.18) thus reduces to

$$(\mathbf{f}_0 \cdot \nabla)\mathbf{v} = 0, \quad (3.20)$$

This means all the flow is uniform in the direction of the rotation axis. In shallow atmospheres we have that $\mathbf{f}_0 = f_0 \mathbf{k}$, implying that $f_0 \partial \mathbf{v} / \partial z = 0$, or

$$\frac{\partial u}{\partial z} = 0, \quad \frac{\partial v}{\partial z} = 0, \quad \frac{\partial w}{\partial z} = 0. \quad (3.21)$$

All three components of velocity are uniform along the axis of vertical rotation.

If we have a stratified atmosphere then we have

$$2\rho\mathbf{\Omega} \times \mathbf{v} = -\nabla p - \rho \mathbf{g} \quad (3.22)$$

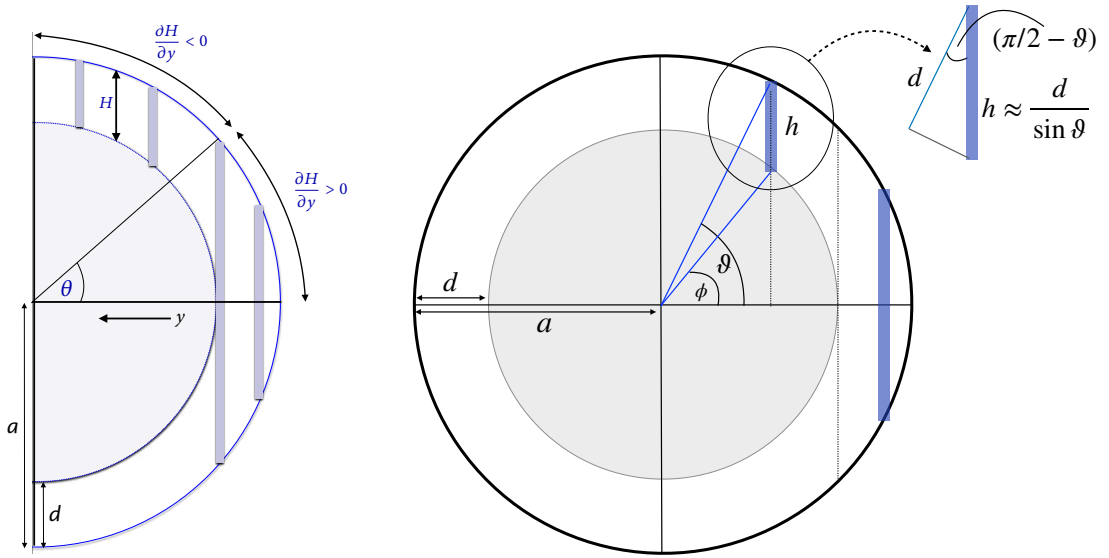


Fig. 3.7: Geometry of columns in the convective zone in a spherical shell. For columns poleward of latitude θ the column height diminishes with latitude and $\partial H/\partial y < 0$, whereas the column height increases with latitude outside the tangent cylinder and $\partial H/\partial y > 0$. The consequence is that the beta parameter, β^* , is positive inside the tangent cylinder (i.e. at higher latitudes) and negative outside the tangent cylinder in the tropical region.

or, approximately

$$2\rho_s \Omega \times \mathbf{v} = -\nabla p' - \rho' \mathbf{g} \quad (3.23)$$

Take the curl

$$2\Omega \cdot \nabla(\rho_s \mathbf{v}) - 2\Omega \cdot \nabla(\rho_s \mathbf{v}) = -\nabla \times (\rho' \mathbf{g}) \quad (3.24)$$

The first term is small because of mass continuity so that

$$2\Omega \cdot \nabla(\rho_s \mathbf{v}) = \nabla \rho' \times \mathbf{g}. \quad (3.25)$$

This is the thermal wind appropriate for giant planets.

3.4 JETS IN DEEP CONVECTION

3.4.1 Column Geometry

Before looking at the dynamics let us look at column geometry, with the goal of determining how a column of fluid, such as the one labeled h in Fig. 3.7, varies with latitude θ .

From the figure, for a column within the tangent cylinder, the following two equations connect the column height, h , its latitudes, θ and ϕ , the radius of the whole sphere, a , and the thickness of the outer region, d :

$$a \sin \theta = (a - d) \sin \phi + h, \quad (3.26a)$$

$$a \cos \theta = (a - d) \cos \phi. \quad (3.26b)$$

From these we easily find that

$$h = a \sin \vartheta - (a - d) \left[1 - \left(\frac{a}{a - d} \right)^2 \cos^2 \vartheta \right]^{1/2}. \quad (3.27)$$

If the shell is sufficiently thin then this expression reduces to

$$h \approx \frac{d}{\sin \vartheta}, \quad (3.28)$$

as can be seen from Fig. 3.7. The rate of change of column height with latitude is, in this approximation,

$$\frac{\partial h}{\partial \vartheta} \approx -\frac{d \cos \vartheta}{\sin^2 \vartheta}. \quad (3.29)$$

This expression reflects the fact that the column shrinks in height as it approaches the pole. We may also note that

$$\frac{1}{h} \frac{\partial h}{\partial \vartheta} \approx \frac{-1}{\tan \vartheta}. \quad (3.30)$$

For columns outside of the tangent cylinder the geometry is easier and we simply have, without approximation,

$$h = 2a \sin \vartheta, \quad (3.31)$$

and

$$\frac{\partial h}{\partial \vartheta} = 2a \cos \vartheta. \quad (3.32)$$

We also have

$$\frac{1}{h} \frac{\partial h}{\partial \vartheta} \approx \frac{1}{\tan \vartheta}, \quad (3.33)$$

which is just the negative of (3.30).

3.4.2 Dynamics

We will model the convective region as a layer of shallow water, obeying the shallow-water equations

$$\text{momentum:} \quad \frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta, \quad (3.34)$$

$$\text{mass continuity:} \quad \frac{Dh}{Dt} + h\nabla \cdot \mathbf{u} = 0, \quad (3.35)$$

$$\text{or} \quad \frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0, \quad (3.36)$$

where $\mathbf{u}$ is the horizontal velocity, h is the total fluid thickness. If there is topography at the bottom then $h(x, y, t) = \eta(x, y, t) - \eta_B(x, y)$. The above equations give the potential vorticity equation

$$\frac{DQ}{Dt} = 0, \quad Q = \left(\frac{\zeta + 2\Omega}{h} \right). \quad (3.37)$$

In the planetary context, quantity h is the thickness of the convecting layer and we write this as $h = H + h'$, where H is the mean shell thickness and h' are small, time-dependent, deviations of that due to fluid motion, and $H \gg h'$. From Fig. 3.7 we see that H varies in the y direction, decreasing toward the pole in the region poleward of the intersection with the tangent cylinder, but decreasing toward the equator in the region equatorward of the intersection with the tangent cylinder. It is this variation with mean thickness, and hence the variation of the background potential vorticity, that gives rise to a ‘topographic beta effect’ and hence to zonal jets. To see this explicitly, we make two more assumptions:

- (i) The small Rossby number assumption, that $|2\Omega| \gg |\zeta|$.
- (ii) The variations in mean height occur on a larger scale than the variations in vorticity.

The potential vorticity is then given by

$$Q = \left(\frac{\zeta + 2\Omega}{h} \right) \approx \left(\frac{\zeta + 2\Omega}{H} \right), \quad (3.38)$$

and, using the assumptions above, its evolution is given by

$$\frac{DQ}{Dt} \approx \frac{1}{H} \frac{D\zeta}{Dt} + 2\Omega \frac{D}{Dt} \left(\frac{1}{H} \right) = \frac{1}{H} \frac{D\zeta}{Dt} - \frac{2\Omega}{H^2} \mathbf{v} \cdot \nabla H, \quad (3.39)$$

and (3.37) becomes

$$\frac{D\zeta}{Dt} + \beta^* v = 0 \quad \text{where} \quad \beta^* = -\frac{2\Omega}{H} \frac{\partial H}{\partial y}, \quad (3.40a,b)$$

where v is the velocity in the y -direction. We see from Fig. ?? that β^* is positive in the region inside the tangent cylinder (the extra-tropics) and negative outside the tangent cylinder, in the tropics. Using (3.30) we obtain, inside the tangent cylinder,

$$\beta^* \approx \frac{2\Omega}{a \tan \vartheta}, \quad (3.41)$$

and the negative of that outside the tangent cylinder.

Tropical super-rotation

In the tropical region the zonal flow on Jupiter is *prograde*, that is, in the direction of the planetary rotation, and the atmosphere super-rotates (Fig. 3.3). The analogous jet on Saturn is also prograde. Why should this be so? As discussed above with respect to Venus, the mechanism must involve eddy motion and a plausible explanation is related to the nature of the beta effect in this region. In the tropical region outside the tangent cylinder the beta parameter, β^* given by (3.40b), is negative because the length of the convecting column *increases* with the y -coordinate. (In this region it is given by $H = 2a \sin \vartheta$, where ϑ is latitude and a is the planetary radius.) To simplify matters, let us suppose that β^* in (3.40) is a constant in which case we can write

An alternative explanation for tropical super-rotation on giant planets posits that it is caused by the propagation of Rossby waves away from the equatorial region, similar to the way eastward flow is formed in the mid-latitudes on Earth (Section 3.2.2). This mechanism can operate in the weather layer alone, provided there is a source for the Rossby waves such as moist convection or a heat flux from the interior, but does not preclude the jets going deeper into the interior.

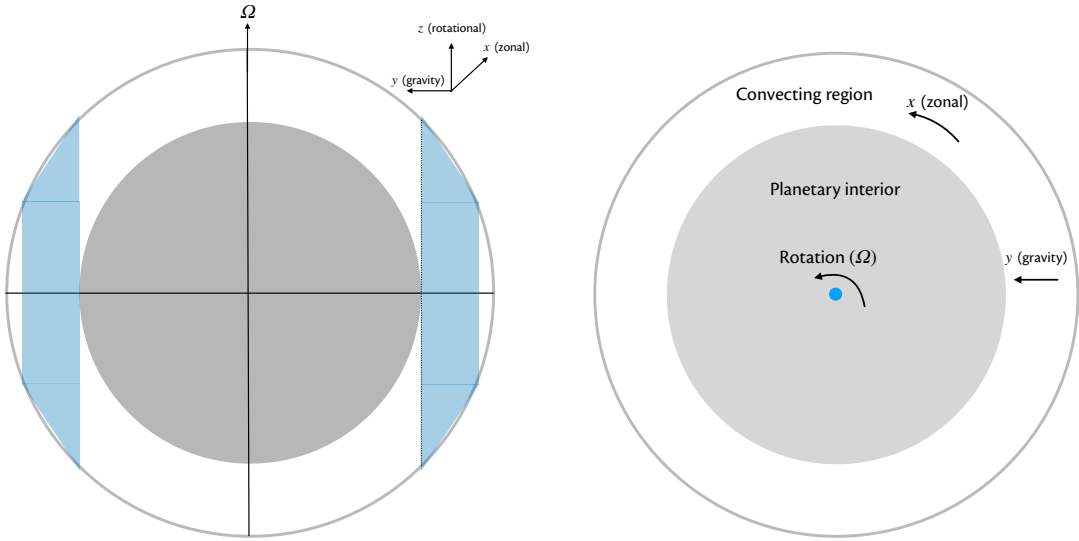


Fig. 3.8: An annulus, outside of the tangent cylinder. Left figure is the side view and the right-hand figure is the view from above the North Pole.

$$\frac{DQ^*}{Dt} = 0, \quad Q^* = \zeta + \beta^* y. \quad (3.42)$$

If the dynamics are turbulent in the tropical region (for example, if they are stirred by convection) then they will seek to homogenize the potential vorticity and the zonally-averaged flow is given by

$$\frac{\partial \bar{u}}{\partial y} + |\beta^*| y = A, \quad \text{or} \quad \bar{u} = Ay - \frac{1}{2} |\beta^*| y^2 + B, \quad (3.43)$$

where A and B are constants. If $\partial \bar{u} / \partial y = 0$ at $y = 0$ then $A = 0$ and $\bar{u}$ takes its *maximum* value at $y = 0$; that is, we expect the flow to be prograde, or super-rotating, at the equator.

If the tropical region is sufficiently wide then alternating jets can form within the tropical region, possible just shallow jets formed by the regular beta effect in the weather layer. (There is in fact some evidence of multiple jets in the tropical region, with distinct eastward maxima at about 7°N and 7°S evident in Fig. 3.3.)

3.4.3 A ‘quasi-geostrophic’ model

Another way to see the importance of the topographic beta effect is via construction of a ‘quasi-geostrophic’ model of the convecting region. We begin with the three-dimensional momentum equation, vertically integrate over the depth of the shell and take the curl (to give the vorticity equation) and specialize to the low Rossby number limit. We use the Boussinesq equations for simplicity of exposition, accepting that it is not necessarily a good approximation.

The Boussinesq equations are

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} + 2\boldsymbol{\Omega} \times \mathbf{v} = -\nabla \phi + b\mathbf{k} + \nu \nabla^2 \mathbf{v}, \quad (3.44)$$

$$\frac{\partial b}{\partial t} + \mathbf{v} \cdot \nabla b = \kappa \nabla^2 b, \quad (3.45)$$

$$\nabla \cdot \mathbf{v} = 0. \quad (3.46)$$

We take x to be the zonal direction, the y -direction is that aligned with gravity and the z -direction to be aligned with the rotation vector, with corresponding components of velocity u, v, w . We assume the flow is nearly geostrophic, that the vertical advection is negligible, and that the flow is outside the tangent cylinder, and so effectively in an annulus, with gravity pointing in the radial direction. We use a Cartesian geometry, as illustrated in Fig. 3.8 and Fig. 3.9, so that the geostrophic x - and y - momentum equation are, omitting the dissipative terms,

$$\frac{\partial u_g}{\partial t} + u_g \frac{\partial u_g}{\partial x} + v_g \frac{\partial u_g}{\partial y} - 2\Omega v = -\frac{\partial \phi}{\partial x}, \quad (3.47)$$

$$\frac{\partial v_g}{\partial t} + u_g \frac{\partial v_g}{\partial x} + v_g \frac{\partial v_g}{\partial y} + 2\Omega u = -\frac{\partial \phi}{\partial y} + b, \quad (3.48)$$

We can write this in the compact form

$$\frac{\partial \mathbf{u}_g}{\partial t} + \boldsymbol{\omega}^* \times \mathbf{u}_g + 2\boldsymbol{\Omega} \times \mathbf{u} = -\nabla \left(\phi + \frac{\mathbf{u}_g^2}{2} \right) + b\mathbf{j}. \quad (3.49)$$

where $\boldsymbol{\omega}^* = \hat{\mathbf{k}} \left(\partial v_g / \partial x - \frac{\partial u_g}{\partial y} \right) = \hat{\mathbf{k}} \zeta_g$ is the vertical component of the vorticity, and $\mathbf{j}$ is the unit vector in the y -direction. Taking the curl gives the vorticity equation, namely

$$\frac{\partial \zeta_g}{\partial t} + \mathbf{u}_g \cdot \nabla \zeta_g + 2\boldsymbol{\Omega} \cdot \left(\frac{\partial \mathbf{u}}{\partial x} + \frac{\partial \mathbf{v}}{\partial y} \right) = \frac{\partial b}{\partial x}. \quad (3.50)$$

or

$$\frac{\partial \zeta_g}{\partial t} + \mathbf{u}_g \cdot \nabla \zeta_g - 2\Omega \frac{\partial w}{\partial z} = \frac{\partial b}{\partial x} \quad (3.51)$$

We now take the vertical average of this equation to give

$$\frac{\partial \zeta_g}{\partial t} + \mathbf{u}_g \cdot \nabla \zeta_g = \frac{2\Omega}{h} [w]_b^t - \frac{\partial b}{\partial x}, \quad (3.52)$$

where t and b denote the top and bottom of the column. Neglecting Ekman layer effects, the vertical velocity is given by the kinematic condition $w = v \partial z_c / \partial y$ where z_c is the co-ordinate of the bounding surface. Given that, (3.52) becomes

$$\frac{\partial \zeta_g}{\partial t} + \mathbf{u}_g \cdot \nabla \zeta_g + \beta^* v = -\frac{\partial b}{\partial x}, \quad (3.53)$$

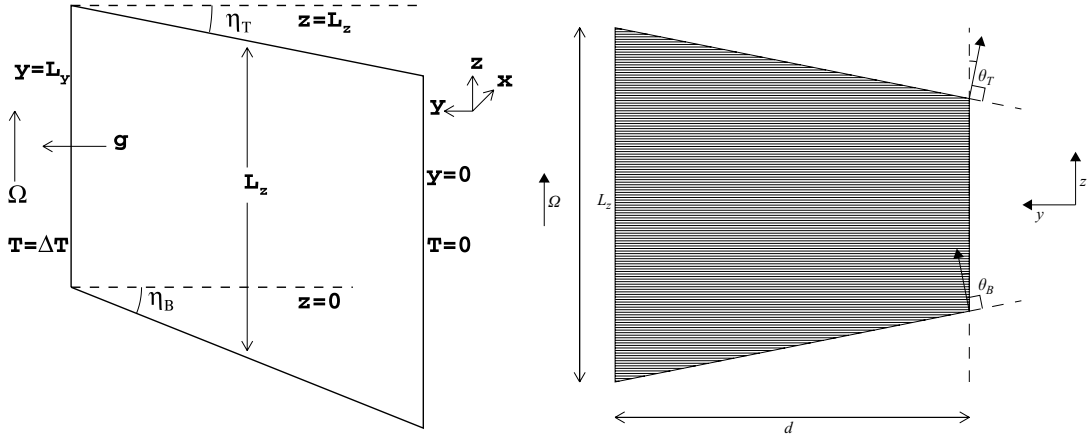


Fig. 3.9: Geometry of an annulus, outside of the tangent cylinder using locally Cartesian coordinates, from two papers by C. Jones.

where

$$\beta^* = -\frac{2\Omega}{h} \frac{\partial h}{\partial y}. \quad (3.54)$$

This is very similar to the usual quasi-geostrophic equations, except that now the beta effect is a topographic one, due to the slope of the spherical surface.

In buoyancy equation we also neglect vertical advection (because the vertical velocity is so small and N^2 is small, assuming near neutral stratification), and this becomes

$$\frac{\partial b}{\partial t} + \mathbf{u}_g \cdot \nabla b = H, \quad (3.55)$$

where we have simply written H to represent any heating or diffusive terms. If we assume that there is a basic-state temperature gradient in the radial direction, so that $b = \bar{b} + b'$, then (3.56) becomes

$$\frac{\partial b}{\partial t} + J(\psi, b) + \frac{\partial \psi}{\partial x} \frac{\partial \bar{b}}{\partial y} = H, \quad (3.56)$$

where $J(\psi, b)$ is the usual Jacobian (and equal to $\mathbf{u}_g \cdot \nabla b$) is the $\partial \psi / \partial x \partial \bar{b} / \partial y = v \partial \bar{b} / \partial y$ is the advection of the mean temperature, and this is the forcing term for the system.

3.4.4 Rossby number and Rhines Scale on Jupiter

On Jupiter, $\Omega \approx 2\pi / (9.925 \times 3600) = 1.76 \times 10^{-4} \text{ s}^{-1}$ so that, in midlatitudes, $f \approx 2\Omega \sin \pi/4 = 2.5 \times 10^{-4} \text{ s}^{-1}$. Its radius is $a \approx 7 \times 10^7 \text{ m}$. A conventional beta parameter in midlatitudes is then

$$\beta \sim \frac{2\Omega}{a} \cos(\pi/4) = 3.5 \times 10^{-12} \text{ m}^{-1} \text{ s}^{-1}, \quad (3.57)$$

with a similar (but not the same) value for the topographic beta parameter. If we take $U \sim 100 \text{ ms}^{-1}$ we get

$$L_R = \left(\frac{U}{\beta} \right)^{1/2} \approx 5300 \text{ km}, \quad (3.58)$$

which is not an unreasonable ball park for the jet width on Jupiter. If we use 5000 km as a scale, then the Rossby number is given by

$$Ro = \frac{U}{\Omega L} = \frac{100}{1.76 \times 10^{-4} \times 5 \times 10^6} \approx 0.1 \quad (3.59)$$

Note that if $f = 2\Omega \sin \vartheta$ then $\Omega \approx f$ in midlatitudes, so using f instead of Ω makes little difference. A bigger difference arises if we look at larger scales, and given Jupiter's radius of 70000 km then a scale of 20000 km would not be amiss, giving $Ro \approx 0.025$. The point is that the Rossby number for the motions we see in the weather layer is small, a little smaller than that in Earth's atmosphere, but not tiny. A thermal Rossby number is not particularly appropriate for Jupiter, since the jets are not primarily forced by the equator–pole temperature gradient.

Convective Rossby number

On Jupiter there is an internal source of energy, F_h , of about 6 W m^{-2} . If we use the Boussinesq approximation (not especially good for Jupiter's convective zone, but it gives us a ballpark) we can convert this to a buoyancy flux, F_b using

$$F_b \approx F_h \times \frac{g\alpha}{\rho c_p} \quad (3.60)$$

For an ideal gas the equivalent expression is

$$F_b \approx F_h \times \frac{g}{\rho c_p T_0} = F_h \times \frac{gR}{p_0 c_p} \quad (3.61)$$

For Jupiter, $g = 24.8 \text{ ms}^{-2}$, $c_p = 1300 \text{ J kg}^{-1} \text{ K}^{-1}$ and, more conveniently, $c_p \approx 7R/2$ and we take $p_0 = 10^5 \text{ Pa}$ or 1 bar. This gives

$$F_b \approx 7 \times 10^{-5} F_h \quad \text{and} \quad F_h = 1.4 \times 10^4 F_b. \quad (3.62)$$

This is about 10^5 times as large a buoyancy flux for a given heat flux as can be found for the ocean on an icy moon, because of the larger gravity and smaller density. For a heat flux of 6 W m^{-2} we obtain $F_b \approx 4 \times 10^{-4} \text{ m}^2 \text{ s}^{-3}$. The velocity then scales as

$$U \sim W \sim (F_b H)^{1/3} \approx 7 \text{ ms}^{-1}. \quad (3.63)$$

This is quite a lot smaller than the velocities observed in the weather layer, suggesting different physics are at work...Also, because this suggests a small Rossby number, other scalings come into play, specifically

$$U \sim W \sim (F_b H)^{1/3} Ro_c^{1/3} \quad (3.64)$$

and

$$L = H Ro_c \quad (3.65)$$

where $Ro_c = (F/(H^2\Omega^3))^{1/5}$ is the convective Rossby number. Putting in numbers gives $Ro_c \approx 0.1$ and $L = 300$ km and $U \approx 3.5$ ms⁻¹. The important point here is that the scale of the convection is smaller than the scale we see.

3.4.5 Thermal wind on a sphere

If we are interested in using the thermal wind relation to obtain velocity from density, then we need:

- To keep sphericity.
- To express thermal wind in terms of density gradients.

Both are actually straightforward.

The full momentum equation is

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} + 2\boldsymbol{\Omega} \times \mathbf{u} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = -\frac{1}{\rho} \nabla p - \mathbf{g}. \quad (3.66)$$

The thermal wind relation follows by using geostrophic and hydrostatic balance. On Earth, we conventionally assume that gravity and the Coriolis term, $\mathbf{f} = f\mathbf{k}$, are both vertical, but that is not necessary. At small Rossby number, and noting that even on giant planets the Centrifugal term is much smaller than gravity, we have

$$2\rho\boldsymbol{\Omega} \times \mathbf{u} = -\nabla p - \rho\mathbf{g}. \quad (3.67)$$

We now let $p = p_s + p'$ and $\rho = \rho_s + \rho'$, where $\nabla p_s = -\rho_s\mathbf{g}$ is hydrostatic balance for the static fields, so that, to a good approximation,

$$2\rho_s\boldsymbol{\Omega} \times \mathbf{u} = -\nabla p' - \rho'\mathbf{g}. \quad (3.68)$$

Take the curl of this to give

$$2\boldsymbol{\Omega} \nabla \cdot (\rho_s \mathbf{u}) - (2\boldsymbol{\Omega} \cdot \nabla)(\rho_s \mathbf{u}) = -\nabla \times (\rho' \mathbf{g}). \quad (3.69)$$

The first term on the left hand side is very small by mass continuity so we have

$$2\boldsymbol{\Omega} \cdot \nabla(\rho_s \mathbf{u}) = \nabla \rho' \times \mathbf{g}. \quad (3.70)$$

This is the thermal wind equation that is used in inverse models of giant planet interiors. The vectors $\mathbf{g}$ and $\boldsymbol{\Omega}$ are not, in general, in the same direction; if they were (as we essentially assume on Earth in the traditional approximation) we have

$$f \frac{\partial}{\partial z}(\mathbf{u} \rho_s) = -g\mathbf{k} \times \nabla \rho'. \quad (3.71)$$

for which the zonal component is $f \partial(\rho_s u) / \partial z = g \partial \rho' / \partial y$. (In terrestrial applications, we would often use an equation of state and/or pressure coordinates to transform density to temperature or entropy.) Despite the

superficial differences, in both the shallow-atmosphere (Earth-like) and deep-atmosphere (giant-planet-like) the equation states that the rate of rotation times the vertical derivative of velocity along the effective axis of rotation is approximately equal to the gravitational acceleration crossed with the variation of density.

In (3.70) Ω and $\mathbf{g}$ are at an angle, and this is plainly an important effect. If we are concerned with zonal winds and how far they might descend into the interior, then that component on (3.70) is

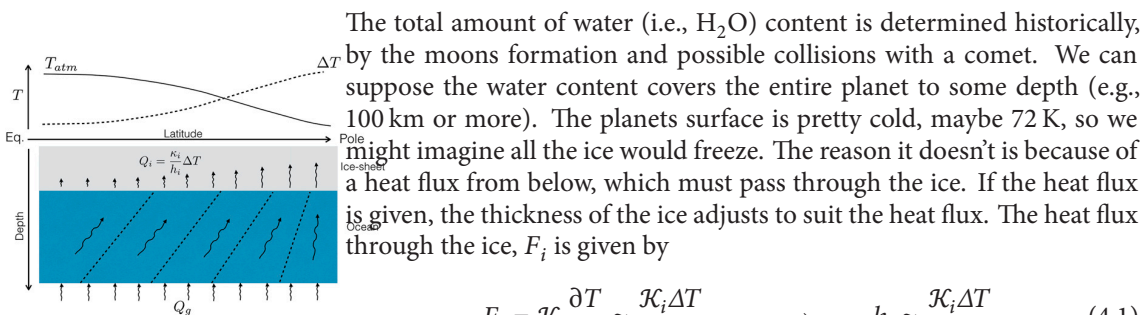
$$2\Omega \frac{\partial \rho_s u}{\partial z} = \frac{g}{r} \frac{\partial \rho'}{\partial \vartheta} \quad (3.72)$$

where $\partial/\partial z = r^{-1} \cos \vartheta \partial/\partial \vartheta + \sin \vartheta \partial/\partial r$ is the derivative in the direction of the axis of rotation. Finally, note that $\mathbf{g}$ is not necessarily a constant, as might normally be assumed from terrestrial applications.

LECTURE 4

Water worlds and Exo-Oceans

4.1 WHY AREN'T THE OCEANS FROZEN?



Sketch of heat transport in an idealized ice-covered ocean, for example the ocean on Earth during a snowball-Earth period, or possibly on some other planetary body that has a given pole–equator temperature gradient atmosphere. (From a paper by M. Jansen et al.)

The total amount of water (i.e., H_2O) content is determined historically, by the moons formation and possible collisions with a comet. We can suppose the water content covers the entire planet to some depth (e.g., 100 km or more). The planets surface is pretty cold, maybe 72 K, so we might imagine all the ice would freeze. The reason it doesn't is because of a heat flux from below, which must pass through the ice. If the heat flux is given, the thickness of the ice adjusts to suit the heat flux. The heat flux through the ice, F_i is given by

$$F_i = \mathcal{K}_i \frac{\partial T}{\partial z} \approx \frac{\mathcal{K}_i \Delta T}{h_i} \longrightarrow h_i \approx \frac{\mathcal{K}_i \Delta T}{F_i}, \quad (4.1)$$

where $\mathcal{K}_i$ is the conductivity of ice (approx $2 \text{ W m}^{-1} \text{ K}^{-1}$), h_i is the ice thickness and ΔT is the thickness across it, which will be approximately equal to the temperature difference of the planetary surface and freezing water (0°C). For Enceladus $\Delta T \approx 200 \text{ K}$ and with $F_i = 0.02 \text{ W m}^{-2}$ we obtain $h_i = 20 \text{ km}$. For Europa with $F_i = 0.05 \text{ W m}^{-2}$ and $\Delta T = 200$ we obtain 8 km and for snowball Earth with $F_i = 0.1 \text{ W m}^{-2}$ and $\Delta T = 50 \text{ K}$ we obtain $h_1 = 1 \text{ km}$.

4.2 SIDWAYS CONVECTION

Imagine an ocean with no overlying atmosphere, or with an atmosphere that is solely composed of water vapour at Earth-like temperature. Then the vapor pressure is the total pressure and at 290 K this is about 25 hPa, as opposed to 100 hPa at the surface of Earth, and there is likely to be very little stress on the top of the ocean. There also will be very little, if any, baroclinic instability. Thus, the forcing on the ocean will be thermal and not mechanical. If all the thermal forcing is exclusively at the top, or at the bottom, and not both (as in Rayleigh–Bénard convection) then the thermal excitation is much weaker and dependent on diffusivity. If there is thermal forcing from the bottom that may dominate, and the situation becomes more like the interior of giant planets. This is quite different from

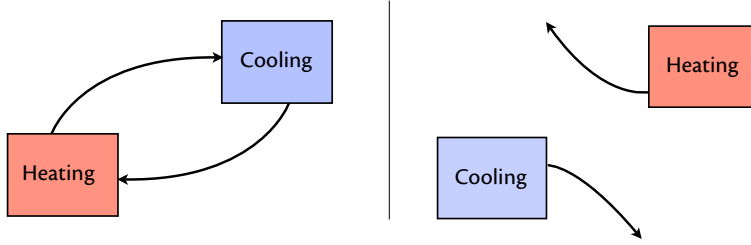


Fig. 4.1: Thermal convection. In the configuration on the left it is easy to imagine a sustained, steady circulation. In the configuration on the right there is nowhere to cool the heated water, or heat the cooled water. The basin will fill up with hot or cold liquid and the circulation will stop.

the situation on Earth where there is strong wind forcing, and tidal forcing. Even on Earth, if the planet were to be frozen (snowball Earth) then there would be very little wind forcing acting on the ocean, both because there would be very little temperature gradient from pole–equator, and the ice won’t transmit the wind to the water, especially in the presence of continents.

4.2.1 Boussinesq energetics

The Boussinesq equations are

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = -\nabla\phi + b\mathbf{k} + \nu\nabla^2\mathbf{v}, \quad \frac{Db}{Dt} = Q = J + \kappa\nabla^2b, \quad \nabla \cdot \mathbf{v} = 0, \quad (4.2a,b,c)$$

where Q is the total heating, and J its non-diffusive part. The boundary conditions are

$$b(x, y, 0, t) = g(x, y), \quad (\text{i.e., at the top}) \quad (4.3)$$

and no flux on the sides and bottom. We rewrite the first two equations as

$$\frac{\partial \mathbf{v}}{\partial t} + 2(\boldsymbol{\omega} + \boldsymbol{\Omega}) \times \mathbf{v} = -\nabla B + b\mathbf{k} + \nu\nabla^2\mathbf{v}, \quad \frac{\partial b}{\partial t} + \nabla \cdot (b\mathbf{v}) = Q, \quad (4.4)$$

where $B = \mathbf{v}^2/2 + \phi$. From the first we get the kinetic energy equation,

$$\frac{1}{2} \frac{\partial \mathbf{v}^2}{\partial t} = -\nabla \cdot (b\mathbf{v}) + wb + \nu \mathbf{v} \cdot \nabla^2 \mathbf{v}. \quad (4.5)$$

and, integrating over the whole domain,

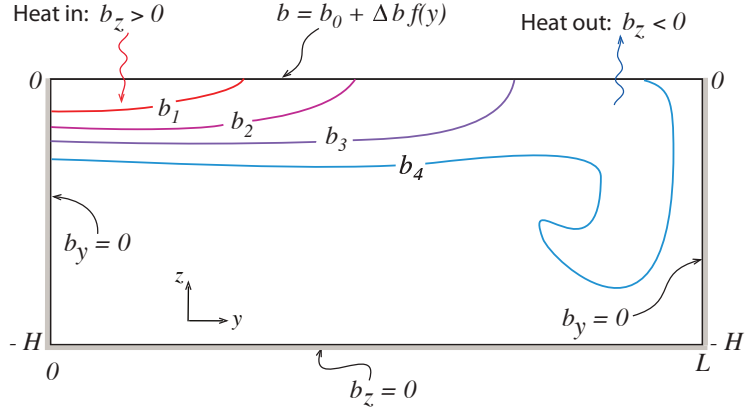
$$\frac{d}{dt} \left\langle \frac{1}{2} \mathbf{v}^2 \right\rangle = \langle wb \rangle - \varepsilon. \quad (4.6)$$

where $\varepsilon = -\nu \langle \mathbf{v} \cdot \nabla^2 \mathbf{v} \rangle = \nu \langle \boldsymbol{\omega}^2 \rangle$ is the average dissipation of kinetic energy. The potential energy budget is obtained by writing

$$\frac{Dbz}{Dt} = z \frac{Db}{Dt} + b \frac{Dz}{Dt} = zQ + bw, \quad (4.7)$$

‘Heating’ means buoyancy forcing in this context.

Fig. 4.2: Sketch of side-ways convection. The fluid is differentially heated and cooled along its top surface, whereas all the other walls are insulating. This gives, typically, a small region of convective instability and sinking near the coldest boundary, with generally upwards motion elsewhere.



so that, integrating over the domain,

$$\frac{d}{dt} \langle bz \rangle = \langle zQ \rangle + \langle bw \rangle. \quad (4.8)$$

and in a steady state

$$\langle zQ \rangle = - \langle bw \rangle. \quad (4.9)$$

Subtracting (4.8) from (4.6) gives the energy equation

$$\frac{d}{dt} \left\langle \frac{1}{2} \mathbf{v}^2 - bz \right\rangle = - \langle zQ \rangle - \epsilon. \quad (4.10)$$

and in a statistical steady state:

$$\langle zQ \rangle = -\epsilon < 0. \quad (4.11)$$

Thus, in order to maintain a circulation in which kinetic energy is dissipated, the heating (including diffusive heating) must occur, on average, at lower levels than the cooling. This is essentially a version of Sandström's effect.

In Earth's ocean the most of the heating occurs predominantly at the surface, except for the small effects of hydrothermal vents. If the diffusion were large case heat could potentially be diffused from the surface to depth, so effectively providing a deep heat source. However, on Earth at least, diffusion is not large and the heating is mainly at the top of the ocean, and Earth's ocean relies on mechanical forcing to drive it.

4.2.2 Height matters

From (4.11) we have

$$\frac{1}{H} \int_z \overline{zQ} dz = -\epsilon. \quad (4.12)$$

where the overbar denotes a horizontal average and H is the total depth of the domain. Suppose there is heating, $Q_h > 0$ over a small thickness centered around one height, z_h and cooling, $Q_c < 0$ at another height, z_c ,

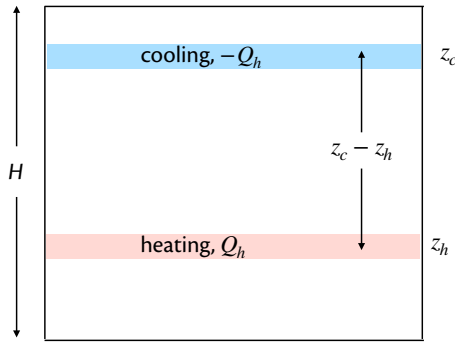


Fig. 4.3: Heating and cooling at two levels in a box.

both being within the domain, and there is no heating elsewhere (Fig. 4.3). Idealizing this with a delta function we write

$$Q = HQ_c\delta(z_c) + HQ_h\delta(z_h), \quad (4.13)$$

so that Q_c and Q_h are the amplitudes of the heating and cooling, and the factor of H simply provides the correct dimensionality. Substituting into (4.12) gives

$$(Q_h z_h + Q_c z_c) = -\varepsilon. \quad (4.14)$$

If Q_h is the heating and is taken to be positive, and $Q_c = -Q_h$, in order that no heat (buoyancy) is added to the system overall, we have

$$\varepsilon = Q_h(z_c - z_h). \quad (4.15)$$

That is to say, the potential energy generation (and consequent conversion to kinetic energy) is proportional to the height difference between heating and cooling.

Since ε is positive, the heating must occur at a lower level than the cooling (as before). Also, supposing that there is some form of energy cascade from the dominant scale with velocity U and scale L , then $\varepsilon \sim U^2/L$ and

$$\frac{U^3}{L} \sim Q_h \times (z_c - z_h). \quad (4.16)$$

This is a rough scaling for the velocity as a function of the height difference between heating and cooling. A little consideration will show that it is dimensionally and physically consistent with the scaling for the ultimate regime, to be found later on in (4.50). Finally, the right-hand side is just the flux of buoyancy between the heating and cooling, so we also have

$$\varepsilon = F_b. \quad (4.17)$$

Let us explore this a bit more.'

4.2.3 Flux and Rayleigh–Bénard convection

A well-studied form of convection is that of a fluid contained between two flat surfaces, a hot one below and a cold one above – this is Rayleigh–Bénard convection. In this section we will suppose the flux through the

boundaries is given, and not the temperature at the boundaries, as that is often the more geophysically meaningful case. Thus, start with (4.8) but replace Q by $-\partial F/\partial z$, where F is the upward diffusive flux of buoyancy, $F = -\kappa \partial b/\partial z$, giving

$$\frac{d}{dt} \langle bz \rangle = - \left\langle z \frac{\partial F}{\partial z} \right\rangle + \langle bw \rangle, \quad (4.18)$$

which in a steady state may be written

$$\frac{1}{H} \int_0^H z \frac{\partial \bar{F}}{\partial z} dz = \langle bw \rangle. \quad (4.19)$$

The overbar denotes an area average and the angle brackets are a volume average, and we may suppose that z extends from 0 to H and that the flux at the top equals that at the bottom, $\bar{F}_0$ say. Integrating by parts leads to

$$F_0 - \int_0^H \bar{F} dz = \langle bw \rangle. \quad (4.20)$$

The right-hand side is the volume-averaged conversion of potential energy to kinetic energy, ε . Thus, similar to the result we obtained previously,

$$\varepsilon = \bar{F}_0 - \int_0^H \bar{F} dz. \quad (4.21)$$

That is, the volume averaged conversion of potential energy to kinetic energy is equal to the area-averaged flux into the fluid, plus a term proportional to the integrated molecular flux over the volume, which will be small if κ is small. Neglecting that term, the total conversion of potential energy to kinetic energy *per unit area*, $PK = \varepsilon H$ (since ε is a volume average, not a volume integral), is thus

$$PK = \bar{F}_0 H, \quad (4.22)$$

and is dependent on the depth of the domain, not just the flux per unit area entering at the bottom. Note that this is not a scaling, it is a calculation. Similarly, if ε scales like U^3/H we have again $U^3 \sim HF$.

4.2.4 Salinity and temperature

In a saline ocean, the buoyancy is affected by both temperature and salinity. The true equation of state is nonlinear, but the main effects can be seen with a linear equation of state of the form

$$b = \alpha T - \beta_S S, \quad (4.23)$$

along with evolution equations

$$\frac{DT}{Dt} = Q_T, \quad \frac{DS}{Dt} = Q_S, \quad (4.24a,b)$$

where the terms Q_T and Q_S represent sources and sinks, including diffusive terms and $Q_b = \alpha Q_T - \beta_S Q_S$. The addition of salt (brine rejection) then represents a negative source of buoyancy. That is:

Ice melting $\implies$ Buoyancy source

Ice freezing $\implies$ Buoyancy sink

The only way that melting and freezing can drive a circulation is if the melting occurs at a lower level than the freezing, which would be unusual.

4.3 SIDEWAYS CONVECTION

4.3.1 Diffusive forcing, non-turbulence result

In the middle of the domain, the potential energy equation, (4.8) gives, in steady state:

$$\frac{\partial \overline{bw}}{\partial z} = \kappa \frac{\partial^2 \overline{b}}{\partial z^2}, \quad (4.25)$$

and integrating once vertically from the bottom, where the flux is zero

$$\overline{bw} - \kappa \overline{b}_z = 0. \quad (4.26)$$

Integrate again

$$\langle wb \rangle = H^{-1} \kappa [\overline{b}(0) - \overline{b}(-H)], \quad (4.27)$$

But the left-hand side is just the flux, equal to the energy dissipation (see (4.6)), so

$$\varepsilon = H^{-1} \kappa [\overline{b}(0) - \overline{b}(-H)]. \quad (4.28)$$

The energy dissipation thus goes to zero as $\kappa \rightarrow 0$ (Paparella & Young; extension to nonlinear EOS by Nycander). This result does not apply to Rayleigh–Bénard convection because in Rayleigh–Bénard convection the vertical flux is not zero, and the diffusive flux goes through a thinner and thinner boundary layer, and as $\kappa \rightarrow 0$, $\kappa \partial^2 \overline{b} / \partial z^2$ stays finite (or so it seems). The result of (4.28) does not necessarily mean that the interior flow is zero; as $\kappa \rightarrow 0$; there could for example be flows with zero Laplacian of vorticity, perhaps isolated vortices. However, numerical simulations do appear to show that the flow gets weaker and weaker.

4.3.2 Scaling estimates for Sideways Convection

This is most easily done for 2D convection, as the lateral dimension then appears more explicitly. Cross differentiate the momentum equations to give

$$\frac{\partial \nabla^2 \psi}{\partial t} + J(\psi, \nabla^2 \psi) = \frac{\partial b}{\partial y} + \nu \nabla^4 \psi, \quad (4.29a)$$

$$\frac{\partial b}{\partial t} + J(\psi, b) = \kappa \nabla^2 b, \quad (4.29b)$$

Nondimensionalize:

$$b = \Delta b \hat{b}, \quad \psi = \frac{\kappa L}{H} \hat{\psi}, \quad y = L \hat{y}, \quad z = H \hat{z}, \quad t = \frac{H^2}{\kappa} \hat{t}. \quad (4.30)$$

All these parameters are external except for H which we will assume is determined by the solution giving

$$\frac{\partial \hat{\nabla}^2 \hat{\psi}}{\partial \hat{t}} + \hat{J}(\hat{\psi}, \hat{\nabla}^2 \hat{\psi}) = Ra \sigma \alpha^5 \frac{\partial \hat{b}}{\partial \hat{y}} + \sigma \hat{\nabla}^4 \hat{\psi}, \quad (4.31)$$

$$\frac{\partial \hat{b}}{\partial \hat{t}} + \hat{J}(\hat{\psi}, \hat{b}) = \hat{\nabla}^2 \hat{b}. \quad (4.32)$$

$$Ra = \left(\frac{\Delta b L^3}{\nu \kappa} \right), \quad (\text{the Rayleigh number}), \quad (4.33)$$

$$\sigma = \frac{\nu}{\kappa}, \quad (\text{the Prandtl number}), \quad (4.34)$$

$$\alpha = \frac{H}{L}, \quad (\text{the aspect ratio}). \quad (4.35)$$

We now demand that the buoyancy term is $\mathcal{O}(1)$, and we get

$$H = L \sigma^{-1/5} Ra^{-1/5} = \left(\frac{\kappa^2 L^2}{\Delta b} \right)^{1/5}, \quad \Psi = Ra^{1/5} \sigma^{-4/5} \nu = (\kappa^3 L^3 \Delta b)^{1/5}. \quad (4.36a,b)$$

This is the scaling of T. Rossby. Interestingly,

$$\frac{H}{L} = \left(\frac{\kappa^2}{\Delta b L^3} \right)^{1/5} = (Ra \sigma)^{-1/5} \ll 1. \quad (4.37)$$

and this is the also the ratio of the horizontal scale of the plumes to the horizontal scale of the upwelling regime.

4.4 SCALING FOR RAYLEIGH–BÉNARD CONVECTION

Back to the 3D equations:

$$\frac{D\hat{\mathbf{u}}}{D\hat{t}} = -\hat{\nabla}_h \hat{\phi} + Pr \nabla^2 \hat{\mathbf{u}}, \quad (4.38)$$

$$\frac{D\hat{w}}{D\hat{t}} = -\frac{\partial \hat{\phi}}{\partial \hat{z}} + Ra Pr \hat{b} + Pr \nabla^2 \hat{w}, \quad (4.39)$$

$$\frac{D\hat{b}}{D\hat{t}} + \hat{w} = \nabla^2 \hat{b}, \quad (4.40)$$

$$\nabla \cdot \hat{\mathbf{v}} = 0. \quad (4.41)$$

and we suppose it is heated from the bottom and cooled from the top, perhaps giving a profile like Fig. 4.4. The classical theory tries to determine what the flux is given the temperature (or buoyancy) difference from top to bottom. There are two main theories, plus theories that try to link them all (Grossman & Lohse, 2000).

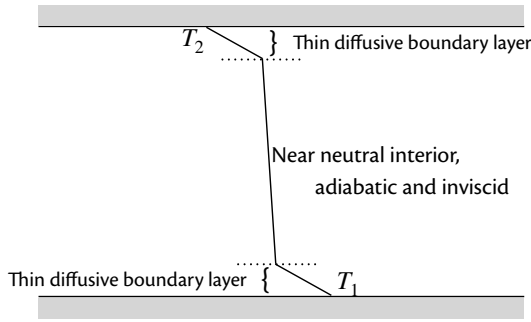


Fig. 4.4: Idealized spectrum in Rayleigh–Bénard convection. Most of the temperature gradient is near the boundaries.

4.4.1 The ultimate regime

The ultimate regime posits that the flux does not depend on the molecular properties of the fluid, because the viscosity and diffusivity are then so small their effects must (*obviously!*) be negligible. The flux in the interior is then the buoyancy flux, w and we suppose this depends only on Δb and H . That is,

$$\overline{wb} = (\Delta b)^\alpha H^\beta, \quad (4.42)$$

meaning, for the scaling variables,

$$\frac{H^2}{T^3} = \left(\frac{H}{T^2} \right)^\alpha H^\beta. \quad (4.43)$$

whence $\alpha = 3/2$ and $\beta = 1/2$. We have then, for the buoyancy flux F_b ,

$$F_b = (\Delta b)^{3/2} H^{1/2} = N_*^3 H^2, \quad (4.44)$$

where $N_* = (\Delta b/H)^{1/2}$, a quantity we use later. The corresponding scalings for horizontal and velocities, U and W , and length scale, L are just

$$U = W = (\Delta b H)^{1/2} = N_* H, \quad L = H. \quad (4.45a,b)$$

It's common to express the scaling in terms of a relationship between the Nusselt number and the Rayleigh number. The Nusselt number (Nu , Wilhelm Nusselt, 1882–1957) is the ratio of the total flux to the conductive or diffusive flux. Obtaining a scaling for the Nusselt number as a function of Rayleigh number is one of the Holy Grails (or Lost Arks) of convection (see Grossman and Lohse, 2000).

$$Nu = \frac{F}{F_d} = F \frac{H}{\kappa \Delta b} = \frac{F}{\beta_b} \quad (4.46)$$

The Nusselt number can be a function *only* of the Rayleigh and Prandtl numbers (assuming the aspect ratio not to be relevant), and a plausible way to produce a scaling for this is to assume that it has a power-law dependence on these quantities, as in

$$Nu = C Pr^\gamma \left(\frac{Ra}{Ra_c} \right)^\alpha. \quad (4.47)$$

Here C is some constant and Ra_c is the critical Rayleigh number for the configuration in question, also a constant (it is kept for convenience, and could be absorbed into C). The flux itself then scales as, in general,

$$F = C \left(\frac{\kappa \Delta b}{H} \right) \left(\frac{\nu}{\kappa} \right)^\gamma \left(\frac{Ra}{Ra_c} \right)^\alpha = \left(\frac{C}{Ra_c^\alpha} \right) \left(\frac{\kappa \Delta b}{H} \right) \left(\frac{\nu}{\kappa} \right)^\gamma \left(\frac{\Delta b H^3}{\nu \kappa} \right)^\alpha. \quad (4.48)$$

(We can write this expression in different ways, but it doesn't matter.) If we assume that the flux does not depend on κ or ν then we must have $\alpha = \gamma$ and $2\alpha = 1$, giving $\alpha = \gamma = 1/2$. Explicitly,

$$Nu = CPr^{1/2} \left(\frac{Ra}{Ra_c} \right)^{1/2}. \quad (4.49)$$

and the Nusselt number varies as the one-half power of the Rayleigh number. This result can also be obtained directly from (4.43).

In some situations we are given the flux, F . In terms of F and H we then have

$$U = W = (FH)^{1/3} \quad B = F^{2/3} H^{-1/3} = (F^2 H^{-1})^{1/3}. \quad (4.50)$$

In the above, the flux is a buoyancy flux. To obtain a genuine heat flux in $W m^{-2}$ we use the relation

$$F_b = F_h \times \left(\frac{g\alpha}{\rho c_p} \right) \quad (4.51)$$

For water, $\alpha = 1.67 \times 10^{-4} K^{-1}$ (smaller, about 1×10^{-4} at freezing point), $\rho = 1000 kg m^{-3}$, $c_p = 3986 J kg^{-1} K^{-1}$ and for Earth $g = 9.8 ms^{-2}$. This gives $g\alpha/\rho c_p \approx 4.1 \times 10^{-10}$, and g diminishes by a factor of about 10 for Europa and Enceladus, respectively.

4.4.2 The one-third scaling

If we allow for the effects of viscosity and diffusivity we open a Pandora's box of possibilities. However, one of these has gained some traction and appears to be supported by experiment, namely the 'one-third' regime. Going back to (4.48) we have

$$F_b = C \left(\frac{\kappa \Delta b}{H} \right) Pr^\gamma Ra^\alpha = C \left(\frac{\kappa \Delta b}{H} \right) \left(\frac{\nu}{\kappa} \right)^\gamma \left(\frac{\Delta b H^3}{\nu \kappa} \right)^\alpha, \quad (4.52)$$

The ultimate regime has $\gamma = \alpha = 1/2$, but there are other possibilities. In particular, Priestley (1959) and then others argued that the turbulent layer effectively insulates one viscous boundary from the other, and this must mean that the flux is (for a given buoyancy difference between top and bottom) independent of H , and the only way this can be achieved is by setting $\alpha = 1/3$, with γ remaining undetermined. We have then

$$F \sim F_c Pr^\gamma Ra^{1/3} = \frac{\kappa^{2/3}}{\nu^{1/3}} \Delta b^{4/3} Pr^\gamma \sim \kappa^{1/3} \Delta b^{4/3} \sim \kappa^{1/3} \beta_b^{4/3} H^{4/3}. \quad (4.53)$$

where the rightmost expressions follow if we set $\nu = \kappa$. For a given buoyancy gradient the heat flux falls as H increases. We equivalently have

$$Nu = CPr^\gamma Ra^{1/3} \quad (4.54)$$

where C is some constant. This is sometime known as the one-third regime for convection.

Malkus argument

Omit this in lectures

Another argument: suppose the thermal boundary layer is close to a state of marginal stability, based on a length scale proportional to the thickness of the boundary layer, λ . (The interior is highly supercritical, based on a Rayleigh number using the entire depth of the domain, H .) In this case the critical Rayleigh number for the system as a whole is equal to the critical Rayleigh number taken for values in the boundary layer, and the boundary-layer Rayleigh number is equal to that (being just critical). Thus

$$Ra_c = Ra_{\lambda c} = Ra_\lambda. \quad (4.55)$$

where Ra_λ is the Rayleigh number calculated using boundary-layer values and $Ra_{\lambda c}$ is the critical Rayleigh number in the boundary layer.

Now, the temperature drop in the boundary layer is half the total drop (assuming similar boundary layers at the top and bottom) so that

$$Ra = 2 \left(\frac{H}{\lambda} \right)^3 Ra_\lambda = 2 \left(\frac{H}{\lambda} \right)^3 Ra_{\lambda c} = 2 \left(\frac{H}{\lambda} \right)^3 Ra_c. \quad (4.56)$$

Now, the Nusselt number in the boundary layer, Nu_λ is equal to 1, since by assumption it is a diffusive layer. The total heat flux in the interior is the same as that in the boundary layer so, but the diffusive flux is lower by a factor of $2\lambda/H$ (the factor of 2 coming from the fact that a single boundary layer carries all the flux but has half the total temperature gradient). Thus, the two Nusselt numbers are related by

$$Nu = \left(\frac{H}{2\lambda} \right) Nu_\lambda = \frac{H}{2\lambda}. \quad (4.57)$$

Eliminating λ from (4.56) and (4.57) gives

$$Nu = \frac{1}{2} \left(\frac{Ra}{2Ra_c} \right)^{1/3}. \quad (4.58)$$

Thus, once again, the Nusselt number is proportional to the Rayleigh number to the one-third power. (Question: are the above two arguments equivalent?)

It should be said that other regimes have been proposed, for example a $2/7$ ths regime. See Grossman and Lohse and probably more recent articles.

4.5 SCALING FOR ROTATING RAYLEIGH–BÉNARD CONVECTION

The scaling is more difficult with rotation since a new timescale is introduced. If viscosity and diffusion are also present then even the linear problem is algebraically complicated, and scaling for the nonlinear problem becomes rather arbitrary and a nontrivial dynamics is needed. Here we will treat the problem by ignoring dissipative effects from the outset and obtain a scaling for what might be called the ‘rotating ultimate regime.’ It turns out that this scaling seems to be better satisfied by numerical simulations than is the ultimate regime in the non-rotating regime, possibly because rotation itself inhibits the heat transfer and the diffusive terms are no longer the bottleneck.

The relevant equations of motion, with the nonlinear terms on the right-hand side, are the momentum equations

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{f} \times \mathbf{u} + \nabla \phi = -(\mathbf{v} \cdot \nabla) \mathbf{u}, \quad \frac{\partial w}{\partial t} + \frac{\partial \phi}{\partial z} - b = -\mathbf{v} \cdot \nabla w \quad (4.59a,b)$$

and the mass continuity and thermodynamic equations,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \quad \frac{\partial b}{\partial t} - w N_*^2 = -(\mathbf{v} \cdot \nabla) b. \quad (4.59c,d)$$

Here we have written the buoyancy as

$$b = \bar{b}(z) + b'(x, y, z, t) = -N_*^2 z + b'(x, y, z, t), \quad (4.60)$$

where $N_*^2 = -\partial \bar{b} / \partial z$, and this is positive if the basic state is statically unstable, as we assume. Henceforth, and in (??) we omit the prime on b . We choose N_*^2 so that $b = 0$ at the top and bottom of the domain. We will be concerned with the rapidly rotating limit of these equations, which we take to be defined by $f \gg N_*$.

4.5.1 Linear Problem

To obtain the associated linear problem we omit the terms on the right-hand side of (4.59) and write $w = \tilde{w} \exp[\sigma t - i(kx + ly + mz)]$, and similarly for the other variables, giving

$$\sigma \tilde{u} - f \tilde{v} = -ik \tilde{\phi} \quad (4.61a)$$

$$\sigma \tilde{v} + f \tilde{u} = -il \tilde{\phi} \quad (4.61b)$$

$$\sigma \tilde{w} = -m \tilde{\phi} + \tilde{b}, \quad (4.61c)$$

$$\sigma \tilde{b} = \tilde{w} N_*^2, \quad (4.61d)$$

$$k \tilde{u} + k \tilde{v} + m \tilde{w} = 0. \quad (4.61e)$$

After some manipulation we obtain the expression for the growth rate, a form of dispersion relation,

$$\sigma^2 = \frac{N_*^2 k_h^2 - f^2 m^2}{k_h^2 + m^2}, \quad (4.62)$$

The ‘rotating mixing-length theory’, or RMLT, stems from Stevenson (1979), with further discussion and numerical experimentation to be found in Barker et al. (2014) and Currie et al. (2020).

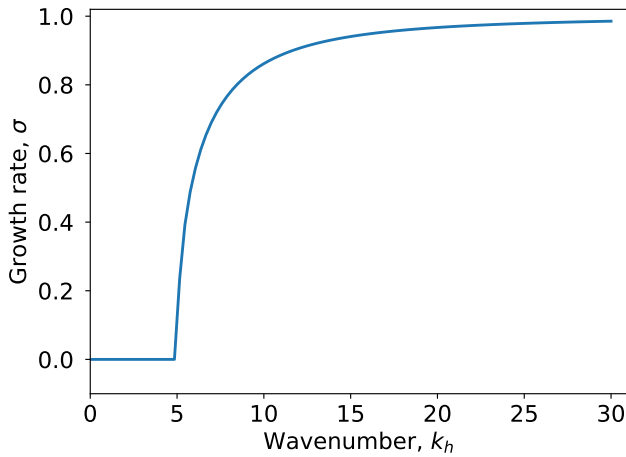


Fig. 4.5: The growth rate, σ , for rotating Rayleigh–Bénard convection with no viscosity or diffusion, with $N_* = 1$, $f = 5$ and $m = 1$. A common scaling for the nonlinear

where $k_h^2 = k^2 + l^2$, and as illustrated in Fig. 4.5. The flow is unstable provided $k_h > fm/N_*$, and so for sufficiently small horizontal scales.

4.5.2 Heuristic scaling

To obtain a scaling for the amplitude of the convection in the rotating regime we two main assumptions:

- (i) That the linear problem remains relevant even in the nonlinear regime, and that the largest unstable scales of motion are responsible for transporting the buoyancy (and so the heat) upward.
- (ii) That the linear growth is balanced by nonlinear effects — some form of nonlinear cascade or eddy diffusivity — so that the linear growth rate is equal to the inverse of the eddy turnover time.

We will make some smaller assumptions along the way.

Scales of motion

From the dispersion relation, (4.62), we see that rotation stabilizes the flow, and that the largest unstable modes occur when the vertical scale is the scale of the domain (and m takes its smallest value) the horizontal scale is such that the rotation term is no larger than the buoyancy term. Thus

$$m \sim \frac{1}{H} \quad \text{and} \quad k_h \sim \frac{\Omega}{N_* H} \quad \text{or} \quad L_h \sim \frac{N_* H}{\Omega}. \quad (4.63)$$

We assume that the buoyancy transport occurs predominantly at the scales associated with these wavenumbers: the horizontal scale of the motion is thus $L_h \sim N_* H/f$, and the vertical scale is H itself. Given that $f \gg N_*$ the horizontal scale is smaller than the vertical scale — the Coriolis force is unable to stabilize motions that are sufficiently elongated in the vertical, and modes with $k_h^2/m^2 > \Omega^2/N_*^2 \gg 1$ grow at a rate proportional to N_* ; that is $\sigma \sim N_*$.

Relative scales of velocity and buoyancy

The linear problem is presumed to provide scalings for the relative amplitudes of the large-scale fields. Using (4.61) with $f \gg \sigma$, and assuming that the horizontal flow has scale N_*H/f , we have

$$fU \sim \frac{f}{N_*H} \Phi, \quad (4.64a)$$

$$\sigma W \sim \frac{\Phi}{H} + B, \quad (4.64b)$$

$$\sigma B \sim WN_*^2, \quad (4.64c)$$

where the capital letters denote scaling values and B is the scale of the buoyancy deviations from the background profile. With $\sigma \sim N_*$ we obtain $B \sim N_*W$ and $U \sim W$ — that is, the horizontal and vertical velocities have similar magnitudes. The scaling for B can also be obtained by supposing that a parcel moves through the background stratification for some distance ℓ_v before losing its identity. If we take $\ell_v = W/N_*$, and the background gradient of buoyancy is just N_*^2 , then $B \sim N_*^2 \ell_v = N_*W$, as before.

Absolute scales of velocity, buoyancy and flux

To obtain the absolute values of buoyancy and velocity we suppose the linear growth is arrested by nonlinear terms that have the same eddy turnover time as the inverse growth rate. Thus, we simply write

$$\sigma U \sim \frac{U^2}{L} \quad (4.65)$$

where $\sigma = N_*$ and $L = N_*H/\Omega$, and similarly for W . This gives

$$U \sim W \sim \frac{N_*^2 H}{\Omega} \quad \text{and} \quad B \sim \frac{N_*^3 H}{f} = Ro \frac{N_*^2}{H}. \quad (4.66)$$

The Rossby number, $U/\Omega L$, is given by

$$Ro = \frac{U}{\Omega L} = \frac{(N_*^2 H/\Omega)}{(N_* H/\Omega)\Omega} = \frac{N_*}{\Omega}. \quad (4.67)$$

The scaling for the flux, F_R is then

$$F_R = WB = \frac{N_*^5 H^2}{\Omega^2} = Ro^2 N_*^3 H^2. \quad (4.68)$$

where $Ro = N_*/\Omega$ is the convective Rossby number.

The velocity and buoyancy variations are related to the flux by

$$U = W \sim \frac{F_R^{2/5} H^{1/5}}{f^{1/5}} \quad \left[= \left(\frac{F_R}{N_*} \right)^{1/2} \right], \quad (4.69)$$

and

$$B \sim \frac{F_R^{3/5} f^{1/5}}{H^{1/5}} \quad [= N_* W]. \quad (4.70)$$

The length scale is given by

$$L = \frac{N_* H}{\Omega} = \left(\frac{F H^3}{\Omega^3} \right)^{1/5}. \quad (4.71)$$

Convective Rossby number

Recall that $Ro = N_*/\Omega$, and in terms of the flux we have

$$N_* = \left(\frac{F \Omega^2}{H^2} \right)^{1/5} \quad (4.72)$$

whence

$$Ro = \left(\frac{F}{\Omega^3 H^2} \right)^{1/5}. \quad (4.73)$$

This is similar to a scaling of Maxworthy & Narimousa (1994). It is precisely equal to the relation

$$Ro = \left(\frac{U}{\Omega L} \right) \quad (4.74)$$

where U is given by (4.69) and $L = N_* H / \Omega = F H^3 / \Omega^3$.

To revert to the non-rotating case we replace f by N_* and obtain $F_{NR} = N_*^3 H^2$, as we have previously obtained by dimensional analysis. The ratio of the flux in the rotating and non-rotating cases is evidently

$$\frac{F_R}{F_{NR}} = \frac{N_*^2}{f^2} = Ro_c^2 \ll 1. \quad (4.75)$$

where $Ro_c = N_*/f$ is the convective Rossby number (see next section), which in the rotating case is less than one. Thus, rotation suppresses the flux due to convection. It is common in the convection literature to write the scaling for the flux as a relation between the flux and the Rayleigh number. In the rotating case this is just

$$Nu_R = \left(\frac{N_* 2}{f^2} \right) Pr^{1/2} Ra^{1/2} = Ro_c^2 Pr^{1/2} Ra^{1/2} \quad (4.76)$$

Geostrophic flow and Mass continuity

Let us write the mass continuity equation and its apparent scales as

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = -\frac{\partial w}{\partial z} \quad (4.77a)$$

$$\frac{U}{L} + \frac{U}{L} = \frac{W}{H}. \quad (4.77b)$$

Table 4.1: Summary of rotating and non-rotating scalings for Rayleigh–Bénard convection in an ultimate regime.

Value	Non-rotating	Rotating
Velocity, U, W	$N_* H = (\Delta b H)^{1/2}$	$N_*^2 H / \Omega = Ro (N_* H) = \Delta b / \Omega$
Buoyancy, B	$\Delta b = N_*^2 H = \Delta B$	$N_*^3 H / \Omega = Ro (N_*^2 H) = (\Delta b)^{3/2} / (H^{1/2} \Omega)$
Length, L	H	$N_* H / \Omega$
Rossby no., $Ro = U / (\Omega L)$	∞	N_* / Ω
Flux, F_{NR} or F_R	$N_*^3 H^2 = (\Delta b)^{3/2} H^{1/2}$	$N_*^5 H^2 / \Omega^2 = Ro^2 (N_*^3 H^2)$
Velocity, U, W	$(F_{NR} H)^{1/3}$	$F_R^{2/5} H^{1/5} / \Omega^{1/5} = (F_{NR} H)^{1/3} Ro^{1/3}$
Buoyancy, B	$(F_{NR}^2 H^{-1})^{1/3}$	$F_R^{3/5} \Omega^{1/5} / H^{1/5} = (F_{NR}^2 H^{-1})^{1/3} Ro^{-1/3}$
Length, L	H	$(F_R H^3 / \Omega^3)^{1/5} = H Ro$
Rossby no., $Ro = U / (\Omega L)$	∞	$(F_R / (H^2 \Omega^3))^{1/5}$

If $L \sim N_* H / f$ and $U \sim W$, as the previous scaling suggests, the scaling in (4.77) is inconsistent unless there is cancellation between $\partial u / \partial x$ and $\partial v / \partial y$. Happily, such cancellation is expected in rapidly rotating flows, because u and v are then in geostrophic balance. Consistency is achieved by writing

$$\left| \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right| = \frac{N_*}{f} \left(\left| \frac{\partial u}{\partial x} \right| + \left| \frac{\partial v}{\partial y} \right| \right). \quad (4.78)$$

so that

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \sim Ro_c \frac{U}{L} = \frac{W}{H}. \quad (4.79)$$

Flux Rayleigh number

The Rayleigh number is conventionally defined as

$$Ra = \frac{\Delta b H^3}{\nu \kappa}, \quad (4.80)$$

Eliminating Δb in favor of F using (4.44), namely $F_b = (\Delta b)^{3/2} H^{1/2}$, gives the flux Rayleigh number

$$Ra_F = \frac{F^{2/3} H^{8/3}}{\nu \kappa}. \quad (4.81)$$

Note

Thus, to summarize, rotation decreases the heat flux for a given temperature gradient, decreases the amplitude of the horizontal and vertical velocities, and decreases the horizontal scale of the flow.

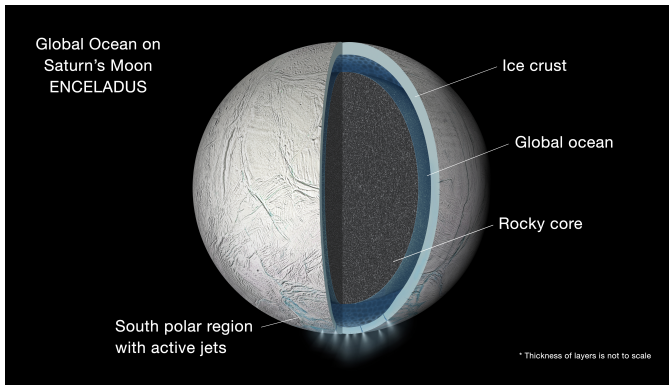


Fig. 4.6: Artist's sketch of possible structure of Enceladus, from Wikipedia.

4.6 JETS AND ROTATING RAYLEIGH–BÉNARD CONVECTION

4.6.1 Rhines scale

If we are in a deep convective regime, the Coriolis parameter is not changing but the thickness of the domain is, and this gives us a topographic β -effect and hence a Rhines scale.

4.7 EXO-OCEANS AND SNOWBALL EARTH

A sketch of the possible interior structure of Enceladus is shown in Fig. 4.6. The possibly global ocean has similarities to the convecting interior of the gas giant planets, as well as to Earth's own oceans. Various questions present themselves.

- (i) Is the circulation mostly like rotating Rayleigh–Bénard convection, and so perhaps like the interior of giant planets, or is it more like Earth's ocean?
- (ii) Is the fact that gravity is tilted relative to rotation vector important?
- (iii) Is the heat transport predominantly vertical or horizontal?
- (iv) Is the upwards heat transport via (vertical) convection, or slantwise convection?
- (v) Is there baroclinic instability?
- (vi) Is salt important?

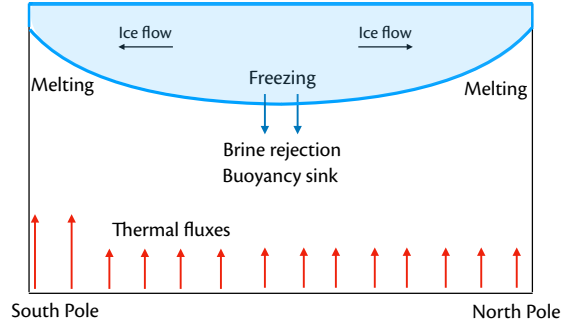
In what follows we will treat the fluid as a Boussinesq one, with the equation of state

$$\rho = \rho_0(1 - \alpha T), \quad (4.82)$$

where α is the coefficient of thermal expansion, and for water $\alpha \approx 1.67 \times 10^{-4} \text{ K}^{-1}$. We won't consider saline effects, although they could be important. If ρ_0 is the reference density then the buoyancy is given by

$$b = -g \frac{\delta \rho}{\rho_0} = g \alpha T. \quad (4.83)$$

Fig. 4.7: Sketch of a generic icy moon. The relative small height differential between freezing and melting, compared to the depth of the ocean, means that the thermal forcing will be more efficient than the saline effects in ‘driving’ the ocean. In this sketch we have imagined the geothermal flux is largest near one of the poles, as on Enceladus.



The fluid is assumed to have constant heat capacity c_p , and the above can then be used to convert between buoyancy, temperature and heat fluxes. For a liquid it is nearly always the case that $c_v \approx c_p$. Some astronomers and planetary scientists have the annoying habit of referring to the buoyancy, b , as temperature, and the buoyancy flux as the heat flux.

Converting buoyancy flux to heat flux, with $c_p = 3986 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\rho = 1000 \text{ kg m}^{-3}$ gives, for Earth with $g = 9.8 \text{ ms}^{-2}$,

$$\text{Heat Flux/Area} \times \frac{g\alpha}{\rho c_p} (\text{W m}^{-2}) = \text{Buoyancy Flux} (\text{m}^2 \text{ s}^{-3}). \quad (4.84)$$

$$\text{Heat Flux/Area} \times 4.1 \times 10^{-10} = \text{Buoyancy Flux} \quad (4.85)$$

$$\text{Heat Flux/Area} = \text{Buoyancy Flux} \times 2.4 \times 10^9. \quad (4.86)$$

For Europa ($g = 1.314 \text{ ms}^{-2}$)

$$\text{Heat Flux/Area} \times 5.5 \times 10^{-11} = \text{Buoyancy Flux} \quad (4.87)$$

$$\text{Heat Flux/Area} = \text{Buoyancy Flux} \times 1.8 \times 10^{10}. \quad (4.88)$$

For Enceladus ($g = 0.113 \text{ ms}^{-2}$)

$$\text{Heat Flux/Area} \times 4.7 \times 10^{-12} = \text{Buoyancy Flux} \quad (4.89)$$

$$\text{Heat Flux/Area} = \text{Buoyancy Flux} \times 2.11 \times 10^{11}. \quad (4.90)$$

Similarly, and related to if there is a heat flux, Q_g , coming in at the bottom of the ocean and out at the top, then the heat flux gives rise to a change in potential energy, P

$$\frac{dP}{dt} = - \int \dot{b} z dV = H \int_A \frac{g\alpha}{\rho c_p} Q_g dA \quad (4.91)$$

where V is its volume, H is its height and A its bottom area. (This will be converted to kinetic energy.)

4.8 SNOWBALL EARTH

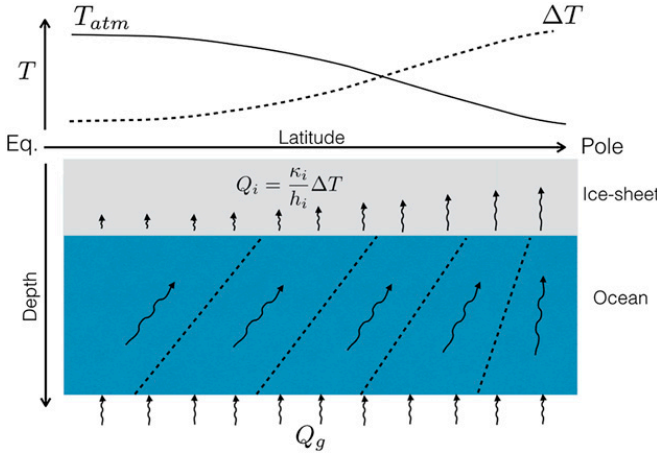


Fig. 4.8: Sketch of heat transport in an idealized ice-covered ocean, for example the ocean on Earth during a snowball-Earth period, or possibly on some other planetary body that has a given pole–equator temperature gradient atmosphere. (From a paper by M. Jansen et al.)

4.8.1 Horizontal and Vertical Heat Transport

Given a geothermal heat flux of 0.1 W m^{-2} we can estimate the associated velocities using $U^3/L = F_B$ so that, using H as the scale,

$$U = W = (F_b H)^{1/3} = \left(\frac{g\alpha}{\rho c_p} F_h H \right)^{1/3} \quad (4.92)$$

for $F_h = 0.1 \text{ W m}^{-2}$ we have $F_b = 4 \times 10^{-11} \text{ m}^2 \text{ s}^{-3}$ and

$$U \sim (4 \times 10^{-11} \times 2000)^{1/3} \approx 0.4 \text{ cm s}^{-1}. \quad (4.93)$$

(A larger estimate of about 2.5 cm s^{-1} was obtained by Jansen (2015) invoking a bottom drag. We can settle on $\sim 1 \text{ cm s}^{-1}$)

Consider an ice-covered ocean as sketched in Fig. 4.8 that is geothermally heated from below and with an ice cover of thickness h_i through which heat is diffused. In this example we suppose that the atmosphere sets the temperature of the top surface and this varies from pole to equator. The heat flux diffusing through the ice, Q_i , is then given by

$$Q_i = \frac{\kappa_i}{h_i} \Delta T = \frac{\kappa_i}{h_i} (T_a(\vartheta) - T_0). \quad (4.94)$$

In this expression, κ_i is the conductivity of the ice, h_i is the thickness of the layer, ΔT is the temperature difference across it, $T_a(\vartheta)$ is the surface temperature and T_0 is the water temperature, which we take to be the freezing temperature of water and hence a constant.

If the heat flux coming in at the bottom, Q_g , is given then in a steady state the horizontal heat transport, Q_h as a function of latitude is given by

$$Q_h(\vartheta) = \int_0^{2\pi} d\lambda \int_{-\pi/2}^{\vartheta} (Q_g(\vartheta) - Q_i(\vartheta)) a^2 \cos \vartheta d\vartheta. \quad (4.95)$$

Latitude ϑ , is assumed to go from $-\pi/2$ to $\pi/2$ between the two poles whereas longitude, λ , goes from 0 to 2π . This expression simply states that

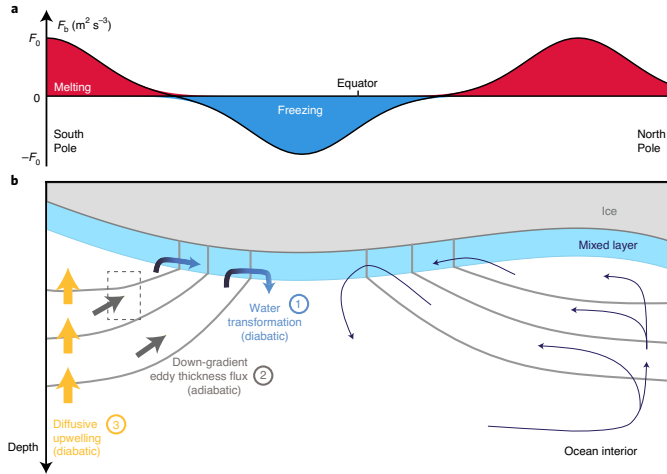


Fig. 4.9: Another view of the ocean circulation on Enceladus, inspired by a numerical simulation. From Lobo et al (2021), but their buoyancy forcing seems very wrong. (See also Fig. 4.8.)

the northwards, vertically integrated horizontal flux across a line of latitude is equal to the difference between the incoming and outgoing fluxes, $Q_g - Q_i$ integrated over all latitudes south of ϑ . If the integral over the globe Q_i is equal to that of Q_g then $Q_h(\pi/2) = 0$ as it should be. If the integral over each hemisphere vanishes the integral over latitude may be taken from 0 to ϑ .

The geothermal heat flux on Earth is estimated to be about 0.1 W m^{-2} , and the atmospheric temperature varies between about -30°C at the equator to -70°C at the poles, so that ΔT accordingly varies between 30 K and 70 K. Regardless of the depth and diffusivity of the ice, the heat flux through the ice will then vary between about 0.6 and 0.14 W m^{-2} . The peak meridional heat transport in midlatitudes may then be estimated to be, very approximately,

$$Q_h(\vartheta) \sim 0.5 \times 0.1 \times (6 \times 10^6)^2 \approx 2 \times 10^{12} \text{ W}. \quad (4.96)$$

or 2 TW. To meaningfully compare with the vertical transport we divide by the cross-sectional area of the ocean, about $2\pi \times 6 \times 10^6 \times 2000 \approx 7.5 \times 10^{10} \text{ m}^2$, giving about 35 W m^{-2} . Even with errors of up to an order or magnitude this is at least an order of magnitude larger than the vertical heat flux per unit area of about 0.1 W m^{-2} .

4.9 ICY MOONS

4.9.1 Why aren't the oceans frozen?

[Repeat of earlier section]

The total amount of water (i.e., H_2O) content is determined historically, by the moons formation and possible collisions with a comet. We can suppose the water content covers the entire planet to some depth (e.g., 100 km or more). The planets surface is pretty cold, maybe 72 K, so we might imagine all the ice would freeze. The reason it doesn't is because of

a heat flux from below, which must pass through the ice. If the heat flux is given, the thickness of the ice adjusts to suit the heat flux. The heat flux through the ice, F_i is given by

$$F_i = \mathcal{K}_i \frac{\partial T}{\partial z} \approx \frac{\mathcal{K}_i \Delta T}{h_i} \quad \longrightarrow \quad h_i \approx \frac{\mathcal{K}_i \Delta T}{F_i}, \quad (4.97)$$

where $\mathcal{K}_i$ is the conductivity of ice (approx $2 \text{ W m}^{-1} \text{ K}^{-1}$), h_i is the ice thickness and ΔT is the thickness across it, which will be approximately equal to the temperature difference of the planetary surface and freezing water (0°C). For Enceladus $\Delta T \approx 200 \text{ K}$ and with $F_i = 0.02 \text{ W m}^{-2}$ we obtain $h_i = 20 \text{ km}$. For Europa with $F_i = 0.05 \text{ W m}^{-2}$ and $\Delta T = 200$ we obtain 8 km and for snowball Earth with $F_i = 0.1 \text{ W m}^{-2}$ and $\Delta T = 50 \text{ K}$ we obtain $h_i = 1 \text{ km}$.

A few points to note:

- (i) For a uniform flux, where the surface is hotter ΔT will be smaller and the ice will be thinner.
- (ii) If the flux increases the ice thickness diminishes.
- (iii) Ice tends to flow from thick to thin, so will tend to melt where ice is thin if in a steady state.

4.9.2 Enceladus:

$$g = 0.113 \text{ ms}^{-2}, \quad Q_g \sim 0.02 \text{ W m}^{-2} \quad \Omega = 5 \times 10^{-5} \text{ s}^{-1}. \quad (4.98)$$

Locally buoyancy flux may be much larger ($\sim 1 \text{ W m}^{-2}$?).

Buoyancy flux, and so KE dissipation (ε):

$$F = \varepsilon \sim 0.02 * 5 \times 10^{-12} = 1 \times 10^{-13} \text{ m}^2 \text{ s}^{-3} \text{ (much larger in places)} \quad (4.99)$$

Typical speed: $U^3/L \sim \varepsilon$ gives, using $L = 10000 \text{ m}$, $U \sim 0.1 \text{ cm s}^{-1}$ or with $L = 30000 \text{ m}$, $U \sim 1 \text{ cm s}^{-1}$. Probably overestimates...

Another way to estimate velocity uses a bottom drag, $F \sim C_d U^2$ where $C_d \sim 2 \times 10^{-3}$ (it is a nondimensional quantity). In that case we use

$$C_d U^3 \sim h \varepsilon \quad \longrightarrow \quad U \sim (h \varepsilon / C_d)^{1/3} \quad (4.100)$$

where h is the depth of the lower turbulent boundary layer. If this is order 2 km we obtain $U \sim 0.4 \text{ cm s}^{-1}$.

Rossby number: Take $L = H = 30 \text{ km}$

$$Ro = \frac{U}{\Omega L} \approx \frac{0.01}{5 \times 10^{-5} \times 30000} \sim 0.006 \ll 1 \quad (4.101)$$

So rotation is important!

For another approach, we go back to the scaling for rotating Rayleigh-Bénard convection for which

$$L \sim \frac{N_* H}{\Omega} = \left(\frac{F H^3}{\Omega^3} \right)^{1/5} \approx 460 \text{ m}. \quad (4.102)$$

Velocity scale:

$$U \sim \left(\frac{F^2 H}{\Omega} \right)^{1/5} \approx 3.6 \times 10^{-4} \text{ m s}^{-1} = 0.36 \text{ mm s}^{-1} \quad (4.103)$$

Both velocity and length-scale seem quite small. The Rossby number is

$$Ro = \left(\frac{F}{\Omega^3 H^2} \right)^{1/5} = \left(\frac{1 \times 10^{-13}}{(5 \times 10^{-5})^3 (3 \times 10^4)^2} \right)^{1/5} = 0.016 \quad (4.104)$$

so still small! (But not absolutely miniscule.)

Flux Rayleigh number:

$$Ra_F = \frac{F^{2/3} H^{8/3}}{\nu \kappa} \approx 1 \times 10^{16} \quad (4.105)$$

using $\nu = 1 \times 10^{-6}$, $\kappa = 0.15 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$ and $H = 30000 \text{ m}$ and $F = 1 \times 10^{-13} \text{ m}^2 \text{ s}^{-3}$. This is large, but not as large as the Earth's atmosphere. The Reynold's number, $Re = UL/\nu \sim 1.5 \times 10^5$ for $L = 450 \text{ m}$ If we use $L = H = 3 \times 10^4 \text{ m}$ we get 1×10^7 , comparable to the square root of the Rayleigh number.

Hotspots

Around the south pole there are hotspots with a sea-floor temperature of about 370 K and fluxes of order 5 W m^{-2} . These gives rise to a buoyancy flux of about $2.5 \times 10^{-11} \text{ m}^2 \text{ s}^{-3}$ Given that flux we estimate a local velocity of, if rotation is not important,

$$U \sim (FH)^{1/3} \approx 0.01 \text{ m s}^{-1} \approx 1 \text{ cm s}^{-1}. \quad (4.106)$$

Could be larger if plumes are concentrated. Could be smaller if rotation affects them, in which case

$$U \sim (F^2 H / \Omega)^{1/5} \approx 0.01 \text{ m s}^{-1} \approx 0.3 \text{ cm s}^{-1}. \quad (4.107)$$

4.9.3 Europa:

$$g = 1.3 \text{ m s}^{-2}, \quad Q_g \sim 0.05 \text{ W m}^{-2} \quad \Omega = 2 \times 10^{-5} \text{ s}^{-1}. \quad (4.108)$$

Locally buoyancy flux may be much larger.

Buoyancy flux, and so KE dissipation (ε):

$$F = \varepsilon \sim 0.05 * 5.5 \times 10^{-11} = 2.7 \times 10^{-12} \text{ m}^2 \text{ s}^{-3} \text{ (much larger in places)} \quad (4.109)$$

Typical speed, go straight to RML theory,

$$U \sim (F^2 H / \Omega)^{1/5} = 0.002 \text{ cm s}^{-1}. \quad (4.110)$$

Length scale:

$$L \sim \left(\frac{FH^3}{\Omega^3} \right)^{1/5} = 3200 \text{ m} \quad (4.111)$$

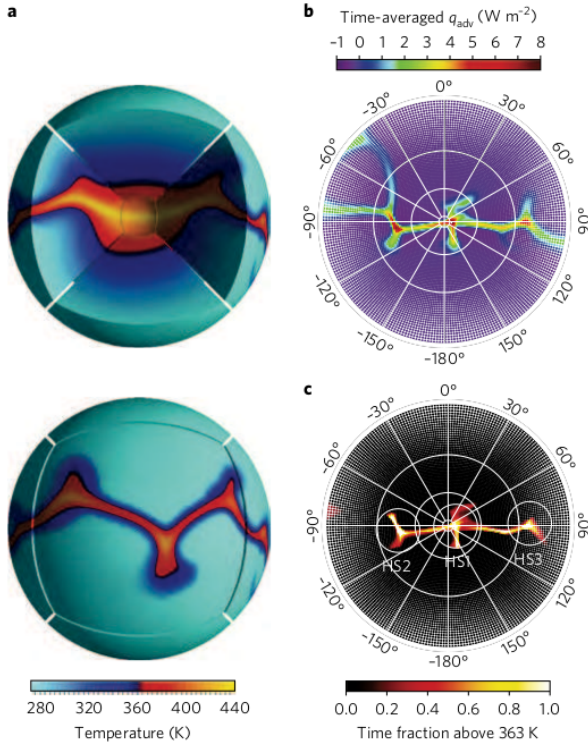


Fig. 4.10: Hotspots at the seafloor from a model of Enceladus interior. from Choblet et al 2017.

Velocity scale

$$U \sim \left(\frac{F^2 H}{\Omega} \right)^{1/5} = 0.002 \text{ ms}^{-1} = 0.2 \text{ cm s}^{-1}. \quad (4.112)$$

Rossby number:

$$Ro \sim \frac{U}{\Omega L} = 0.03. \quad (4.113)$$

4.9.4 Nature of the convection and lateral transport

Given the small Rossby number, convection is probably aligned along the axis of rotation, and not radially, except in regions of intense plumes. Mis-aligned gravity and rotation.

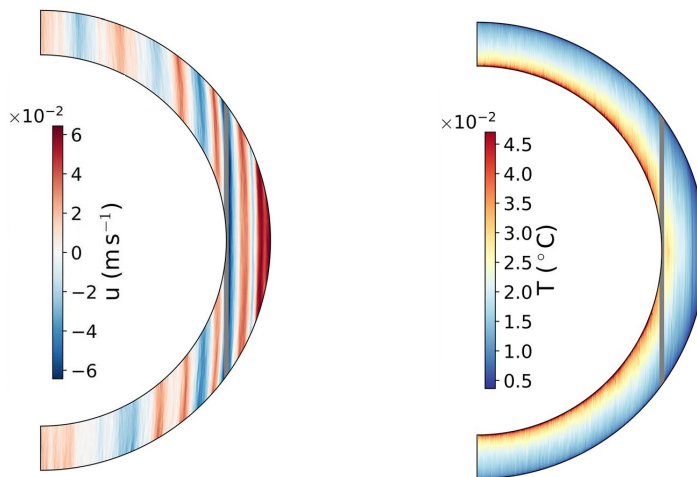
4.10 SALINE EFFECTS

Active are of research. Melting and brine rejection. Lets look at the equations again, supposing that buoyancy is a linear function of temperature and salinity and that both of those are quasi-conserved

$$\frac{DT}{Dt} = Q_T, \quad (4.114)$$

$$\frac{DS}{Dt} = Q_S, \quad (4.115)$$

Fig. 4.11: Zonal velocity (left) and temperature (right) from a numerical simulation of the ocean in icy moon, with a small Rossby number of about 0.0001. Bire et al.



$$b = g\alpha T - \beta_S S. \quad (4.116)$$

Here Q_T and Q_S are source terms of temperature and salinity, corresponding to heating (positive Q_T and ice melting or water freezing (negative and positive Q_S , respectively). Saline effects are likely to be important in affecting the shape of the circulation, but can't drive the circulation.

Notes

LECTURE 5

Condensibles and Habitability

Condensibles and radiative effects. Runaway greenhouse.

5.1 HABITABILITY

We will take the conventional definition that the temperature at the surface is amenable to the presence of liquid water. If the planet is too far from host star the water freezes.

As well as planets with oxygen/nitrogen/carbon-dioxide atmospheres there possibly are ones with hydrogen atmospheres, overlying an ocean. These are called hycean (hydrogen-ocean) planets, and possibly carbon planets (Cancri 55e).

But even if you are close enough to, in principle, allow liquid water there are issues:

- (i) Runaway greenhouse.
- (ii) Water gets cold-trapped at the cold point on the planet and does not return, dry elsewhere. This would be the case on a planet without oceans.

5.2 CONVECTIVELY-NEUTRAL LAPSE RATES

5.2.1 A few Thermodynamic Reminders

In the absence of chemical effects (such a salt contributing to the internal energy, the internal energy changes if work is done on it or if heat is added to it. In undergraduate textbook notation this is sometimes written as $du = Tds - p dV$ where u is internal energy, s in entropy and V is volume. We write this is

$$dI = Td\eta - p d\alpha \quad (5.1)$$

where $\alpha = 1/\rho$, η is entropy, and all values are specific (i.e., per unit mass).

Enthalpy is defined by $h = I + p\alpha$. Thus, in absence of chemical effects

$$\begin{aligned} dh &= dI + \alpha dp + p d\alpha \\ &= Td\eta + \alpha dp, \end{aligned} \quad (5.2)$$

having used (5.1) An adiabatic profile is one in which the entropy is constant. If the environmental profile follows that which an adiabatically displaced parcel takes, then the profile is convectively neutral: a parcel that is adiabatically displaced will have the same density as its new surroundings. If the displacement is vertical and hydrostatic, then $dp = -g dz$ and the convective neutrality condition is

$$d(h + gz) = 0. \quad (5.3)$$

Sometimes the quantity $h + gz$ is called a generalized entropy and then denoted h^* .

The internal energy of a simple ideal gas (or perfect gas) is given by $I = c_v T$, and from the ideal gas equation (i.e., $p = \rho RT$) we have that $p\alpha = RT$. Thus, $h = I + p\alpha = c_p T$. When potential energy is released (because a parcel descends, for example) it is converted to enthalpy, not internal energy. This is because the work needs to be done against pressure forces and enthalpy accounts for that.

5.2.2 Dry Adiabatic Lapse Rate

The dry adiabatic lapse rate has

$$d(c_p T + gz) = 0, \quad (5.4)$$

whence

$$\left(\frac{\partial T}{\partial z} \right)_{ad} = -\frac{g}{c_p}, \quad (5.5)$$

which is a classic result.

5.2.3 Lapse Rate with a Condensible

Now consider an ideal gas made up of two components, which we will call dry air and vapor (i.e., the condensible, which on Earth is water vapor), as well as some condensible in liquid form. The total enthalpy, H of the mixture is

$$H = h^d M^d + h^v M^v + h^l M^l. \quad (5.6)$$

The ‘latent heat of condensation’ is just the difference between the enthalpy of the vapor and the liquid, so that $L = h^v - h^l$, whence

$$H = h^d M^d + h^v M^v + (h^v - L) M^l = h^d M^d + h^v M^c - L M^l, \quad (5.7)$$

where $M^c = m^v + m^l$ is the total amount of condensate. Suppose now that the parcel is displaced, and some vapor is converted to liquid, so that $\delta M^c = -\delta M^v$ and $\delta M^d = 0$. We have

$$\delta H = \delta h^d M^d + \delta h^v M^c + L \delta M^v. \quad (5.8)$$

From this point on the liquid content of the atmosphere plays no role: it may be considered to have been removed by precipitation. Now divide by the total mass of the parcel, $M^t = M^d + M^v$ to give

$$\delta h^* = \delta h^d q^d + \delta h^v q^v + L \delta q^v + g \delta z = 0. \quad (5.9)$$

In this expression, q is the specific amount of the quantity, which when applied to the vapor is called the specific humidity.

We will now consider two limits of this expression: (i) The conventional, Earthly, limit in which the condensible is just a small fraction of the gas; (ii) the steam limit, in which the condensible is everything.

The Earthly limit

In the conventional limit the dry air dominates and we take $q^d = 1$ and $h^d = c_p T$ where c_p is a constant. The derivation follows that given in textbooks, and (5.9) becomes, to a good approximation,

$$c_p \delta T + L \delta q^v + g \delta z = 0, \quad (5.10)$$

or

$$c_p \frac{dT}{dz} = -L \frac{dq^v}{dz} - g. \quad (5.11)$$

If the air is moist but not saturated then an ascending parcel will follow the dry adiabatic lapse rate (because $dq^v/dz = 0$) but if it is saturated (and $q^v = q_s$) then as a parcel ascends it will cool and some water vapour will condense. Since $q_s \approx \epsilon e_s/p$, and so is a function of temperature and pressure, we have

$$\frac{dq_s}{dz} = \left(\frac{\partial q_s}{\partial T} \right)_p \frac{dT}{dz} + \left(\frac{\partial q_s}{\partial p} \right)_T \frac{dp}{dz} = \left(\frac{\partial q_s}{\partial T} \right)_p \frac{dT}{dz} + \left(\frac{q_s}{p} \right) \rho g, \quad (5.12)$$

using hydrostasy. Using the Clausius–Clapeyron equation and the ideal gas equation of state gives

$$\frac{dq_s}{dz} = \frac{Lq_s}{R^v T^2} \frac{dT}{dz} + \frac{gq_s}{R^d T}. \quad (5.13)$$

Using (5.12) and (5.13) we obtain an expression for the lapse rate of an adiabatically ascending saturated parcel,

$$\Gamma_s = - \left. \frac{dT}{dz} \right|_{ad} = \frac{g}{c_p} \frac{1 + Lq_s/(R^d T)}{1 + L^2 q_s/(c_p R^v T^2)}. \quad (5.14)$$

The steam limit

In the steam limit we are concerned with an atmosphere made of a single gas that also condenses. First some elementary thermodynamics. Suppose we have a mixture of gas and liquid, and the two are in initially in equilibrium. We then suppose some change occurs that converts a small amount of mass, ΔM , from liquid to vapor. The amount of heat input required to do this, ΔQ , is equal to the change in internal energies of the two subsystems, plus the work that needs to be done, with the changes occurring at the same temperature and pressure. The heat input is related to the entropy change so that

$$\Delta Q = T \Delta M (\eta_v - \eta_l) = \Delta M (I_v - I_l) + \Delta M p (\alpha_v - \alpha_l), \quad (5.15)$$

or, more simply,

$$\Delta Q/\Delta M = T(\eta_v - \eta_l) = (I_v - I_l) + p\alpha_v - p\alpha_l. \quad (5.16)$$

The quantity $I + p\alpha$ is the enthalpy, h , so we obtain a well-known expression

$$L \equiv T(\eta_v - \eta_l) = h_v - h_l, \quad \text{or equivalently} \quad h_v - T\eta_v = h_l - T\eta_l \quad (5.17a,b)$$

where L is called the latent heat of condensation, and is almost constant (which is a decent assumption, to a few percent accuracy). This equation means that when a mixture of gas and liquid are in equilibrium, the above relationship between enthalpy, pressure, temperature and entropy holds.

Now imagine some other change, caused for example by a displacement of a parcel of vapor to another pressure level and another temperature. From (5.17) we must have

$$\delta h_v - T\delta\eta_v - \eta_v\delta T = \delta h_l - T\delta\eta_l - \eta_l\delta T. \quad (5.18)$$

Since $h = I + p\alpha$ so and $\delta I = T\delta\eta - p\delta\alpha$ we have

$$\delta h = \delta I + p\delta\alpha + \alpha\delta p = T\delta\eta + \alpha\delta p \quad (5.19)$$

and using this in (5.18) gives

$$\alpha_v\delta p - \eta_v\delta T = \alpha_l\delta p - \eta_l\delta T \quad (5.20)$$

or

$$(\alpha_v - \alpha_l)\delta p = (\eta_v - \eta_l)\delta T, \quad (5.21)$$

or

$$\frac{dT}{dp} = \frac{T(\alpha_v - \alpha_l)}{(\eta_v - \eta_l)} \quad (5.22)$$

or

$$\frac{dT}{dp} = \frac{T}{L\rho} = \frac{RT^2}{Lp}. \quad (5.23)$$

neglecting the volume that the liquid occupies ($\alpha_l \ll \alpha_v$) and using $L = T(\eta_v - \eta_l)$. This is the Clausius–Clapeyron equation!

Using the hydrostatic approximation $\delta p = -\rho g \delta z$ gives the remarkably simple expression

$$\frac{dT}{dz} = -\frac{gT}{L}. \quad (5.24)$$

The steam limit directly from Clausius–Clapeyron

The Clausius–Clapeyron equation is, in its commonly used form,

$$\frac{de_s}{dT} = \frac{Le_s}{R^v T^2}. \quad (5.25)$$

where R^v and L are the gas constant and latent of condensation for the gas in question and e_s is the saturation vapor pressure, which we just denote as p in what follows. If L is constant the solution of this equation is

$$p = p_0 \exp \left[\frac{L}{R^v} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right], \quad (5.26)$$

which we can invert to give an expression for T , namely

$$T(p) = \frac{T_0}{1 - (RT_0/L) \log(p/p_0)}. \quad (5.27)$$

The pressure lapse rate is, from (5.25),

$$\frac{dT}{dp} = \frac{R^v T^2}{Lp} = \frac{(RT_0^2/Lp)}{[1 - (RT_0/L) \log(p/p_0)]^2}. \quad (5.28)$$

Using hydrostasy, $dp = -\rho g dz = -(p/R^v T) dz$ we obtain

$$\frac{dT}{dz} = -\frac{gT}{L} = -\frac{gT_0}{L} \left(\frac{1}{1 - (RT_0/L) \log(p/p_0)} \right). \quad (5.29)$$

and $T = T_0 \exp(-gz/L)$. The scale height is just L/g . If Earth had a pure water-vapor atmosphere with $L = 2.5 \times 10^6 \text{ J kg}^{-1}$ and $g = 9.8 \text{ ms}^{-2}$ then the scale height would be approximately 250 km and the lapse rate at 300 K would be about 1.1 K km^{-1} . Compare this with a dry isothermal atmosphere for which density scale height is about 8 km, and a convectively neutral atmosphere with a lapse rate of about 10 K km^{-1} .

Pressure and density in a Clausius–Clapeyron atmosphere

The general case

In the non-trivial limit the condensate is an appreciable fraction of total gas. The lapse rate in this case is left as an exercise. (Lee and Ingersoll, Graham et al.)

5.3 THERMAL WIND IN A STEAM ATMOSPHERE

The thermal wind relation is of foundational importance in geophysical co-ordinates. It combines geostrophic balance and hydrostatic balance with the result that the vertical shear of the zonal wind is proportional to the meridional temperature gradient. To refresh ourselves we do the standard dry case first.

5.3.1 Dry thermal wind

Height co-ordinates

Geostrophic wind is

$$fu = -\frac{1}{f}\rho \frac{\partial p}{\partial y} \approx -\frac{\partial}{\partial y} \left(\frac{p}{\rho_0} \right) \quad (5.30)$$

Hydrostasy is

$$\frac{\partial p}{\partial z} = -\rho g \quad \text{or approximately} \quad \frac{\partial p'}{\partial z} = -\rho' g. \quad (5.31)$$

Combine the above

$$f \frac{\partial u}{\partial z} = \frac{\partial}{\partial y} \left(\frac{g\rho'}{\rho_0} \right) = -\frac{\partial b}{\partial y}. \quad (5.32)$$

Pressure co-ordinates

Geostrophic wind is

$$fu = -\left(\frac{\partial \Phi}{\partial y} \right)_p. \quad (5.33)$$

Hydrostasy is

$$\frac{\partial \Phi}{\partial p} = -\frac{RT}{p}. \quad (5.34)$$

Combine the two

$$f \frac{\partial u}{\partial p} = \frac{\partial}{\partial y} \left(\frac{RT}{p} \right)_p, \quad (5.35)$$

or

$$f \frac{\partial u}{\partial p} = \frac{R}{p} \left(\frac{\partial T}{\partial y} \right)_p. \quad (5.36)$$

5.3.2 Steam thermal wind

As reminders, Clausius–Clapeyron can be written as $dp/dT = L\rho/T$ and the lapse rate in hydrostatic steam atmosphere is $\partial T/\partial z = -gT/L$.

Height co-ordinates

Geostrophic wind is

$$fu = -\frac{1}{f} \frac{\partial p}{\partial y} = -\frac{1}{f} \frac{\partial p}{\partial T} \frac{\partial T}{\partial y} = \frac{L}{T} \frac{\partial T}{\partial y} = L \frac{\partial \ln T}{\partial y}. \quad (5.37)$$

Hydrostasy in a steam atmosphere

$$\frac{\partial T}{\partial z} = -gT/L \quad \text{or} \quad \frac{\partial \ln T}{\partial z} = -\frac{g}{L}. \quad (5.38)$$

Combine the above two equations:

$$f \frac{\partial u}{\partial z} = L \frac{\partial}{\partial y} \frac{\partial \ln T}{\partial z} = -L \frac{\partial}{\partial y} \left(\frac{g}{L} \right) = 0. \quad (5.39)$$

Thus, a horizontal gradient of temperature does not induce a vertical shear of the zonal wind!

Pressure co-ordinates

Thermal wind in pressure co-ordinates is, as in (5.40),

$$f \frac{\partial u}{\partial p} = \frac{R}{p} \left(\frac{\partial T}{\partial y} \right)_p. \quad (5.40)$$

Without further ado we note that T is a function of p alone, so that the right-hand side is, immediately, zero, and there is no thermal wind.

5.4 ASPECTS OF A GLOBAL CIRCULATION IN STEAM ATMOSPHERES

We don't really know what the global circulation would be in a steam atmosphere, as we have no observations and very few simulations. Here we just make a few remarks.

5.4.1 Horizontal temperature and pressure gradients

Although there is no thermal wind there is likely to be a strong geostrophic wind. In Cartesian co-ordinates we have the familiar expression

$$fu = -\frac{1}{\rho} \frac{\partial p}{\partial y}. \quad (5.41)$$

If p is the saturation vapor pressure then

$$\frac{\partial p}{\partial y} = \frac{dp}{dT} \frac{\partial T}{\partial y} = \frac{Lp}{T} \frac{\partial T}{\partial y}, \quad (5.42)$$

so that the geostrophic wind is

$$u = \frac{L}{Tf} \frac{\partial T}{\partial y}. \quad (5.43)$$

Although there is no shear here, it is instructive to compare this to the familiar thermal wind. In pressure co-ordinates this is

$$f \frac{\partial u}{\partial p} = \frac{R}{p} \frac{\partial T}{\partial y} \quad \text{or} \quad f \frac{\partial u}{\partial Z} = -\frac{R}{H} \frac{\partial T}{\partial y} \quad (5.44)$$

where $Z = H \ln p/p_0$ and H (which is actually arbitrary) may be taken as a scale height. Thus, over a scale height, the thermal wind, u_T is given by

$$u_T \sim \frac{R}{f} \frac{\partial T}{\partial y}. \quad (5.45)$$

Comparing this with the geostrophic wind in a steam atmosphere with the same rotation rate and temperature gradient, we see

$$\frac{u_g}{u_T} \sim \frac{L}{TR} \approx 20, \quad (5.46)$$

using $L = 2.5 \times 10^6 \text{ J kg}^{-1}$, $R = 462 \text{ K kg}^{-1} \text{ K}^{-1}$ and $T_s = 300 \text{ K}$. Thus, the geostrophic wind on a steam planet is much larger than the thermal wind on a comparable Earth-like planet. If we take a temperature gradient of $60 \text{ K}/10000 \text{ km}$, similar to Earth, we find

$$u_g \sim \frac{2.5 \times 10^6}{300 \times 10^{-4}} \frac{60}{10^7} = 500 \text{ ms}^{-1}. \quad (5.47)$$

This result arises because $L \gg RT$ for most planetary values.

5.4.2 Barotropic adjustment

Such a strong wind is likely to be barotropically unstable. The Rayleigh-Kuo condition for that is that the potential vorticity not change sign, that is

$$\frac{\partial q}{\partial y} = \beta - \frac{\partial^2 u}{\partial y^2} \quad (5.48)$$

is of one sign. If u increases monotonically from either the pole or equator (better the former) we can estimate that the magnitude of the wind is limited by the value

$$u \sim \beta a^2. \quad (5.49)$$

This in turn gives us an estimate of the temperature gradient

$$\frac{1}{T} \Delta T \sim \frac{U f a}{L} \sim \frac{\beta a^3 f}{L} \sim \frac{\Omega^2 a^2}{L}. \quad (5.50)$$

if $\Omega = 7 \times 10^{-5} \text{ s}^{-1}$ then $\Delta T/T \sim 0.08$ and $\Delta T \sim 300 \times 0.08 = 24 \text{ K}$.

5.4.3 Convective Overturning

In a conventional (Earth-like) atmosphere there is rising motion where there is heating and descending elsewhere, if only to satisfy the mass continuity equation. There is now no such mass-continuity requirement for compensating subsidence; the vapor could in principle be ascending everywhere with mass continuity being satisfied by condensation and rain. On Earth's atmosphere, regions of subsaturation arise because of downwelling, either between convective updrafts (compensating subsidence) or in large-scale downwelling regions like the subtropics. In a steam atmosphere

5.5 SIMPLE RADIATIVE TRANSFER

We will assume a gray atmosphere (not a particularly good assumption, but it illustrates the point) and make a two-stream approximation. The upward, U , and downward, D , streams of longwave radiation are governed the radiative-transfer equations,

$$\frac{dD_L}{d\tau} = B - D_L, \quad \frac{dU_L}{d\tau} = U_L - B. \quad (5.51a,b)$$

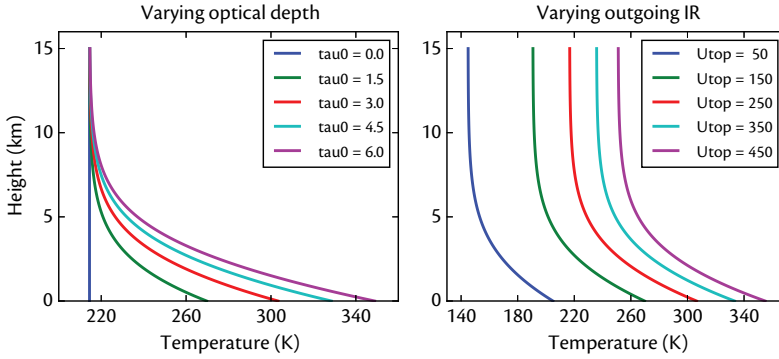


Fig. 5.1: Radiative equilibrium temperature calculated using (1.61), with $H_a = 2 \text{ km}$. Left: outgoing IR (and incoming solar) radiation is 240 W m^{-2} and surface optical depth varies from 0 to 6. Right: the optical depth is 3.0 and outgoing IR radiation varies from 50 to 450 W m^{-2} .

Here, $B = \sigma T^4$ is the radiative flux emitted by a black body, τ is the optical depth and σ is Stefan's constant. The optical depth is related to the geometric height z by $d\tau = -e_L dz$ where e_L is the longwave emissivity. The net flux of longwave radiation is $N_L = U_L - D_L$ and the longwave heating is proportional to the net flux divergence, $-\partial N_L / \partial z$. (Various definitions of τ can be found in the literature, and sometimes a factor of π is to be found in front of B , leading to slightly different forms of these equations.)

A *radiative equilibrium* state has, by definition, no radiative heating. If the atmosphere is transparent to solar radiation then the condition implies that the vertical divergence of the longwave radiation is zero:

$$\frac{\partial(U_L - D_L)}{\partial z} = 0 \quad \text{implying} \quad \frac{\partial(U_L - D_L)}{\partial \tau} = 0. \quad (5.52a,b)$$

The boundary conditions on the radiative transfer equations at the top of the atmosphere are

$$D_L = 0, \quad U_L = U_{Lt} \quad \text{at} \quad \tau = 0. \quad (5.53)$$

and, if the column is in overall radiative equilibrium, U_{Lt} is equal to the incoming solar radiation, so is given.

To obtain a solution we rewrite (5.51) as

$$\frac{\partial}{\partial \tau}(U_L - D_L) = U_L + D_L - 2B, \quad \frac{\partial}{\partial \tau}(U_L + D_L) = U_L - D_L. \quad (5.54a,b)$$

and a little algebra reveals that a solution of these equations that satisfies (1.57b) is

$$D_L = \frac{\tau}{2} U_{Lt}, \quad U_L = \left(1 + \frac{\tau}{2}\right) U_{Lt}, \quad B = \left(\frac{1 + \tau}{2}\right) U_{Lt}, \quad (5.55a,b,c)$$

as can be easily verified by substitution back into the equations. Now suppose that τ has an exponential profile,

$$\tau(z) = \tau_0 \exp(-z/H_a), \quad (5.56)$$

where H_a is the scale height of the absorber, about 4 km perhaps (depending on whether it is water vapour or CO_2). From (1.59c) the temperature then varies as

$$T^4 = \frac{U_{Lt}}{2\sigma}(1 + \tau_0 e^{-z/H_a}). \quad (5.57)$$

Of specific interest to us is the relation between surface temperature and outgoing IR, namely

$$T_s^4 = \frac{U_{Lt}}{2\sigma}(1 + \tau_0) = \frac{1}{2}T_e^4(1 + \tau_0). \quad (5.58)$$

where T_e is the emission temperature. The ground temperature is slightly different and is given by

$$T_g^4 = \frac{U_{Lt}}{\sigma}(1 + \tau_0/2) = T_e^4(1 + \tau_0/2). \quad (5.59)$$

A slightly different form was written down in a paper by Komabayasi (1967), namely

$$\sigma T_g^4 = U_{Lt}(1 + 3\tau/4) \quad (5.60)$$

Neither of these equations are quantitatively accurate, but both have similar behavior as regards the runaway greenhouse effect.

Runaway greenhouse is a termed coined by Andrew Ingersoll in his famous paper of 1967.

5.5.1 Runway Greenhouse Effects

The simplest model of Earth's temperature is simply $\sigma T^4 = S(1 - a)$ where S is the incoming solar flux and a is the albedo, and for later use we define $S_0 = S(1 - a)$. The next simplest model is to take into account the presence of greenhouse gases by supposing that the outgoing IR flux, F is related to surface temperature using one of the formulae above. We'll use (??) and will suppose that the optical depth is a function of the water vapour content of the atmosphere, plus a constant due to a non-condensable greenhouse gas, so that $\tau = \tau_v + \tau_c$ where τ_c is a constant and

$$\tau_v = \kappa m_v = \kappa e/g, \quad (5.61)$$

where κ is the mass absorption coefficient, m_v is the mass per unit area of the a column of the absorber, and e_s is the vapor pressure at the surface.

We can re-arrange (5.59) and use (5.61) to give

$$e = e_r \equiv \frac{2g}{T_e^4 \kappa}(T^4 - (1 + \tau_c/2)T_e^4). \quad (5.62)$$

where T_e is such that $\sigma T_e^4 = U_{Lt}$ and in radiative equilibrium $U_{Lt} = S_0$ and thus is given. For convenience we will call e_r the radiative vapor pressure. We now suppose that e obeys the Clausius–Clapeyron solution which is, as in (5.26)

$$e_s(T) = e_0 \exp \left[\frac{L}{Rv} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right] = e_{00} \exp \left(\frac{-L}{RvT} \right), \quad (5.63)$$

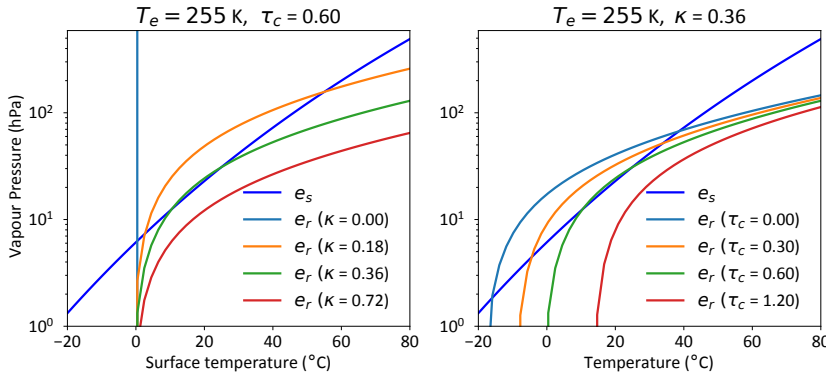


Fig. 5.2: Saturation vapor pressure, e_s and radiative vapor pressure from (5.62) and (5.63) for various values of κ and τ .

where e_0 is the pressure when $T = T_0$ and $e_{00} = e_0 \exp(-L/R^v T_0)$.

Equilibrium solutions are found when $e_s = e_r$. Graphical solutions to this for a given value of incoming solar flux corresponding to $T_e = 255$ K are plotted in Fig. 5.2 for a given value of incoming solar flux (i.e., a given T_e). There are two solutions in the plots shown. For some values there are two solutions and for some there are none. The solution at the higher temperatures is unstable, as may be determined with a little thought: a small positive perturbation of temperature will lead to a greenhouse effect that is larger than the increased emission, and the perturbation will grow and run away. If the perturbation is a cooling one, the system will go to the cooler state. As the solar constant increases, then beginning from the cool state the system warms up, but at some critical temperature there is no solution; that is, it is impossible for the surface to reach a sufficiently high temperature to provide enough outgoing longwave radiation for the system to find an equilibrium.

Aside: If we are given the emitting temperature then we can only find the surface temperature numerically or by the use of special functions (specifically the Lambert W function), for we must solve the following equation for T :

$$e_{00} \exp\left(\frac{-L}{R^v T}\right) = \frac{2g}{T_e^4 \kappa} (T^4 - (1 + \tau_c/2)T_e^4). \quad (5.64)$$

However, it is easy to solve it the other way round, for the solution is

$$T_e^4 = \frac{T^4}{1 + \tau_c/2 + \kappa e_s(T)/2g}. \quad (5.65)$$

where E_s is given by (5.63).

5.6 HEIGHT OF THE TROPOPAUSE

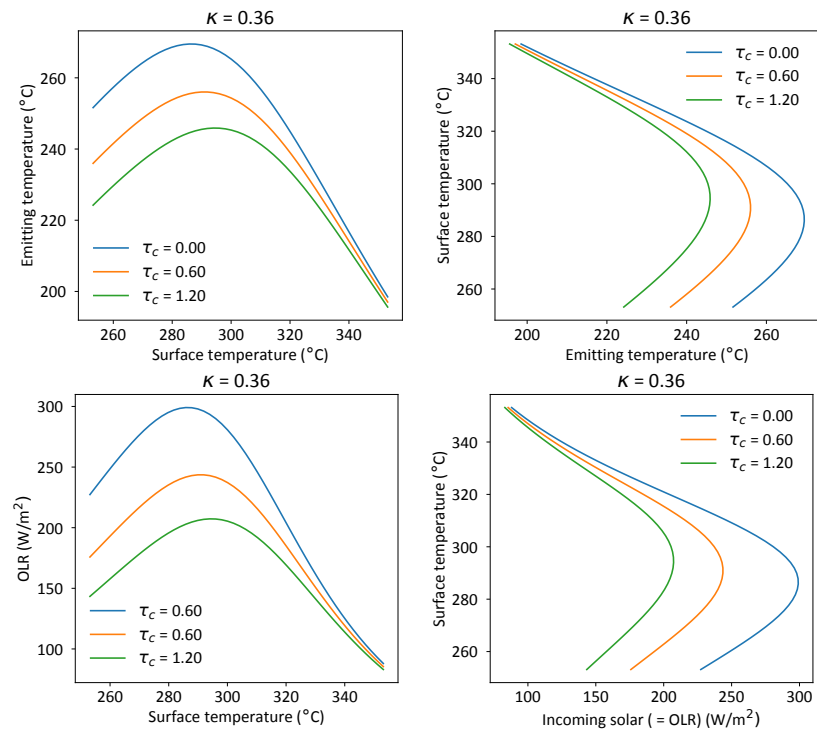


Fig. 5.3: Outgoing IR radiation or emitting temperature plotted against surface temperature, and vice versa, for various values of κ and τ .

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